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Evaluating Metrics for Machine Translation Evaluation: From Paraphrases to Semantic Embeddings

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I declare that I carried out this doctoral thesis independently, and only with the cited sources, literature and other professional sources.

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In Prague on 11 January 2026 signature of the author

I would like to dedicate this thesis to my grandfather Bob, who was always so proud of me for trying to follow in his academic footsteps. I can imagine him shouting with joy upon seeing that I finally managed to submit this thesis. He would certainly reward it with a cheeky poem.

Many thanks go to my parents, who have always supported me – first financially and emotionally, and later with babysitting and supplies of amazing home-made food.

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Abstract: Automatic evaluation of machine translation remains a challenging problem, as commonly used metrics often correlate imperfectly with human judgment and are sensitive to surface-level variation. This thesis investigates two complementary approaches to improving and analyzing machine translation evaluation. The first part explores targeted paraphrasing of reference translations as a means of increasing the robustness of string-based evaluation metrics. Several paraphrasing strategies are examined, including simple substitution-based methods as well as approaches based on statistical and rule-based machine translation systems.

The second part of the thesis focuses on the evaluation of sentence embeddings and their suitability for machine translation evaluation. We analyze the geometric properties of English and multilingual sentence embedding spaces and introduce a new dataset for evaluating Czech sentence embeddings. Furthermore, we examine whether performance on intrinsic embedding evaluation tasks can predict the effectiveness of embedding-based evaluation metrics in downstream machine translation evaluation.

Keywords: machine translation evaluation, automatic evaluation metrics, paraphrasing, sentence embeddings

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1. Introduction

> *'E's not pinin'! 'E's passed on! This parrot is no more! He has ceased to be! 'E's expired and gone to meet 'is maker! 'E's a stiff! Bereft of life, 'e rests in peace! If you hadn't nailed 'im to the perch 'e'd be pushing up the daisies! 'Is metabolic processes are now 'istory! 'E's off the twig! 'E's kicked the bucket, 'e's shuffled off 'is mortal coil, run down the curtain and joined the bleedin' choir invisible!! THIS IS AN EX-PARROT!!*
>
> - Mr. Praline, *Idle et al. [1969]*

The paragraph above is an iconic scene from Monty Python's Dead Parrot sketch, which is arguably one of the most famous examples of paraphrasing in popular culture. It vividly illustrates the extensive variety found within natural languages, showcasing the numerous ways in which a simple fact of a bird's demise can be expressed. It highlights the richness and diversity inherent in language usage.

In fact, according to Wikipedia [Wikipedia contributors, 2023], there are more than a hundred expressions related to death, including idioms, euphemisms, and slang. While humans may appreciate and enjoy the linguistic diversity in various contexts such as poetry, humor, or everyday speech; it makes interpretation and comprehension of the intended meaning more difficult for automated systems. Despite substantial progress in recent years, even advanced models struggle to capture semantic equivalence when paraphrases involve structural reordering, idiomatic usage, or culturally dependent formulations [Michail et al., 2025, Zhou et al., 2024]. For natural language processing (NLP) applications, the ability to recognize that diverse expressions have the same underlying concept remains a persistent challenge.

Let us demonstrate this issue in more detail in two NLP tasks: natural language inference (NLI) and machine translation (MT) evaluation. In NLI, models assess relationships between two texts, a *premise* and a *hypothesis*. The task is to determine whether the meaning of the hypothesis can be inferred from the premise based on their semantic similarity. The relationship is typically categorized as *entailment*, *contradiction*, or *neutral*, depending on whether the hypothesis logically follows from, contradicts, or is independent of the premise. [Dagan et al., 2005].

A major difficulty in NLI is the variability of linguistic expression: two sentences may convey the same meaning using entirely different lexical or syntactic structures (See Table 1.1). Lexical overlap–based methods or surface-level similarity metrics often fail to recognize such paraphrastic relationships, as they rely primarily on word matching rather than deeper semantic understanding. Consequently, models must rely on contextual representations capable of capturing meaning beyond exact wording in order to correctly identify entailment and semantic equivalence. However, even modern embedding-based models might fail at this task, as we show in further detail in Section 4.4.9.

Similarly, automatic machine translation evaluation faces challenges due to the variability and diversity of valid translations for a single source sentence. During

Premise	**Hypothesis**
Less loud.	Please be quiet.
Stop blaming John.	Quit holding John responsible.
And no one will be nicer.	And not a single person will act in a more kind manner.

Table 1.1: Examples of entailment from MultiNLI [Williams et al., 2018a], a large dataset designed to facilitate the development and evaluation of NLI models.

Source	*Banks are testing payment by mobile telephone*					
Hyp.	*Banky*	*zkoušejí*	*platbu*	*pomocí*	*mobilního*	*telefonu*
	Banks	are testing	payment	by	mobile	phone
	Banks are testing payment by mobile phone					
Ref.	*Banky*	*testují*	*placení*	*mobilem*		
	Banks	are testing	paying	by mobile phone		
	Banks are testing paying by mobile phone					

Table 1.2: In this example from Workshop on Machine Translation (WMT) 2012 [Callison-Burch et al., 2012], despite the hypothesis being grammatically correct and having the same meaning as the source sentence, there is only a single overlapping unigram between them. As a result, the hypothesis would receive a very low score when evaluated using n-gram overlap-based automatic evaluation metrics.

evaluation, a *hypothesis* (the system-generated translation) is typically compared to a *reference* translation produced by a human expert. However, there are often hundreds of correct ways to translate the same sentence [Bojar et al., 2013c], and many commonly used evaluation metrics, such as BLEU [Papineni et al., 2002], fail to consider synonymous expressions or variations in word order. As a result, even a perfectly adequate and fluent translation may be assigned a low score, reflecting the metric's reliance on surface-level string matching rather than semantic equivalence.

Figure 1.2 illustrates this issue, showing how a correct and semantically accurate translation can nevertheless receive a lower score due to the lexical difference from the reference. This problem is particularly pronounced in morphologically rich languages with relatively free word order, such as Czech [Bojar et al., 2010]. Consequently, the ability to handle paraphrasing and meaning variation becomes crucial for a reliable MT evaluation. In particular, the role of paraphrasing within MT evaluation constitutes the central motivation of this thesis.

First of all, it should be noted that this thesis was originally proposed a decade ago, when the state-of-the-art in NLP was considerably different from today. At that time, paraphrasing was considered a key strategy for addressing the variability of correct translations in machine translation evaluation. Metrics such as METEOR-NEXT [Denkowski and Lavie, 2014] or TERP [Snover et al., 2009b] incorporated paraphrase tables and synonym matching to reduce the dependence

on exact word overlap. Another approach, how to make string-based metrics might more reliable, was the automatic paraphrasing of reference sentences to cover a wider range of acceptable translation variants. The first part of this thesis revisits this line of research, exploring ways to improve string-based evaluation through the use of paraphrasing resources and targeted reference expansion.

Over the past decade, the field of machine translation evaluation has undergone a significant transformation with the emergence of embedding-based metrics, which capture semantic similarity more effectively than traditional surface-level approaches. Evaluation models based on sentence embeddings or large language models (LLMs) consistently demonstrate a much higher correlation with human judgments than traditional string-based metrics, largely eliminating the need for explicit paraphrasing. Nevertheless, despite the clear advantages of modern evaluation metrics, string-based metrics remain widely used in research and shared tasks.

In a large study, Marie et al. [2021] showed that between 2010 and 2021 more than 100 new evaluation metrics were proposed, yet approximately 99% of MT studies still relied on BLEU for system comparison. Among these newer metrics, only `RIBES` [Isozaki et al., 2010] and chrF [Popović, 2015] have gained modest adoption, while BLEU, TER [Snover et al., 2006b] and Meteor-next [Denkowski and Lavie, 2014] remained the most widely used.

This situation has changed substantially since 2021. It has become more common to publish results using modern metrics such as BLEURT [Sellam et al., 2020a] and COMET [Rei et al., 2020, 2022b], each accumulating more than a thousand citations. There is also a strong pressure to use other metrics in addition to BLEU, as both recent analyzes [Mathur et al., 2020, Freitag et al., 2022] confirm that BLEU and other surface-level metrics often correlate poorly with human evaluation and urge the community to move beyond them.

However, BLEU remains widely used for several practical reasons - the simplicity and explainability of the metric, the comparability with previous work and the lack of transparency and trust in "black-box" neural evaluation models [Leiter et al., 2023a]. Furthermore, reviewers sometimes continue to require BLEU scores for publication, reinforcing its persistence [Marie et al., 2021].

This thesis reflects both perspectives, the old and the new one. The earlier chapters revisit paraphrasing as a once-key strategy for overcoming the limitations of surface-based metrics, which is actually even today still applicable to make BLEU more reliable, even though more modern approaches for creating sentence paraphrases exist. The later chapters shift toward evaluation of sentence embeddings, since metrics based on sentence embedding represent the current state-of-the art in MT evaluation. We created a dataset for thorough testing of Czech sentence embeddings and explored which of them provides the strongest foundation for evaluating Czech translations. We also examine whether it is possible to identify a quality embedding for translation evaluation simply by testing how well it captures the meaning and similarity of sentences. Taken together, the thesis follows the transition in the field: from paraphrase-driven fixes to string-based evaluation to embedding-driven approaches that capture to some extent semantic similarity — while also acknowledging the ongoing reliance on older metrics for practical and historical reasons.

2. Machine Translation Evaluation

2.1 Human Evaluation

From the very beginning of machine translation (MT), researchers have contemplated how to measure the quality of the systems they build. What makes one translation better than another? Can we find a method that allows us to evaluate MT systems in an objective and reliable way?

Direct human evaluation is considered the gold standard – no automatic metric can fully capture the nuances of meaning, style, or naturalness that human readers perceive. However, human evaluation comes with its own challenges – evaluations can be costly, time-consuming, and sometimes inconsistent across annotators.

Even deciding how to conduct a human evaluation is not straightforward. Over the years, various methods have been proposed to manually judge the quality of a hypothesis, ranging from evaluating how well a translation preserves meaning or reads naturally in the target language to the ability to perform a task based on the automatically translated instructions. For example, Sinaiko [1979] examined how the quality and readability of translated technical manuals (including early computer translations) affect technicians' ability to perform maintenance tasks.

Judging machine translation quality is not a simple task. In Table 2.1, we present an example from the WMT15 [Bojar et al., 2015] Russian to English translation task, which shows a Russian source sentence that was translated differently by all thirteen translation systems. This variability in translation demonstrates the difficulty of human evaluation. Consider the task of comparing the aforementioned translations. Which undesirable attribute should be considered the most crucial? Nonsensical sentences, untranslated words, inaccurate translations, shifts in sentence meaning, grammatical errors, or omitted articles? Determining which of these attributes should be considered more severe is subjective and varies among annotators based on their linguistic sensibility.

Furthermore, after seeing several complex translations and making the difficult choice in their comparison, making consistent judgments over time is difficult, which negatively affects reproducibility. Additionally, it is worth noting that a person's assessment of quality evolves during the evaluation process. As evaluators review numerous good and bad translations, their judgments tend to shift, resulting in a low inter-annotator agreement.

As a result, crowd workers tend to be not as reliable [Graham et al., 2017a] as experts, and even automatic metrics based on pre-trained embeddings can outperform human crowd workers [Freitag et al., 2021a], which is in contrast to previous work by Bentivogli et al. [2011].

In Sections 2.1.1 to 2.1.5, we discuss several manual evaluation methods used in the Workshop on Machine Translation general translation and metric tasks that subsequently served as gold standards for the WMT Metrics shared tasks.

Source: Митинг закончился возложением цветов к закладному камню.
Ref.: The rally ended with the placing of flowers at the foundation stone.
1. The meeting ended with the laying of flowers of the chapel stone.
2. The meeting ended with the laying of flowers to stone.
3. Rally ended Bill of sale laying flowers to stone.
4. The meeting ended with the laying on of colours to stone.
5. The rally ended with laying flowers to the zakladnomu stone.
6. The meeting ended with the laying of flowers to the foundation stone.
7. Meeting ended with the placing of colors to the mortgage stone.
8. The rally ended with laying flowers to the memorial stone.
9. Meeting ended with flower-laying to the foundation stone.
10. The meeting ended with laying flowers to clad stone.
11. The rally ended laying flowers to slaid stone.
12. Rally ended laying flowers to закладному stone.
13. The meeting was laying flowers to закладному stone.

Table 2.1: Example from WMT15 [Bojar et al., 2015], Russian to English translation task.

2.1.1 Adequacy and Fluency

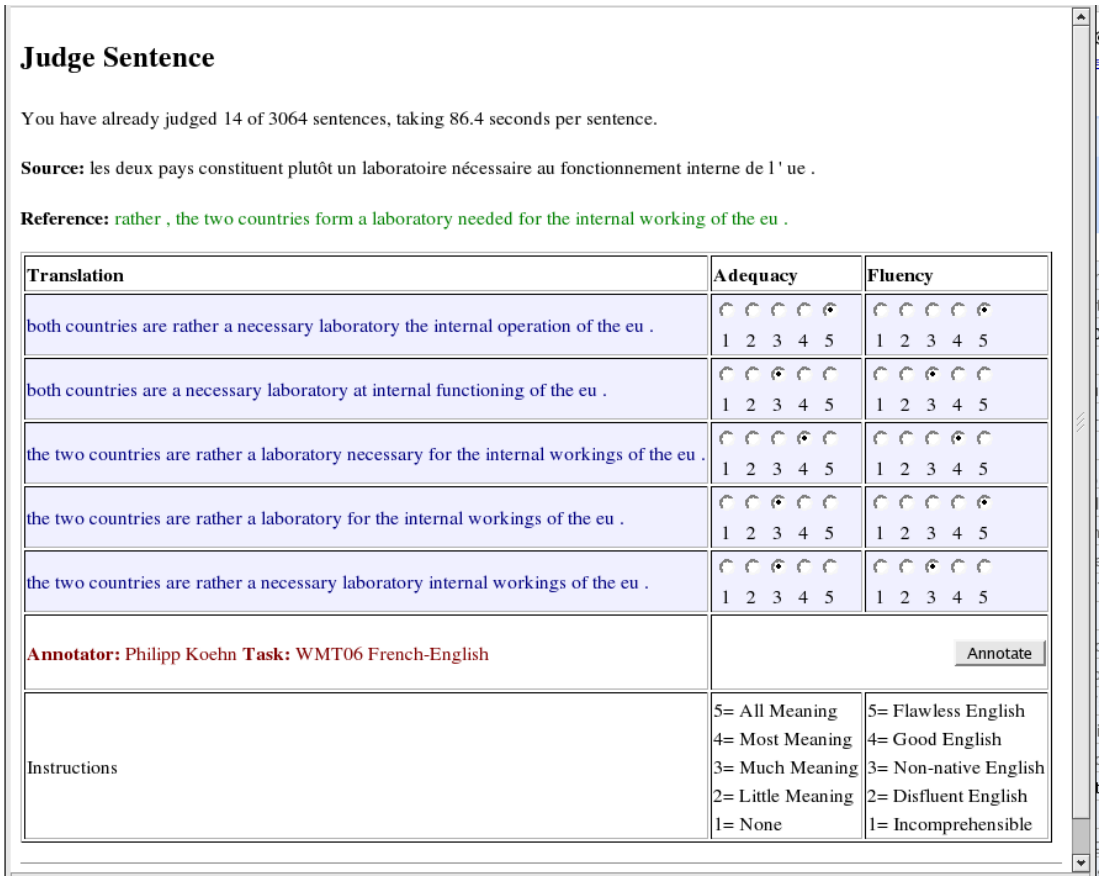
In the first year of the Workshop on Machine Translation (WMT) shared tasks [Koehn and Monz, 2006], the evaluation of machine translation quality was approached from two distinct dimensions: fluency and adequacy. This evaluation methodology is much older than the WMT shared tasks and was already employed in large-scale machine translation evaluations conducted by the Defense Advanced Research Projects Agency (DARPA) in the early 1990s [White et al., 1994].

Fluency captures how natural and well-formed a translation sounds in the target language, independently of the source text. It reflects whether the translation reads like authentic text written by a native speaker. As such, it can be judged by monolingual evaluators who focus on the grammatical correctness, clarity, and naturalness of the hypothesis.

In contrast, adequacy measures how well the translated sentence preserves the meaning of the source text. This dimension requires access to both the source and the target sentences and the evaluators must be fluent in both languages. Adequacy assesses the extent to which the informational content of the source is accurately and completely conveyed in the translation.

In these evaluations, a human judge is typically presented with five translation hypotheses for a given source sentence and asked to rate each on a five-point scale. There is one scale for fluency and another for adequacy. We present an example of the evaluation interface in Figure 2.1

Subsequent studies, including Callison-Burch et al. [2007a] and Bojar et al. [2016] reported a high correlation between annotators' adequacy and fluency scores. In other words, it is difficult to derive meaningful content from a sentence lacking fluency, and similarly, complete understanding of a sentence often requires a high level of fluency, since even minor morphological variation can substantially modify meaning of an entire sentence – a phenomenon particularly



Judge Sentence

You have already judged 14 of 3064 sentences, taking 86.4 seconds per sentence.

Source: les deux pays constituent plutôt un laboratoire nécessaire au fonctionnement interne de l ' ue .

Reference: rather , the two countries form a laboratory needed for the internal working of the eu .

Translation	Adequacy	Fluency
both countries are rather a necessary laboratory the internal operation of the eu .	1 2 3 4 5	1 2 3 4 5
both countries are a necessary laboratory at internal functioning of the eu .	1 2 3 4 5	1 2 3 4 5
the two countries are rather a laboratory necessary for the internal workings of the eu .	1 2 3 4 5	1 2 3 4 5
the two countries are rather a laboratory for the internal workings of the eu .	1 2 3 4 5	1 2 3 4 5
the two countries are rather a necessary laboratory internal workings of the eu .	1 2 3 4 5	1 2 3 4 5
Annotator: Philipp Koehn **Task:** WMT06 French-English		Annotate
Instructions	5= All Meaning 4= Most Meaning 3= Much Meaning 2= Little Meaning 1= None	5= Flawless English 4= Good English 3= Non-native English 2= Disfluent English 1= Incomprehensible

Figure 2.1: Annotation interface used for the manual evaluation of adequacy and fluency. For each task, translations from five randomly selected systems are displayed for a randomly chosen source sentence, no additional information is provided. The screenshot is originally from Figure 3 in Koehn and Monz [2006].

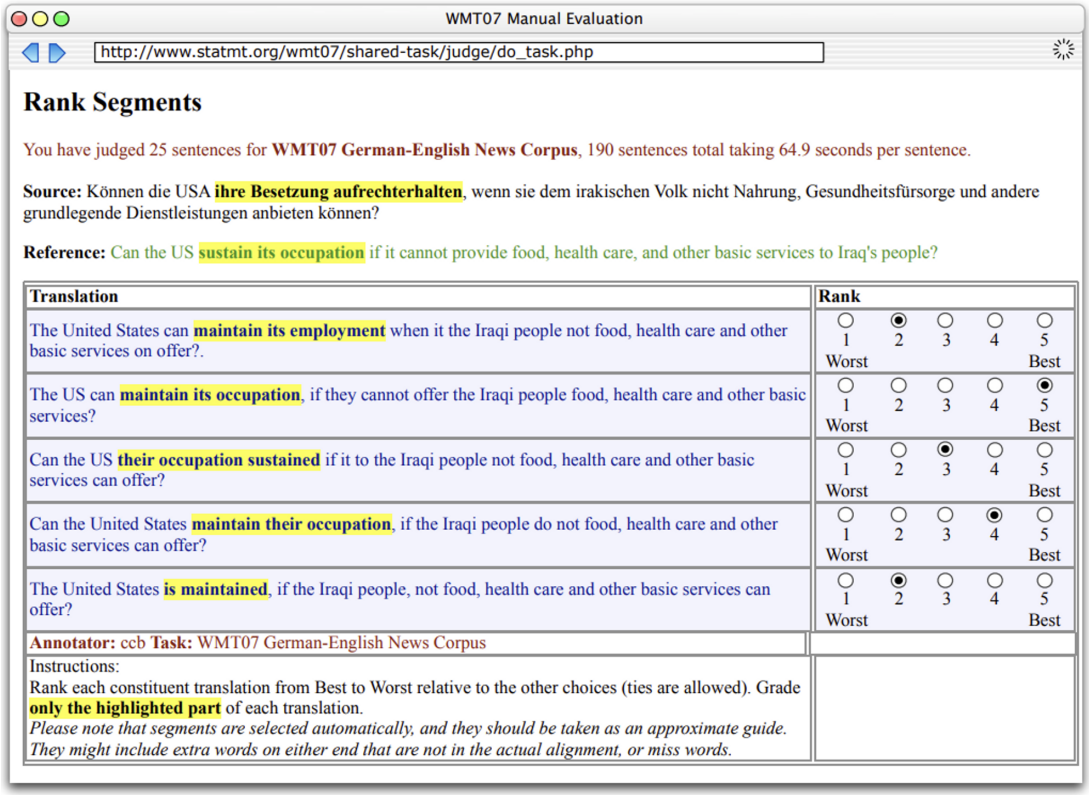


Figure 2.2: Annotation interface for the relative ranking evaluation method. Reprinted from Figure 3 Callison-Burch et al. [2007a]

evident in highly inflected languages such as Czech.

Furthermore, separating adequacy and fluency introduces the complication of recombining the two scores in a meaningful way for tasks like tuning automatic metrics. This challenge motivated later attempts to unite translation quality into a single scale [Denkowski and Lavie, 2010a, Przybocki et al., 2010].

2.1.2 Relative Ranking (RR)

Between 2008 and 2016, the primary technique of human evaluation in WMT was the *relative ranking* method (RR) [Callison-Burch et al., 2007a]. This approach, first proposed by Vilar et al. [2007], presents human annotators—fluent in both the source and target languages—with five translation hypotheses for the same source sentence. Annotators are asked to rank the translations from 1 (worst) to 5 (best), where the numerical ranks do not signify an absolute measure of quality but rather a *relative ordering*. Ties in rankings are allowed.

Presenting five translations simultaneously enables ten pairwise comparisons per single annotation, thus potentially providing more comparative information in a shorter amount of time. Figure 2.2 illustrates this method. However, this setup also increases the cognitive load on annotators and may result in less reliable rankings, especially when differences between translations are subtle. Consequently, some translation tasks limit the comparison to only two translations at a time to reduce complexity and annotator fatigue [Braslavski et al., 2013].

In particular, judges are not given explicit instructions to evaluate specific aspects of the translation system, such as fluency or adequacy. Instead, they are only instructed to assess overall translation *quality*, leaving the precise definition of quality to their own judgment. This absence of exact rules is well motivated – it acknowledges that the annotator, being a fluent speaker of the evaluated language, is capable of making a choice based on their own subjective judgment without requiring explicit guidelines.

Relative ranking yields higher inter-annotator and intra-annotator agreement compared to absolute adequacy-fluency judgments [Ye et al., 2007, Graham et al., 2017a]. This improvement is largely attributed to the variability in how different annotators interpret numerical scales (e.g., what constitutes a score of '3' vs. '4'). Song et al. [2025] further note that relative ranking methods tend to reduce noise and increase consistency in translation judgments.

However, Popović [2021] report that annotator agreement diminishes when translations contain severe errors or when translations are highly similar – highlighting the limitations of subjective ranking in edge cases. Furthermore, Bojar et al. [2011] demonstrate that five-fold rankings can be susceptible to bias introduced by exceptionally high- or low-quality translations.

Aggregating Pairwise Rankings

However, a key limitation of relative ranking is the absence of an absolute measure of quality. Although this method can reliably distinguish better and worse translations, it does not indicate *how much* better one translation is compared to another.

Several aggregation methods have been proposed over time to infer absolute scores from comparative judgments. One of the first approaches was the *win-rate* method, in which individual systems were scored based on how often their output was judged to be better than or equal to other competing systems [Callison-Burch et al., 2010]. This method, also known as the $\geq$ *others* score [Bojar et al., 2011], is calculated as follows:

$$\frac{wins + ties}{wins + ties + loses}$$

Bojar et al. [2011] argue that the $\geq$ *others* score may overestimate the performance of systems that are very similar to others due to issues related to the luck-of-the-draw. This approach could lead to a system with less favorable matchups that consistently compete against stronger machine translation systems, resulting in unfairly low scores. To address this issue, they propose the $>$ *others* score, which became the official aggregation method for WMT12 [Callison-Burch et al., 2012]. The $>$ *others* aggregation method excludes ties and it is computed in the following way:

$$\frac{wins}{wins + loses}$$

However, Lopez [2012] shows that $>$ *others* does not significantly outperform $\geq$ *others*, and instead recommends using a ranking derived from a *Minimum Feedback Arc Set* (MFAS). In this approach, all pairwise preferences between systems are represented as directed edges in a graph (e.g., A $\rightarrow$ B if A is judged

better than B). Because human judgments are often inconsistent, this graph may contain cycles such as A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ A. The MFAS algorithm identifies the smallest set of edges whose removal breaks all such cycles, producing a consistent, acyclic ordering that agrees with the maximum possible number of pairwise comparisons. In other words, MFAS yields a ranking that best fits the observed human preferences while minimizing contradictions.

To mitigate the *luck-of-the-draw* effect, a Monte Carlo model called *Expected Wins* Koehn [2012] was introduced in WMT13 [Bojar et al., 2013a]. This metric is both intuitive and statistically grounded: it estimates the probability that a given system's translation would be ranked higher than that of a randomly selected competing system for a randomly chosen sentence and judge.

Formally, for a system S_i evaluated among a set of systems $\{S_j\}$, and given a set of pairwise rankings summarized by $\text{win}(A, B)$ - the number of times system A is ranked higher than system B - the expected wins score is defined as:

$$\text{score}(S_i) = \frac{1}{|\{S_j\}_{j \neq i}|} \sum_{j, j \neq i} \frac{\text{win}(S_i, S_j)}{\text{win}(S_i, S_j) + \text{win}(S_j, S_i)}$$

This formulation provides a smoothed and probabilistic measure of the relative performance of a system, reducing the influence of outlier comparisons and making it better suited for system-level comparisons across large evaluation sets. Note that this score ignores the ties.

In Bojar et al. [2014], Expected Wins was compared against two alternative latent-variable ranking approaches: a Bayesian model proposed by Hopkins and May [2013] and Microsoft's TrueSkill[Sakaguchi et al., 2014], a Bayesian ranking algorithm originally developed for online gaming [Herbrich et al., 2006]. All three methods produced nearly identical system rankings, with TrueSkill exhibiting slightly lower variance and better robustness to sparse comparisons. Because of these practical advantages, TrueSkill was subsequently adopted as the official aggregation method in WMT14–WMT16 until WMT16, because in WMT17 RR was completely replaced by absolute scoring method and no method of combining the score was needed anymore.

2.1.3 Direct Assessment (DA) and Scalar Quality Metric (SQM)

The Direct Assessment (DA) framework [Graham et al., 2013, 2014, 2017a] was introduced to address key limitations of previous approaches, particularly the lack of absolute quality scores and the cognitive demands of ranking multiple translations simultaneously. Instead of comparing several outputs, DA requires annotators to evaluate each translation independently on a a fine-grained discrete scale from 0 to 100, typically implemented as a slider.

In the DA framework, annotators assess how well a system translation expresses the meaning of a reference text. Depending on the setup, DA can be performed in two main configurations:

- **Bilingual DA:** Annotators are shown the *source sentence* and the *system translation*. They judge how accurately and completely the system translation conveys the meaning of the source. Figure 2.3 shows an example of the annotation interface.

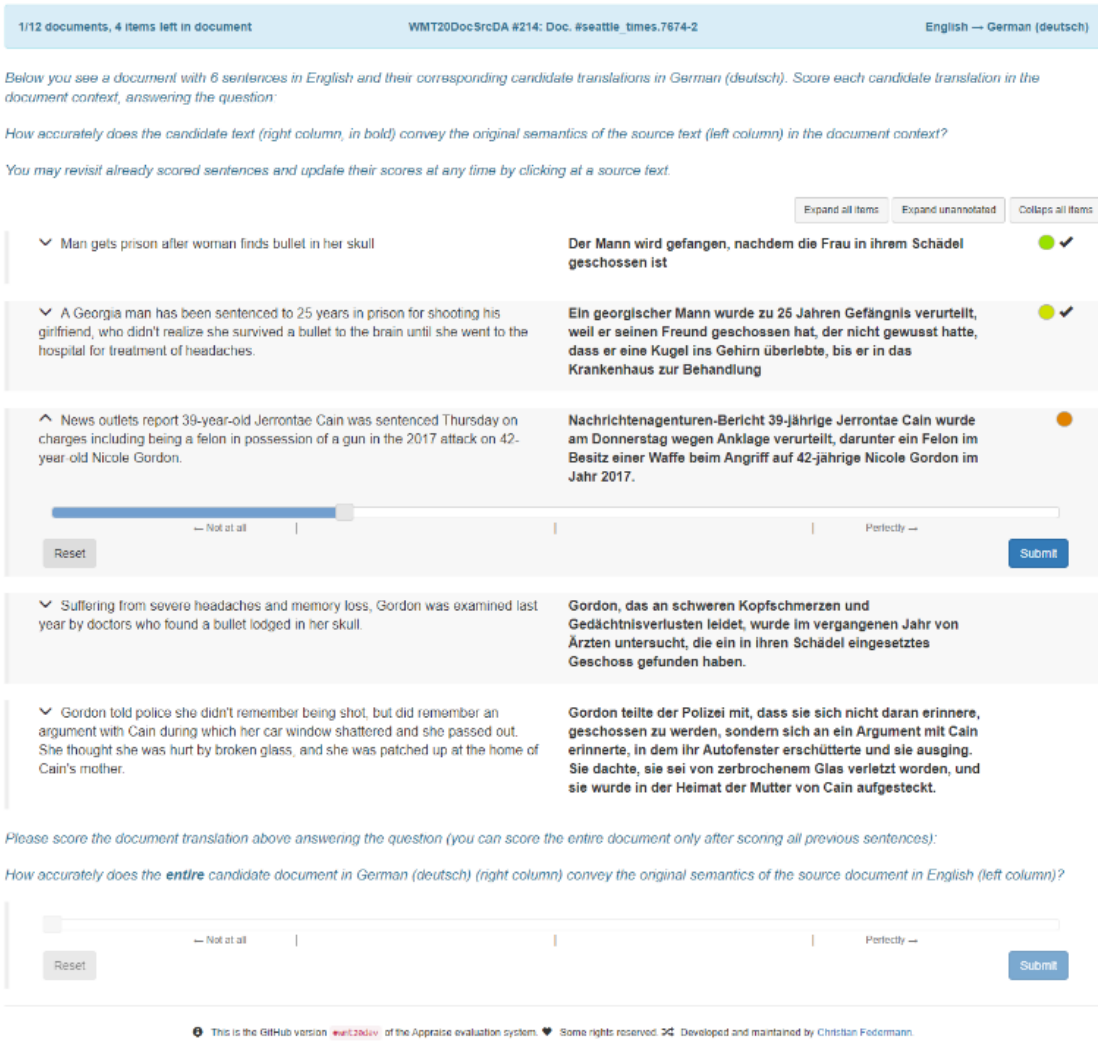


Figure 2.3: Example of a bilingual Direct Assessment (DA) setup. Annotators are shown a full translated document, randomly drawn from anonymized system outputs, and asked to evaluate the quality of each segment as well as the overall document using sliding scales. Reprinted from Figure 6 in Barrault et al. [2020].

- **Monolingual DA:** Annotators see only the *reference translation* and the *system translation*. The reference serves as a meaning proxy for the source, so this setup still targets adequacy, not mere fluency. It allows non-bilingual annotators to participate without loss of validity.

In both cases, annotators assign a score between 0 and 100 to indicate the perceived adequacy of the translation. This scale, proposed by Graham et al. [2013], offers greater granularity and absolute score compared to relative ranking or adequacy-fluency scoring. Individual annotator scores are first *z-normalized* by subtracting each annotator's mean score and dividing by their standard deviation, accounting for differences in how annotators use the rating scale. The system-level scores are then computed by averaging these normalized adequacy scores across all sentences.

Fluency scores were collected using monolingual DA without references; how-

ever, it was not included in the final score, as a translation can appear highly fluent while not conveying the source meaning [Bojar et al., 2016].

DA was piloted in WMT16, initially complementing the official RR scores/ The WMT16 evaluation compared bilingual and monolingual variants and found that both produced reliable adequacy judgments, with monolingual DA offering a scalable alternative for crowdsourcing. Subsequent analysis demonstrated high agreement between crowd-sourced DA and researcher-based RR evaluations [Bojar et al., 2016] and showed that DA produces greater inter-annotator agreement and stronger correlation with expert judgments, especially when there is a large number of judges per sentence. [Graham et al., 2017b].

Based on these findings, DA was adopted as the official human evaluation method for adequacy in WMT from 2017 onward. In WMT18 [Bojar et al., 2018], bilingual DA evaluation was introduced and became the official method for language pairs not translated into English, using native speaker evaluators highly fluent in English, while the monolingual setting was retained for into-English evaluation.

A crucial feature of the DA methodology is the use of *quality control* via calibration items. Each annotator is shown a mix of high- and low-quality reference-based translations with known quality levels (typically generated by introducing artificial degradation). Annotators whose scores do not correlate with the expected quality levels of these control items are filtered out, improving the reliability of collected data. This mechanism allows DA to be deployed on a scale through crowdsourcing, while maintaining high annotation quality [Graham et al., 2017a].

However, DA is not without drawbacks. It relies on subjective interpretation of the scale 0 to 100, and since annotators assess adequacy without specific reference to fluency or grammar, translations that are semantically accurate but disfluent or ungrammatical may still receive high scores. For that reason, some hybrid methods have been proposed to combine DA-style scalar judgment with separate fluency assessments [Freitag et al., 2021a]. *Scalar Quality Metric* (SQM augments DA with seven tick marks with descriptive scale labels (e.g., "perfect translation," "minor errors," "major errors"). DA+SQM was found to correlate well with more advanced Multidimensional Quality Metrics (MQM) annotations [Freitag et al., 2021a], which we introduce in more detail in the following section.

Despite these problems, Direct Assessment has become the standard method for human evaluation in WMT and is widely regarded as a reliable and scalable solution for large-scale MT evaluation. Since WMT22 [Kocmi et al., 2022], DA has been used with SQM [Freitag et al., 2021a] as the official human evaluation protocol for the general task.

2.1.4 Multidimensional Quality Metrics (MQM)

Multidimensional Quality Metrics (MQM) [Lommel et al., 2014] provides a comprehensive and systematic framework for translation quality assessment, originally developed to meet the demand of the professional translation industry for robust and standardized quality assurance [Lommel et al., 2013].

Unlike earlier methods focused on linguistic accuracy or grammatical correctness, MQM incorporates multiple quality dimensions through a hierarchical

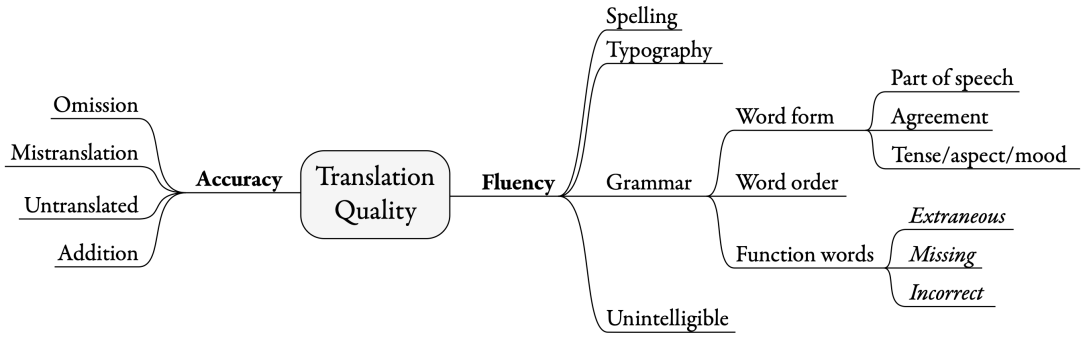


Figure 2.4: Original MQM hierarchy of errors. Reprinted from Figure 1 in Lommel et al. [2014]

taxonomy of error categories. The framework requires evaluators to identify specific error types – such as mistranslations, omissions, additions, word-order issues, and grammatical mistakes, while also assessing their *severity* levels (Minor, Major, Critical). The original taxonomy is presented in Figure 2.4.

This fine-grained approach enables a more interpretable analysis of translation quality compared to simple numeric scores. Freitag et al. [2021a] showed that MQM-based rankings by expert annotators correlate more strongly with professional human judgments than DA or RR, and often reveal system-specific weaknesses masked by aggregate scores.

Although MQM has proven to be highly effective and is widely used for quality assurance in the translation industry, this level of detail comes at a cost. The annotation process is considerably more time-consuming and expensive than other evaluation methods, requiring trained experts capable of applying the error taxonomy consistently. This limits scalability, especially in large-scale, multilingual settings like WMT. Aggregating structured MQM annotations into system-level scores also introduces additional complexity.

Due to its high quality Freitag et al. [2022], MQM has been the gold standard for the WMT Metrics shared task since WMT21, however due to its complexity, they are collected only for three language pairs each year.

2.1.5 Error Span Annotation (ESA)

More recently, WMT24 [Kocmi et al., 2024a] introduced the *Error Span Annotation* (ESA) framework [Kocmi et al., 2024b]. ESA represents a hybrid inspired by DA and MQM: DA is simple and fast, but less informative and may be less reliable, while MQM is very informative and reliable, but expensive and time-consuming. ESA combines them both. Annotators first mark error spans with just *major* or *minor* severity, then assign a scalar overall quality score. Example of the interface for ESA evaluation is presented in Figure 2.5.

In the official WMT24 evaluation setup, only the scores are used to compute the final system scores, following the standard z-normalization and averaging procedure used for DA+SQM. The error spans serve as a preliminary step that encourages annotators to identify specific issues before assigning an overall score,

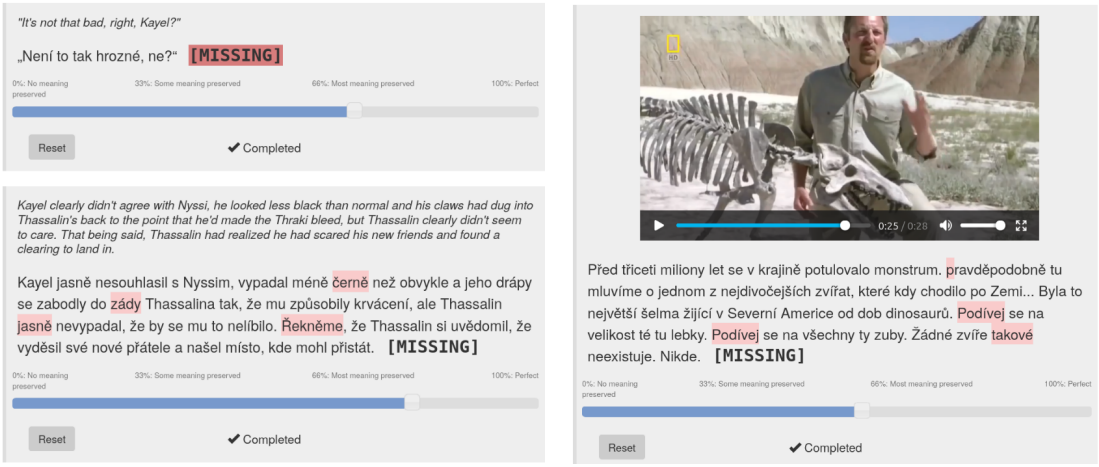


Figure 2.5: Two screenshots illustrating the ESA interface and the corresponding annotator instructions. ESA displays multiple segments from a document simultaneously, along with any associated video sources. After highlighting the individual error spans, the annotator assigns a final segment-level score on a 0–100 scale. Reprinted from Kocmi et al. [2024a].

which improves consistency and provides diagnostic information, but they are not numerically aggregated into the final metric. Kocmi et al. [2024b] proposes optional MQM-like span-weighted scoring formulas, but they are not used for systems ranking in WMT tasks.

The authors claim ESA reduces fluency bias compared to DA+SQM, though direct human parity with MQM remains untested on shared data. Freitag et al. [2024] confirm that top automatic metrics yield near-identical system-level rankings for MQM and ESA.

The 20-year evolution of WMT human evaluation methods is summarized in Table 2.2.

Year	Citation	Primary Method	Notes
2006	Koehn and Monz [2006]	Adequacy & Fluency	segment-level averages.
2007	Callison-Burch et al. [2007b]	Adequacy & Fluency	experimental RR
2008	Callison-Burch et al. [2008]	RR	adequacy & fluency as supporting metrics.
2009	Callison-Burch et al. [2009]	RR	"≥ others"; post-editing trialed
2010	Callison-Burch et al. [2010]	RR	"≥ others"
2011	Callison-Burch et al. [2011]	RR	"≥ others"
2012	Callison-Burch et al. [2012]	RR	"> others"
2013	Bojar et al. [2013a]	RR	"expected wins".
2014	Bojar et al. [2014]	RR	TrueSkill
2015	Bojar et al. [2015]	RR	TrueSkill
2016	Bojar et al. [2016]	RR	TrueSkill; exp. monolingual DA
2017	Bojar et al. [2017b]	monolingual DA	
2018	Bojar et al. [2018]	monolingual DA	bilingual DA piloted
2019	Barrault et al. [2019]	DA*	
2020	Barrault et al. [2020]	DA*	
2021	Akhbardeh et al. [2021]	DA*	MQM in metrics task
2022	Kocmi et al. [2022]	DA*+SQM	MQM in metrics task
2023	Kocmi et al. [2023]	Bilingual DA+SQM	MQM in metrics task.
2024	Kocmi et al. [2024a]	ESA	MQM in metrics task
2025	Kocmi et al. [2025]	ESA + MQM hybrid	AUTORANK: combines ESA, MQM, LLM-as-judge

Table 2.2: Evolution of primary methods used for human evaluation methods in the WMT General MT (News) Task (2006–2025). DA* indicates that evaluations were conducted as bilingual DA for translations from English into other languages, and as monolingual DA for translations into English or between non-English language pairs. This setup reflects the fact that there are many annotators who are native speakers of non-English languages with strong English proficiency, contrasted with the relative scarcity of native English speakers with high proficiency in other languages.

2.2 Automatic Translation Evaluation

In Section 2.1, we discuss several methods of human evaluation. Each method has its distinct advantages, but all share one significant limitation: human evaluation is slow and expensive. Due to this constraint, it is practically impossible to rely on human evaluation for incremental development and tuning of machine translation systems. Reliable automatic MT evaluation metrics are, therefore, essential for these tasks.

Compared to human judgment, automatic evaluation metrics offer significant advantages, particularly in terms of speed, cost-efficiency, and reproducibility. Given the fast pace and dynamic nature of machine translation research, a large-scale and timely human evaluation is unfeasible. As a result, development and refinement of automatic evaluation metrics has become a highly active and productive area of research over the past two decades.

The objective of an automatic evaluation metric is to produce a numerical score that can be regarded as a measure of the quality of an MT system. Typically, the absolute score generated by a metric is not meaningful on its own. It is primarily used to rank machine translation systems. A system with a higher score or lower error rate is considered superior.

Although automatic measures are an invaluable tool for the day-to-day development of machine translation systems, they are an imperfect substitute for human assessment of translation quality. We evaluate the quality of an automatic metric by comparing it to scores that were produced by human judges: the better the metric, the higher it should agree (correlate) with manually made human judgment.

Similarly to a human annotator, automatic MT metrics typically receive the following inputs:

- The translation *hypothesis* generated by the MT system,
- At least one human-produced *reference* translation, and
- Optionally, the original *source* sentence.

Traditionally, MT evaluation has focused on comparing a system output to reference translations, often ignoring the source sentence entirely. However, recent metrics based on neural embeddings and large language models are increasingly incorporating source-side information to better evaluate meaning preservation. Moreover, while *reference-free* evaluation, also known as *quality estimation*, historically performed worse than reference-based metrics, this distinction is becoming less clear. Recent advances in neural and LLM-based metrics have shown that reference-free approaches can, in some cases, rival or even outperform reference-based methods [Freitag et al., 2021b, Rei et al., 2021, Moosa et al., 2024].

Traditional metrics are constrained by their reliance on surface-level string matching, making them sensitive to variations in wording, even when those variations preserve the underlying meaning. As a result, correct phrase variants in a system output are often penalized, leading to a mismatch between metric scores and human judgments of translation quality.

This chapter reviews related work from two complementary perspectives. First, we examine machine translation evaluation metrics starting with traditional

string-matching metrics such as BLEU, TER, and chrF, which depend on an n-gram or character overlap and thus struggle with semantic equivalence beyond lexical similarity. We then consider metrics that extend these approaches with paraphrase support, such as Meteor and TERp, which incorporate linguistic resources to better accommodate lexical variation. After mentioning probability-based metric, we explore recent metrics based on sentence embeddings or LLMs, which leverage contextual language models to capture deeper semantic similarity and are inherently more robust to paraphrastic differences.

Second, we review methods for generating paraphrases of reference sentences, to increase the diversity of reference sets and reduce evaluation bias. These methods include human-authored paraphrases, rule-based systems, and neural paraphrase generation techniques such as back-translation and transformer-based models. By creating multiple reference variants, these approaches help reduce evaluation bias against valid but non-literal translations and improve alignment with human preferences.

Together, these two perspectives, automatic evaluation metrics and paraphrased references, show how paraphrasing affects the evaluation of machine-translated output into more robust and reliable assessment frameworks.

2.3 Metrics for Automatic Evaluation

2.3.1 String-based

Traditional machine translation evaluation metrics are based on string similarity between a reference translation and a translation hypothesis. Their main advantages are:

- **Low computational cost:** traditional metrics often rely on efficient string-matching algorithms at the character and/or token level. While some metrics might involve token shifting, which could be more resource-intensive for long translations, their calculation is easily parallelizable and does not require GPU processing.
- **Explainability:** Traditional metrics are straightforward to compute and understand, particularly for short segments. This transparency is helpful in error analysis and debugging. However, ease of interpretation does not necessarily imply that the scores reliably reflect overall translation quality.
- **Language Independence:** Most traditional metrics are designed to be largely language-agnostic and can be applied without modification across different language pairs, with only minor exceptions.

However, traditional metrics also come with substantial disadvantages:

- **Poor Correlation with Human Judgments:** This is the most significant drawback compared to modern neural metrics. Recent large-scale comparisons [Mathur et al., 2020, Freitag et al., 2022] consistently show that lexical metrics perform poorly compared to neural alternatives.

- **Tokenization and Implementation Inconsistencies:** Traditional metrics often rely on pre-tokenized input, and tokenization is handled outside the metric itself. As a result, the scores can vary depending on the specific tokenization pipeline used. Furthermore, multiple implementations of these metrics exist, some of which differ subtly or significantly in behavior. These inconsistencies can lead to misleading or non-reproducible results. To address this, Post [2018] introduced the SacreBLEU[1] framework, which standardizes tokenization and metric computation for reproducible evaluation.

BLEU

BLEU (BiLingual Evaluation Understudy; Papineni et al. [2002]) is one of the earliest and most widely adopted automatic metrics for machine translation evaluation. It is based on the hypothesis that even though there are many different ways to correctly translate a sentence, high-quality translations will tend to share overlapping unigrams and n-grams (i.e., contiguous word sequences of length n) with reference translations. The metric was originally designed with multiple reference translations in mind, under the assumption that a set of references would approximate a more complete representation of valid translation variants. However, in practice, most MT evaluations use only a single reference, severely limiting the effectiveness of BLEU.

BLEU is computed as modified n-gram precision: the proportion of n-grams in the candidate translation that also appear in the reference(s). BLEU does not incorporate recall, as recall is difficult to define when evaluating against multiple references simultaneously. To compensate for missing recall, BLEU includes a brevity penalty (BP), discouraging too-short hypotheses. BLEU is formally defined as:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right),$$

where p_n is the modified n-gram precision for n-grams of size n, w_n is the weight assigned to each precision (typically $w_n = \frac{1}{N}$ for uniform weighting, N is the maximum n-gram length, typically $N = 4$) and BP is the brevity penalty, defined as:

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-\frac{r}{c})} & \text{if } c \leq r \end{cases}$$

This makes BLEU biased toward precision, potentially ignoring parts of the reference translation that are missing from the hypothesis. Recall is a crucial component to capture the completeness of content coverage, and its absence in BLEU can lead to underestimating the quality of longer, more content-rich hypotheses [Banerjee and Lavie, 2005]. Despite these limitations, BLEU remains widely used due to its simplicity, language independence, and computational efficiency. It has become the de facto standard for the evaluation and optimization

[1] `https://github.com/mjpost/sacrebleu`

of MT systems and is still used as a primary tuning objective in many statistical and neural MT systems, although there are currently more reliable metrics.

However, the reliability of BLEU has been increasingly questioned. Callison-Burch et al. [2006] and Koehn and Monz [2006] observed that the correlation of BLEU with human judgment is weaker than previously assumed, especially at the sentence level [Blatz et al., 2004]. Moreover, comparing systems based solely on BLEU scores can be misleading Post [2018]. In particular, BLEU is sensitive to surface-level variation and often fails to reward valid paraphrases or alternative phrasings. This issue is magnified in morphologically rich and syntactically flexible languages such as Czech, where even grammatically and semantically correct translations may be penalized for differing word order or inflectional forms [Bojar et al., 2010].

In addition, generating high-quality reference translations is a resource-intensive process. For example, the creation of reference translations by the Linguistic Data Consortium involves professional translation agencies, the adherence to detailed guidelines, and rigorous quality control Strassel et al. [2006]. Due to these costs, MT evaluations frequently rely on only one reference sentence per source sentence. As a result, BLEU is often applied in contexts for which it was not originally designed, further reducing its reliability.

In summary, BLEU's influence on the field of machine translation is undeniable. However, its reliance on exact n-gram overlap, lack of recall, and sensitivity to word order make it less suitable for evaluating the full range of acceptable translations, especially when only a single reference is available. This has motivated the development of alternative metrics, including those that incorporate paraphrase support or semantic similarity through neural embeddings.

In a large scale comparison, Marie et al. [2021] showed that despite the development of more than 100 new evaluation metrics between the years 2010-2021, a staggering 99% of the research papers in machine translation continue to rely on BLEU to evaluate translation quality and rank systems. However, in recent years, experts Mathur et al. [2020], Freitag et al. [2022] have proven that clinging to BLEU can mislead research conclusions and mask true progress (or regress) in translation quality. Consequently, they urge the community to reconsider the use of BLEU and other overlap-based metrics for MT evaluation and use more reliable metrics based on neural networks that demonstrate a substantially better correlation with human evaluations, especially when these evaluations are carefully performed (e.g., expert MQM annotations, which are described in more detail in **??**).

While BLEU can still be computed for legacy reasons, meaningful evaluation of MT quality in 2026 requires metrics or methods that account for semantic adequacy. BLEU's enduring presence is now mostly a matter of convention and convenience, not because it is theoretically or empirically the most adequate measure.

TER

Translation Error Rate (TER) [Snover et al., 2006b] evaluates the quality of machine translations by measuring the amount of editing operations needed to make a machine-translated text match a human-translated reference. Essentially, it quantifies the effort required to 'fix' the machine translation output.

Formally, given a machine translation hypothesis H and a reference translation R, the edit distance E is defined as the sum of the following four editing operations required to transform H into R:

- Insertion: Adding a single token to H.
- Deletion: Removing a single token from H.
- Substitution: Replacing one token in H with another.
- Shift: Translocating a contiguous block of tokens within H to better align with R.

TER is then computed as:

$$\text{TER} = \frac{E}{|R|},$$

where $|R|$ denotes the number of tokens in the reference (or the average across multiple references). Lower TER indicates better translation quality, directly corresponding to a reduced editing effort. If there are multiple references, the minimum error rate across references is typically used.

Snover et al. [2006b] show that the single-reference variant of TER correlates as well with human judgments of MT quality as the four-reference variant of BLEU. Agarwal and Lavie [2008] explore metric variants tuned for better human correlation in the WMT'08 shared task. They compare Meteor, *M-BLEU* (BLEU with improved preprocessing), and *M-TER* (TER with the same preprocessing) and find that M-TER achieves the highest system-level correlation with human rankings, surpassing both M-BLEU and Meteor

chrF

chrF (Character n-gram F-score) Popović [2015] is a metric that computes the F-score over character n-gram matches between the hypothesis and reference translations. It captures both precision and recall and is particularly effective in scenarios where word-level matches are sparse or unreliable. Unlike BLEU, which focuses on precision and uses a brevity penalty, chrF combines precision and recall symmetrically, making it more sensitive to partial matches and morphological variations.

Formally, the chrF score is calculated as the harmonic mean of character n-gram precision and recall:

$$\text{chrF} = (1 + \beta^2) \cdot \frac{P \cdot R}{\beta^2 \cdot P + R},$$

where P is the precision, R is the recall, and β is a weight that gives β times more importance to recall than to precision – if $\beta = 1$, they have the same importance, it is commonly set to 2, favoring recall.

chrF++ [Popović, 2017] extends chrF by incorporating word-level n-gram F-scores alongside the character-level n-grams. This hybrid approach captures both local character-level similarity and broader syntactic or lexical patterns at the word level, providing a more comprehensive assessment of translation quality.

The inclusion of word n-grams allows CHRF++ to better reflect fluency and adequacy in translation, improving correlation with human judgments in a variety of languages and systems.

The combined metric leverages a weighted average of the character- and word-level F-scores:

$$\text{CHRF++} = \lambda \cdot \text{CHRF}_{\text{char}} + (1 - \lambda) \cdot \text{CHRF}_{\text{word}},$$

where $\text{CHRF}_{\text{char}}$ is the character-level F-score, $\text{CHRF}_{\text{word}}$ is the word-level F-score and $\lambda \in [0, 1]$ controls the relative importance of character and word components.

2.3.2 String-based with paraphrase support

To address the limitations of traditional metrics, particularly their inability to capture true semantic equivalence, second-generation metrics incorporate linguistically motivated resources, typically in the form of paraphrase tables, while still largely relying on an n-gram overlap.

These metrics show better correlation with human judgment, but their widespread adoption is limited by reliance on language-specific resources, typically developed for English. This dependence limits one of the key advantages of n-gram overlap metrics: language independence and universal applicability across languages. As a result, there is a trade-off between the linguistic-rich strategy for better performance and the applicability of simple string level matching.

String matching is hardly discriminative enough to reflect human perception and there are many metrics that compute their score based on rich linguistic features and matching based on parse trees, POS tagging, or textual entailment, for example, [Liu and Gildea, 2005, Owczarzak et al., 2007, Amigó et al., 2009, Padó et al., 2009, Macháček and Bojar, 2011, Han et al., 2012]. In the following sections, we introduce TERP and METEOR-NEXT in more detail.

TERp

TERP (Translation Error Rate plus[2], Snover et al. [2009b]) extends the basic Translation Error Rate. TERP uses the same editing operations as TER - Insertion, Deletion, Substitution, and Shift, but introduces the following new editing operations:

- Stem Substitutions: words are substituted if they share the same stem based on the Porter stemming algorithm [Porter, 1980]
- Synonym Substitutions: words are substituted if they are synonyms according to WordNet [Fellbaum, 1998]
- Phrase Substitutions: phrase substitutions are identified by pre-computed paraphrase table, extracted automatically using the pivot-based method [Bannard and Callison-Burch, 2005] on English-Arabic newswire parallel corpus.

[2]The metric is named after the Terrapin the mascot of University of Maryland.

Because these new edit operations cannot be treated in the same way as the original TER operations, TERp introduces distinct weights for each edit type. This allows the metric to be customized in terms of its sensitivity to different types of translation errors. For instance, the cost assigned to a synonym substitution is deliberately set lower than that of an unrelated word replacement, reflecting the relative severity of the error in terms of meaning preservation.

TERp is then computed as:

$$\text{TER}_\text{P} = \frac{\sum_i c_i \cdot e_i}{|R|},$$

where e_i is the count of the edit operation i, c_i is the cost (weight) of operation i and $|R|$ denotes the number of tokens in the reference (or the average across multiple references). Lower TERp indicates better translation quality, directly corresponding to a reduced editing effort.

Snover et al. [2009a] demonstrate that TERp significantly outperforms the original TER metric, highlighting the effectiveness of the enhancements incorporated into TERp.

Meteor-next

In 2005, Banerjee and Lavie introduced Meteor (Metric for Evaluation of Translation with Explicit Ordering), explicitly designed to address weaknesses of the BLEU metric. These shortcomings included BLEU's lack of recall, reliance on high-order n-grams, absence of explicit word-level matching between the hypothesis and reference sentences, and its use of geometric averaging for combining n-gram scores Banerjee and Lavie [2005].

In this section, we provide a detailed description of a subsequent version of the Meteor metric, known as Meteor-next[3], which was introduced together with the Meteor paraphrase tables in Denkowski and Lavie [2010b]. We present the Meteor paraphrase tables in more detail in Section 3.1.2.

The Meteor-next evaluates a machine translation hypothesis against a reference translation by computing a similarity score based on the alignment of words between the two sentences. In the case of multiple references provided, the hypothesis is compared against each reference separately and the highest score is selected.

In the first step, the search space of possible alignments is constructed. For a single hypothesis-reference pair, the space of possible alignments is constructed by identifying all possible matches of words and phrases between the two sentences, using the following matching criteria:

- **Exact matching:** Words match only if their surface forms are identical.
- **Stem matching:** Words are matched if they share identical stems after being processed by a language-specific Snowball Stemmer [Porter, 2001].
- **Synonym matching:** Words match if they belong to the same synset according to WordNet [Miller and Fellbaum, 2007].

[3]There is a newer version 1.5 of Meteor-next introduced in Denkowski and Lavie [2014], which is computed slightly differently - it also distinguishes between content and function words. However, we describe in more detail the previous version 1.4 as it was used in our experiments.

- **Paraphrase matching:** Phrases match if they are listed as paraphrases in the METEOR paraphrase tables.

In its basic configuration for evaluating Czech sentences, METEOR-NEXT supports two types of matching: exact and paraphrase. However, as discussed in Section 3.3, Czech paraphrase tables contain a lot of noise (pairs that do not actually represent true paraphrases), which harms the performance of the metric. These incorrect paraphrases can even lead to rewarding mistranslated or untranslated words.

The METEOR-NEXT score is calculated as follows. First, the metric builds a space of possible alignments and then chooses the alignment that maximizes coverage between a hypothesis and a reference sentence. Given an alignment, METEOR-NEXT computes precision (P) and recall (R):

$$P = \frac{\text{Number of aligned words}}{\text{Number of words in hypothesis}}$$

$$R = \frac{\text{Number of aligned words}}{\text{Number of words in reference}}$$

Precision and recall are combined into an F-score, typically with more emphasis added on recall (α is usually set to 0.9):

$$F_{\text{mean}} = \frac{P \times R}{\alpha \times P + (1 - \alpha) \times R}$$

To account for word-order differences, Meteor penalizes alignments based on the number of chunks (contiguous matched segments). Let ch be the number of chunks (continuous segments of aligned words) and m be the total number of matched words. METEOR-NEXT computes a penalty Pen:

$$Pen = \gamma \times \left(\frac{ch}{m}\right)^{\beta},$$

where the default parameter values are typically $\gamma = 0.5$, $\beta = 3$. The penalty is higher if words are matched but not in the correct order (more chunks). Finally, the Meteor score combines the F-score and the penalty:

$$\text{METEOR-NEXT} = (1 - Pen) \times F_{\text{mean}} = (1 - \gamma \times \left(\frac{ch}{m}\right)^{\beta}) \times \frac{P \times R}{\alpha \times P + (1 - \alpha) \times R}$$

This results in a single value, ranging from 0 (no similarity) to 1 (perfect match).

METEOR-NEXT generally performs better than BLEU in WMT evaluations; however, it has its own issues. For example, it can be overly permissive in its matching, aligning words that are technically synonyms but inappropriate in a context or even not synonyms due to noise in its paraphrase tables, leading to inflated scores for poor translations. Guo and Hu [2019] extends METEOR-NEXT with syntactic paraphrases from PPDB [Ganitkevitch and Callison-Burch, 2014] and shows it further improves correlation with human judgment.

2.3.3 Embedding based metrics

Marie et al. [2021] reported that in 2019 and 2020, approximately 82% of the MT papers published on the ACL Anthology evaluated their systems only using BLEU scores, without employing any additional metric or human evaluation. Their analysis also shows that 89% of the newly proposed MT evaluation metrics have never been reused since their introduction. However, following the widespread adoption of pre-trained neural embeddings, the field of MT evaluation has slowly begun to experience a gradual shift from relying solely on traditional string-based metrics towards incorporating neural embedding-based metrics, although BLEU remains extensively used.

Metrics such as COMET [Rei et al., 2020], BERTscore [Zhang et al., 2020], and BLEURT [Sellam et al., 2020a] have gained considerable popularity in the community, each receiving more than a thousand citations. These newer metrics are now increasingly used in MT research papers, reflecting a growing recognition of their better alignment with human judgments.

Embedding-based neural metrics use pretrained language models such as BERT Devlin et al. [2019], RoBERTa Liu et al. [2019b], or XLM-R Conneau et al. [2019] to generate contextual word or sentence embeddings. Unlike traditional metrics that rely on surface-level word overlap or paraphrase tables, these models compare representations in a semantic vector space, allowing a more nuanced evaluation of meaning preservation in translations. Unfortunately, this is also their weakness - because the embeddings spaces tend to cluster semantically or syntactically similar tokens, embeddings-based metric might be vulnerable to critical errors involving numerical values and named entities. Such elements can have similar embeddings to incorrect alternatives, potentially leading to overly optimistic evaluations of flawed translations.

Embedding-based evaluation metrics differ in the methodology for using semantic representations. Some metrics, such as BERTscore, YiSi [Lo, 2019] or UniTE [Wan et al., 2022], compute similarity scores directly between embedding representations using measures like cosine similarity or contextual alignment derived from attention-based models. These approaches typically rely on pre-trained multilingual models and operate in an unsupervised manner, without task-specific fine-tuning. In contrast, other metrics—such as COMET [Rei et al., 2020] and BLEURT [Sellam et al., 2020b]—are fine-tuned on human-annotated data to predict translation quality scores that closely align with human evaluations.

More recently, *preference-trained* evaluation metrics—also known as *reward-model-style* metrics—have gained increasing attention in the machine translation (MT) community. These do not rely on prompting an LLM for evaluation. Instead, they train a smaller (often open-weight) language model explicitly on human preferences (e.g., "translation A is better than B"), rather relying on DA or MQM annotations. This design choice has made it particularly robust on low-quality translations, where absolute scoring may be noisy or inconsistent. These metrics include MT-Ranker [Moosa et al., 2024] or ReMedy [Tan and Monz, 2025], which recently achieved the current state-of-the-art on both segment- and system-level evaluation surpassing much larger models.

Embedding-based metrics are typically open-source, reproducible, and efficient to compute compared to LLMs, which make them better suited for integration into development workflows. In the following sections, we present an

overview of the most widely used and best performing embedding-based neural metrics for machine translation evaluation.

BERTscore

BERTscore [Zhang et al., 2020] is a reference-based metric that does not require a source sentence. Specifically, for each token in a hypothesis, BERTscore identifies the most similar token in the reference, and vice versa. These similarity scores are then aggregated to compute precision, recall, and F1 scores that reflect how well the hypothesis preserves the meaning of the reference. In this formulation, recall captures how much of the reference content is covered and precision assesses how much of the hypothesis is semantically relevant.

It is fully unsupervised and does not involve any fine-tuning on human judgment data, which makes it fast and easy to apply across various domains and language pairs. It also supports multilingual evaluation when used with appropriate multilingual models like XLM-R.

Empirical evaluations in the original article and subsequent benchmarks [Hanna and Bojar, 2021, Freitag et al., 2022] demonstrate that BERTscore correlates better with human judgments than BLEU and other traditional metrics, especially at the segment level.

Despite its advantages, one limitation of BERTscore is its sensitivity to token alignment and strong reliance on embedding similarity, which may not always reflect nuanced translation errors or pragmatic changes. However, it remains a popular and accessible metric, particularly useful for tasks that emphasize semantic adequacy.

COMET

The COMET (Crosslingual Optimized Metric for Evaluation of Translation; Rei et al. [2020]) framework is built upon a multilingual transformer architecture, XLM-RoBERTa [Conneau et al., 2019] is used as the default encoder. COMET incorporates contextual representations of the source sentence, the machine-translated hypothesis, and the reference translation. These representations are pooled and passed into a shallow feedforward neural network trained to predict human quality judgments.

The standard training objective is regression[4], optimizing the model to align its predictions with human-assigned DA and MQM scores collected through WMT evaluation campaigns.

There are several pretrained models available for the COMET framework, such as the reference-based `Unbabel/wmt22-comet-da` and the referenceless `Unbabel/wmt22-co` both trained on DA data from WMT17 to WMT20. These models predict scores ranging from 0 to 1, where 1 denotes a perfect translation. In particular, the COMET framework is fully open source[5], allowing researchers to train custom models for specific domains or objectives.

Empirical evaluations, particularly in recent WMT Metrics shared tasks [Freitag et al., 2021b, 2022, 2023], have shown that COMET models consistently

[4]In addition to regression-based models, there are also ranking and unified objectives.

[5]`https://github.com/Unbabel/COMET.git`

outperform traditional lexical-based metrics in terms of correlation with human judgments across a wide range of language pairs and domains. As a result, COMET is becoming one of the most popular evaluation metrics in machine translation research.

BLEURT

BLEURT (Bilingual Evaluation Understudy with Representations from Transformers) is a neural machine translation evaluation metric introduced by Sellam et al. [2020a]. It combines contextual embeddings pre-trained from BERT [Devlin et al., 2019] with supervised fine-tuning on human-annotated quality data to produce a scalar score that reflects translation adequacy. BLEURT is a reference-based metric - it compares a machine-generated hypothesis to one or more reference translations but does not use the source sentence.

The BLEURT framework begins with a BERT-based encoder pre-trained in general language understanding tasks and is further pre-trained in large-scale synthetic data that simulate translation-like noise. It is then fine-tuned on human ratings of MT quality (e.g., Direct Assessment scores from WMT shared tasks), enabling the model to learn subtle patterns of semantic variation, fluency issues, and adequacy mismatches that go beyond simple token-level similarity.

BLEURT outputs a real-valued score, typically between 0 and 1, where higher values indicate greater semantic similarity and translation quality. It has been shown to outperform traditional metrics and several other neural approaches in terms of correlation with human judgments, especially at the sentence level [Sellam et al., 2020a]. As a result, BLEURT is often used in model evaluation, tuning, and even as an auxiliary training signal in MT systems.

A key strength of BLEURT is its ability to capture nuanced differences in meaning and phrasing. However, like other fine-tuned metrics, it is only as good as the data on which it was trained; its generalization may be limited to low-resource languages or domains far from its training distribution.

2.3.4 Probabilistic models

Prism

Prism [Thompson and Post, 2020] is an evaluation metric based on using a NMT system for paraphrasing as a zero-shot translation task. Instead of relying on surface-level overlap with reference translations, Prism uses a multilingual sequence-to-sequence model trained for paraphrasing. At evaluation time, it computes the probability of generating a hypothesis given a reference (or vice versa), effectively measuring semantic adequacy through conditional likelihood. Because the underlying model is multilingual, Prism can be applied across many language pairs without task-specific training data. Prism outperformed or matched all other metrics on WMT 2019 segment-level shared metrics task [Thompson and Post, 2020].

2.3.5 LLM as Judge

In recent years, large language model (LLM)-based metrics have emerged as strong alternatives to neural metrics. These approaches prompt general-purpose LLMs (e.g., GPT-4, LLaMA-3) to act as judges, either by assigning absolute scores or ranking translation pairs. Metrics such as GEMBA [Kocmi and Federmann, 2023a], MetricX-24 [Juraska et al., 2024] achieved both state-of-the-art system-level correlations in WMT23, often surpassing fine-tuned neural metrics.

Despite their strong performance, LLM-based metrics have several important limitations. They have historically performed poorly at the segment level [Kocmi and Federmann, 2023b], even with extensive fine-tuning [Fernandes et al., 2023]. Furthermore, they are also computationally expensive, typically operate as black boxes, which makes them difficult to audit or deploy efficiently in production pipelines.

Another major drawback of LLM-based evaluation metrics is their limited reproducibility and temporal stability. Many metrics depend on proprietary APIs (e.g., OpenAI, Anthropic) whose underlying models may change without notice. Consequently, scores produced today may differ from those obtained weeks or months later, even with identical inputs. This undermines their reliability for benchmarking or system-tuning workflows. Although setting deterministic decoding parameters (e.g. temperature = 0) and using static prompts can reduce some variability, LLM-based metrics remain inherently unstable unless deployed locally using frozen open-weight models, an option that is computationally demanding and not always feasible.

Recent works mitigate these issues via *pairwise ranking* [Kocmi et al., 2024c] or *distillation into smaller reward models* [Tan and Monz, 2025]. While neural metrics fine-tuned on human judgment, such as COMET, currently dominate WMT benchmarks [Freitag et al., 2024], general-purpose large language models are promising for zero-shot, multilingual, or explanation-rich evaluation [Kocmi and Federmann, 2023a, Fernandes et al., 2023, Leiter et al., 2023b, Xu et al., 2023, Leiter and Eger, 2024]. For example, `Prometheus` [Kim et al., 2024] utilizes frozen `LLaMA-2-70B-Chat` model with a simple static pairwise prompt: "Compare these two translations and explain which is better." This setup achieves 85% pairwise accuracy on WMT23 data while ensuring full reproducibility via Hugging Face. Similarly, **`AutoMQM`** [Fernandes et al., 2023] uses `FLAN-T5-XXL` for zero-shot error detection, reporting stable segment-level correlations.

2.4 Generating additional references

The quality of human reference translations plays a critical role and can influence the outcome of the evaluation more significantly than the choice of metric itself [Freitag et al., 2020]. To address the limitations of relying on a single or limited set of references, researchers have explored methods to generate additional references through both manual and automatic approaches.

2.4.1 Manual Paraphrasing Approaches

There is an evidence that paraphrasing reference translations improves the MT evaluation. Bojar et al. [2013c] demonstrated this with the *Deprefset* dataset, where annotators created exhaustive paraphrases for 50 sentences from the WMT11 evaluation dataset [Callison-Burch et al., 2011]. Although correlation with human judgments improved, the process was highly labor intensive, and annotators were allowed to spend up to two hours per sentence. However, the annotators managed to generate hundreds of thousands of paraphrases per reference sentence, illustrating both the richness of linguistic variation and the computational challenges of exhaustive paraphrasing.

To reduce annotation costs, Bloodgood and Callison-Burch [2010] used crowdsourcing platforms such as Amazon Mechanical Turk, enabling scalable paraphrase collection while maintaining acceptable quality. More recently, Freitag et al. [2020] showed that human paraphrases lead to evaluation scores that correlate more strongly with expert judgments and can even improve system tuning. Interestingly, however, such systems may score lower under BLEU, highlighting a persistent misalignment between traditional n-gram–based metrics and human preferences.

2.4.2 Automatic Paraphrasing

The topic of this work is inspired by Kauchak and Barzilay [2006], who introduced the concept of targeted paraphrasing for MT evaluation, generating synthetic references closer to system outputs while retaining semantic equivalence. Rather than generating all possible paraphrases, their approach focuses on creating synthetic references that are lexically closer to the system output while maintaining semantic equivalence with the original reference. Their approach is straightforward; it consists of the following four steps:

1. **Candidate Selection**: The algorithm identifies pairs of words from the reference and the system output that could potentially form paraphrases, i.e. all pairs of words for which one word appears in a reference sentence, the second in a hypothesis, but none of them in both.

2. **Candidate Filtering**: Only pairs of synonymous words, i.e. words appearing in the same WordNet Miller [1995] synset, were kept from the candidate pairs.

3. **Contextual Classification**: A classifier tests whether each candidate paraphrase is admissible in the reference sentence, because words from a synset are not interchangeable in every context.

4. **Contextual Substitution**: For every confirmed synonym, a new reference sentence is created by placing it to the reference sentence on the position of its synonym.

Their experiments demonstrated that the use of such synthetic reference sentences improves the precision of automatic evaluation. Furthermore, they established a strong connections between the quality of automatic paraphrases, as assessed by humans, and their impact on automatic evaluation.

Other studies confirm these findings. Madnani et al. [2007] show that paraphrased references, even noisy ones, reduce the number of human references needed to tune an SMT system, without significantly affecting MT quality.Tamchyna and Barančíková [2016] proposed automatic dataset creation methods and compared manual versus automatic paraphrases for evaluation.

Another common approach is to use machine translation to create paraphrases by translating the source sentence. Albrecht and Hwa [2008] pioneered the use of off-the-shelf MT systems (including Systran, Google Translate, and Moses) to generate *pseudo-references*, used both in direct evaluation and as features in learned metrics. This method showed improvements for most language pairs, though the results varied by target language. This method was improved more recently by Yoshimura et al. [2019], who again used standard MT systems (here Google Translate, Bing Translate) to translate the source sentences, and then filtered the translation using a BERT-based classifier trained on Microsoft Research Paraphrase Corpus (MRPC) [Dolan and Brockett, 2005].

Recent advances in large language models have opened up new possibilities. Instead of only scoring system outputs, LLMs can generate paraphrastic reference-like variants and then evaluate them through semantic similarity measures, a *generation-based evaluation* paradigm that outperforms many direct scoring approaches [Cui, 2025].

2.5 Automatic Metrics Evaluation

The performance of an automatic machine translation evaluation metric is usually computed as the Pearson correlation between automatic metric scores and human judgments. The correlation estimates the linear dependency between two sets of values.

The Pearson correlation coefficient ρ is defined by the following formula:

$$\rho(H, M) = \frac{\sum_{i=1}^{n} (H_i - \tilde{H})(M_i - \tilde{M})}{\sqrt{\sum_{i=1}^{n} (H_i - \tilde{H})^2} \sqrt{\sum_{i=1}^{n} (M_i - \tilde{M})^2}}$$

where H is the vector of human scores and M is the vector of corresponding scores predicted by a certain metric. $\tilde{H}$ a $\tilde{M}$ are their means, respectively.

Alternatively, the Spearman correlation coefficient r_s is defined as the Pearson correlation coefficient between the rank variables – all human scores and metric's scores of MT systems are converted into ranks $r(H)$ and $r(M)$, i.e., their order from the best to the worst.

$$r_s(H, M) = \rho(r(H), r(M))$$

Both correlations measure the strength of association between two sets of values and range from -1 (perfect negative linear relationship) to 1 (perfect linear correlation).

In our experiments, we use the Pearson correlation coefficient, as it takes into account the distances between the system scores, thus it should be more reliable for similarly evaluated systems [Macháček and Bojar, 2014].

3. Targeted Sentences Paraphrasing for MT Evaluation

Automatic evaluation of machine translation systems is sensitive to lexical overlap between system outputs and reference translations. As discussed in previous chapters, this sensitivity leads to systematic bias against valid translations that differ lexically or syntactically from the reference, particularly in morphologically rich languages such as Czech. One way to mitigate this bias is to increase the diversity of reference translations by introducing paraphrastic variants that preserve meaning while varying the surface form.

This chapter investigates *targeted paraphrasing* of reference sentences as a means of improving machine translation evaluation. Unlike general-purpose paraphrasing, which aims to generate a wide range of alternative formulations, targeted paraphrasing focuses specifically on producing reference variants that are lexically closer to existing system outputs while remaining grammatical and semantically equivalent. The goal is not to generate all possible paraphrases, but to reduce evaluation bias by selectively bridging the gap between hypotheses and references.

We present several techniques for the targeted paraphrasing of Czech sentences to improve machine translation evaluation. It is organized as follows. Before exploring different paraphrasing methods, Section 3.1 introduces the datasets used across all experiments in Chapter 3.

The initial method, described in Section 3.2, adopts a straightforward substitution strategy directly inspired by Kauchak and Barzilay [2006]. This approach involves the substitution of words and phrases in reference sentences using paraphrase tables, while ensuring grammatical correctness and the original meaning of the resulting sentences. In Section 3.3 we utilize this method to propose a new machine translation evaluation metric.

The subsequent sections explore the use of machine translation systems for targeted paraphrasing. There are notable similarities between translation and paraphrasing; both aim to preserve the meaning of a sentence while changing its form. While translation focuses on converting text from one language to another, paraphrasing achieves the same objective within a single language by using a different choice of words or syntactic structures. Both processes require a thorough understanding of the original content and the ability to accurately express the intended message while considering linguistic nuances, cultural context, and the target audience.

Machine translation, however, benefits from a much wider variety of accessible tools and resources compared to the relatively limited availability of paraphrasing tools. Given this disparity, it appears to be only logical to repurpose existing machine translation systems for targeted paraphrasing within a single language.

Section 3.4 presents experiments conducted using the freely available statistical machine translation (SMT) system Moses. Although its results are inconclusive, an analysis of the generated reference sentences reveals systematic issues that motivate an alternative approach. Section 3.5 therefore introduces targeted paraphrasing with components of a rule-based translation system implemented

year	systems	sentences	gold standard	publication
WMT11	14/10	3003	"$>=$ others"	Callison-Burch et al. [2011]
WMT12	13	3003	"$>$ others"	Callison-Burch et al. [2012]
WMT13	14/12	3000	*Expected Wins*	Bojar et al. [2013a]
WMT14	10	3003	*TrueSkill*	Bojar et al. [2014]

Table 3.1: Overview of the WMT English–Czech datasets. The number of systems translating from English to Czech (total systems / manually evaluated systems), the number of source sentences, and the official method used to compute absolute human judgment scores ("gold standard"). For implementation details, see Section 2.1.2.

in the NLP framework Treex, which successfully mitigates many of the problems observed with Moses.

3.1 Data

We use two types of data: (i) data from the Workshop on Statistical Machine Translation (WMT) to evaluate our experiments, and (ii) paraphrase tables, which provide the source material for targeted paraphrasing.

3.1.1 Test Data

Our experiments are conducted on several datasets from the English-to-Czech translation task of the WMT shared translation campaigns held between 2011 and 2014.

Each dataset consists of one file containing the original English source sentences, several files with Czech hypotheses (outputs produced by various MT systems), and one file with corresponding reference translations. The datasets vary across years in the number of participating MT systems and the official evaluation methodology, as shown in Table 3.1. All hypotheses and reference sentences were morphologically analyzed and tagged using MorphoDiTa [Straková et al., 2014].

As mentioned in Section 2, during this period the manual evaluation of translated sentences was performed using the relative ranking method. However, the procedure for computing the absolute system scores varied from year to year as shown in Table 3.1. We refer to these official WMT scores as the *gold standard*. Therefore, to make our evaluation internally consistent, we additionally compute a common absolute score for every year. Specifically, we adopt the "$>$ others" method [Bojar et al., 2011], which served as the WMT12 official system score. Throughout the whole Chapter 3, we refer to this method as the *silver standard*.

3.1.2 Czech Paraphrase Tables

To the best of our knowledge, we considered all major available sources of Czech paraphrases. These include the *Czech WordNet* [Pala and Smrž, 2004], which contains manually annotated, high quality paraphrases. However, the coverage

of Czech WordNet is limited and insufficient for the scale required in our experiments. To address this limitation, we adopt another resource – *the METEOR paraphrase tables* [Denkowski and Lavie, 2010b], which are substantially larger but inherently noisy due to their automatic construction. Notably, we have not integrated *PPDB* [Ganitkevitch and Callison-Burch, 2014], as it was released shortly after the publication of our experiments.

Czech WordNet

The Czech WordNet, developed by the NLP Centre at the Faculty of Informatics, Masaryk University, is part of the EuroWordNet project [Vossen, 1998]. Its primary objective is to create a semantic network that connects multiple languages through a shared core structure.

The Czech WordNet contains 28,201 synsets, each representing a set of words or phrases that share an associated meaning within a given context. While some synsets include glosses (clarifying their meanings), the majority does not, as the initial focus was on establishing semantic relations rather than providing exhaustive lexical definitions. Efforts to fill these gaps have been reported by researchers such as Rambousek et al. [2017], who involved linguistics students in drafting missing definitions. However, these definitions have not yet been fully integrated into the released version of Czech WordNet.

The original, large version of Czech WordNet is available only under a closed, paid license distributed by the ELDA/ELRA agency. Therefore, in our experiments we use a smaller, corrected version – Czech WordNet 1.9 PDT [Pala et al., 2011], which is publicly available in the LINDAT/CLARIN repository.[1] This version contains 23,094 synsets covering nouns, verbs, adjectives, and partly adverbs; 7,028 synsets include multiple entries, i.e., paraphrases.

Czech WordNet was derived from Princeton WordNet [Miller, 1995, Fellbaum, 1998] through automatic translation, followed by manual correction. This method has several disadvantages [Rambousek et al., 2018]. Although most of the paraphrases it contains are of high quality, there are instances of multi-word expressions that are neither fixed nor idiomatic phrases, e.g., *hřebíček na papír* (a little nail for paper), *kára s jablky* (cart with apples) or *zapékaná klobása* (baked sausage).

At the same time, Czech WordNet includes many useful paraphrases for frequently used words (such as "*ambulance*" , "*sanitka*", "*pohotovost*", "*záchranka*", "*sanita*", which all can be translated as an ambulance). However, it also contains archaic or rarely used phrases, which may limit its applicability in contemporary MT evaluation. Furthermore, certain synsets — particularly those referring to proper names (e.g., the apostles) — were left untranslated. Examples of synsets containing rarely used or even erroneous entries are presented in Figure 3.1.

Due to its limited size and many rarely used words, Czech WordNet proves insufficient for the automatic paraphrasing of reference sentences. In a simple experiment focused on identifying paraphrases between a translation hypotheses and their corresponding reference sentences from WMT12 and WMT13, Czech WordNet produces, on average, only one paraphrase per four sentence pairs (see Table 3.2). To address this limitation, we turn to a larger, automatically created

[1] `http://hdl.handle.net/11858/00-097C-0000-0001-4880-3`

Czech WordNet paraphrases	
monocykl (0) (unicycle)	*jednokolový velocipéd* (0) (single-wheeled velocipede)
celeróza (0) (celerosis)	*kukuřičný cukr* (0) (corn sugar)
cheeseburger (59) (cheeseburger)	*smažený sýr v housce* (1) (fried cheese in a bread roll)
sponka (329) (clip)	*hřebíček na papír* (0) (small paper nail)

Figure 3.1: Examples of the rare paraphrases in the Czech WordNet. The number in parentheses represents a frequency of the word or phrase in the Czech National Corpus SYN2020 [Křen et al., 2020].

	WMT12	**WMT13**	**average**
Czech WordNet 1.9 PDT	0.26	0.22	0.23
filtered METEOR	1.53	1.29	1.41
their union	1.59	1.34	1.46

Table 3.2: The average number of word pairs identified as paraphrases per sentence pair consisting of a hypothesis and a corresponding reference sentence.

paraphrase database, namely the Czech METEOR tables.

Meteor paraphrase tables

The METEOR paraphrase tables were created as a resource for the METEOR-NEXT machine translation evaluation metric [Denkowski and Lavie, 2010b]. They are available for five languages: Czech, English, German, French, and Spanish. For languages lacking direct support, Denkowski and Lavie [2014] introduced an automatic tool for extracting paraphrase tables from parallel corpora.

All METEOR paraphrase tables are automatically generated from parallel data via the *pivot phrase method* [Bannard and Callison-Burch, 2005]. This method is based on a simple principle: if two phrases in a source language frequently translate to the same phrase in another language (the *pivot language*, typically English), they likely convey similar meanings and can be considered paraphrase candidates.

More formally, two source-language phrases s_1 and s_2 are potential paraphrases if they both align to the same phrase f in the pivot language. The method estimates the paraphrase probability as:

$$P(s_2 \mid s_1) = \sum_f P(f \mid s_1)\, P(s_2 \mid f),$$

where the sum aggregates over all pivot phrases f that co-occur with both

the Czech METEOR tables paraphrases		PS
svého názoru (its opinion)	*šermovat rukama a mlátit neviditelného* (to flail arms and to beat an invisible one)	0.01
vědecký pracovník v oboru (researcher in a field)	*z vnitřku almary ozval zvuk* (a sound came from the wardrobe)	0.03
nemoudré (unwise)	*okrouhlá skla* (round glasses)	0.01
důsledků takového (implications of such)	implications of such an	0.4
na poloostrově (in a peninsula)	*šimpanzím mlékem* (with a milk of a chimpanzee)	0.01

Figure 3.2: Example of the noise in the Czech METEOR tables. The PS column represents the pivot score (PS).

source phrases in the parallel data. The individual probabilities $P(f \mid s_1)$ and $P(s_2 \mid f)$ are estimated from phrase alignment frequencies in the training corpus.

Each phrase pair in the resulting tables is assigned a *pivot score* (PS), which corresponds to this aggregated probability $P(s_2 \mid s_1)$. Higher scores indicate stronger evidence that the two phrases are valid paraphrases: the more frequently both s_1 and s_2 align to the same pivot phrase(s), and the more consistent these alignments are across the corpus, the higher the pivot score.

Let us now examine the specific characteristics of the Czech METEOR tables. The Czech METEOR tables are substantially larger than Czech WordNet, containing 756,113 paraphrase pairs. They were generated using the pivot phrase method with English as the pivot language on the Europarl [Koehn, 2005] and CzEng 1.0 [Bojar et al., 2012] corpora.

However, the automatic construction method introduces considerable noise into the resulting paraphrase pairs. For instance, 28% of all pairs consist merely of different word forms of the same lemma, rather than true lexical alternatives. This may be useful for the MT evaluation metric's flexible matching, however it also means the tables contain less lexical diversity.

Several factors contribute to genuine noise in the tables, including misaligned sentences in the source corpora, errors in phrase alignment, and lexical or semantic ambiguity. The latter is particularly evident in pairs such as *znásilnění* (rape; as a form of sexual assault) vs. *řepka* (rape; as a cultivated plant and oil source) or *netopýr* (bat; as a flying mammal) vs. *pálka* (bat; as a sport equipment), where words with identical surface forms in the pivot language but different meanings are incorrectly identified as paraphrases.

Noise is particularly evident among paraphrases with low pivot scores and multi-word expressions, we present examples of it in Figure 3.2. Although noise is sparser among single-word paraphrases, there are still instances like "1873" – *pijavice* (leech) or *afghánci* (Afghans) – *šťastně* (happily) identified as synonyms.

We attempted to minimize non-paraphrastic pairs in the Czech METEOR tables using two different methods. The first, applied specifically to pairs of single-words, is based on manual error analysis, and we refer to the resulting tables

as *filtered* METEOR. The second method is fully automatic and applicable to all data; however, its results are inconclusive. This automatic filtering method is detailed further in Section 3.2.2 as it depends on paraphrasing methods that need to be introduced first. In this section, we present only the first method, which is applied consistently in all our experiments.

Filtered Meteor

Based on our observations, we perform the following operations on pairs of one-word paraphrases from the METEOR paraphrase tables:

1. **morphological analysis:** We perform morphological analysis using Morče [Spoustová et al., 2007], a tool for morphological disambiguation of Czech text, and replace word forms with their lemmas.
2. **identical lemmas removal:** Pairs with identical lemmas are eliminated.
3. **different POS removal:** Pairs with different parts of speech are removed.
4. **unknown words removal:** Pairs involving unknown words, typically foreign, are eliminated.

However, an exception to the two last rules was made for paraphrases consisting of numerals and corresponding digits (e.g., *osmnáct* (eighteen) and *18*). These pairs, denoted by the part of speech C for numerals and X for unknown words, were preserved due to their prevalence and quality as paraphrases.

By applying these rules, we successfully reduced the initial pool of over 160,000 pairs of one-word paraphrases to a refined set of 32,154 pairs of lemmas, effectively addressing all examples of inadequate one-word paraphrases mentioned earlier.

3.2 Simple substitution method

In this section,[2] we present a simple method for generating additional reference sentences by substituting lexical and phrasal paraphrases into existing references. The approach mirrors Kauchak and Barzilay [2006] but differs in three key aspects:

1. **Multiple substitutions in one sentence:** In Kauchak and Barzilay [2006], only one word per sentence is replaced by its paraphrase. However, this is not possible in the Czech language. Czech is an inflected language with rich morphology, a word typically has many forms and the correct form depends heavily on its context, e.g., the (morphological) case of a noun depends on the particular verb valency frame. Therefore, we allow multiple substitutions per sentence to produce one well-formed additional reference, rather than many partially broken ones.

2. **No context classifier:** Unlike previous work, we do not use a classifier to evaluate the contextual fit. Instead, we focus on grammatical correctness and preserving meaning by applying Depfix [Rosa et al., 2012], an automatic post-editing system capable of correcting grammatical errors in Czech sentences.

3. **Noisy sources of paraphrases:** Whereas Kauchak and Barzilay relies on the English WordNet as a clean source of paraphrases, the Czech WordNet [Pala and Smrž, 2004] is considerably smaller. To overcome this limitation, we utilize alternative sources of Czech paraphrases, namely the Meteor paraphrase tables, a large but noisy paraphrase table for several languages including Czech. Because of the noise in the Czech Meteor tables, we experiment with adding alignment between a hypothesis and its corresponding reference sentence.

This section is structured as follows. First, we present several variants of the algorithm proposed by [Kauchak and Barzilay, 2006] for lexical and phrase substitutions (Section 3.2.1). These algorithms are further utilized in Section 3.2.2 in an experiment with automatic filtering of the noise from the Meteor paraphrase tables. Our approach is metric-agnostic, focusing on generating additional reference sentences that can be assessed using any machine translation evaluation metric. To evaluate our results, we employ the BLEU score due to its extensive usage. The results are presented in Section 3.2.3.

In particular, we concentrate on answering the following research questions:

- **Does the simple substitution of paraphrases improve the correlation between BLEU scores and human judgments?**
- **Are multi-word paraphrases beneficial?**

[2]This chapter is based on the article [Barančíková et al., 2014], a joint work with Rudolf Rosa and Aleš Tamchyna. Rudolf applied Depfix to all paraphrased sentences and Aleš conducted all experiments with alignments. My responsibilities included paraphrasing, filtering the noisy paraphrase tables, and evaluating the experiments.

- **Does constraining substitutions to aligned word or phrase pairs improve the quality of the generated references?**
- **Does post-editing with Depfix enhance grammatical correctness and overall evaluation results?**
- **Can noisy paraphrase tables be automatically filtered using pivot or lexical scores?**

3.2.1 Core Algorithms and Its Variants

We experiment with several variants of the algorithm for paraphrasing reference sentences. All of them consist of the following three steps:

(1) **Candidate selection:** selecting candidates, i.e, pairs of words or phrases from a hypothesis and its corresponding reference sentence,

(2) **Paraphrasing:** substituting the candidates that are synonymous based on paraphrase tables,

(3) **Depfix:** applying Depfix to fix grammatical errors caused by paraphrasing

The variants of the algorithm differ in all three steps: (1) how potential paraphrase pairs are selected, (2) how identified paraphrase candidatesare applied, and (3) how Depfix is used.

(1) Candidate Selection

We use two different methods to select potential paraphrase pairs. The first is a simple greedy search similar to Kauchak and Barzilay [2006]; the second relies on automatic word alignment to select the corresponding segments of a reference sentence and a hypothesis.

a. Simple Greedy Method
All sentences are used in their lemmatized forms. There are two reasons for this approach: (i) we want to avoid selecting different inflected forms of the same lemma as paraphrases, and (ii) paraphrases in Czech WordNet and *filtered* METEOR are represented in the form of lemmas.

As the first step, we search for one-word paraphrase candidates, as described below. Let H_L, R_L be the sets of lemmas from a hypothesis and the corresponding reference sentence, respectively. The set of one-word paraphrase candidates C_L is then defined as:

$$C_L = \{(r, h) \,|\, r \in R_L \setminus H_L \wedge h \in H_L \setminus R_L\}$$

Second, multi-word candidates C_M are selected as the Cartesian product of all contiguous word sequences in the reference sentence and in the hypothesis. The maximum sequence length is set to seven words, which corresponds to the length of the longest paraphrases in the Czech METEOR tables.

In contrast to C_L, C_M is chosen directly from word forms, not lemmas. Formally, let $h_1, ..., h_m$, $r_1, ..., r_n$ be the word forms in the hypothesis and reference sentence, respectively. Then the set of multi-word paraphrase candidates is:

Source	*The location alone is classic.*
Hypothesis	*Samotné místo je klasické .* alone place is classic The place alone is classic.
Reference	*Už poloha je klasická .* Already position is classic . The position itself is classic.
$\mathbf{H_L}$	[samotný, místo, být, klasický]
$\mathbf{R_L}$	[už, poloha, být, klasický]
$\mathbf{C_L}$	{(už, samotný), (už, místo), (poloha, samotný), (poloha, místo)}
$\mathbf{C_M}$	{(už, samotné místo), (už, samotné místo je), . . . , (klasická, místo je klasické), (klasická, je klasické)}

Figure 3.3: Example of the candidate selection methods on a hypothesis and a reference sentence from WMT11. Note that C_M is substantially shortened – there are 84 sequence pairs in the full version of C_M.

$$C_M = \{\, (r_i \dots r_{i+x}, h_j \dots h_{j+y}) \,|1 \leq i \leq n-x,\ 1 \leq j \leq m-y, \\ 0 \leq x, y \leq 6,\ (x \neq 0 \vee y \neq 0)\,\}.$$

An example of one-word and multi-word candidates is presented in Figure 3.3.

b. Word and Phrase Alignments

A potentionally more reliable way to identify paraphrase candidates is to restrict substitutions to pairs of words or phrases that are aligned with each other. We compute word alignment between reference translations and the sets of hypotheses using GIZA++ [Och and Ney, 2000], a widely adopted implementation of IBM Models and the HMM alignment model for word-level alignment in statistical machine translation. GIZA++ learns translation probabilities between tokens in two parallel corpora and induces word alignment links by iteratively improving a generative model of word correspondences.

Although GIZA++ typically performs best with large parallel corpora, the available data consisting solely of hypotheses paired with their corresponding reference translations is too small to yield high-quality alignments.[3] To overcome this limitation, we take advantage of the fact that all system outputs translate the same source sentences. We therefore augment the training corpus with all possible pairs of system outputs, effectively treating every hypothesis as a *translation* of every other. This results in over one million artificial Czech–Czech sentence pairs,

[3]Hypotheses with corresponding reference sentences constitute only 75,039 sentence pairs (13 MT systems × 3003 sentences from WMT12 and 12 MT systems × 3000 sentences from WMT13). This dataset is insufficient for training high-quality alignment.

greatly improving the statistical support available to GIZA++.[4] For example, the parallel data for WMT12 then looks as follows:

Source	Target
system 1	reference
system 1	system 2
system 1	system 3
...	...
system 1	system 13
system 2	reference
system 2	system 1
system 2	system 3
...	...
system 13	system 12

GIZA++ is trained bidirectionally (reference → hypothesis and hypothesis → reference), and the resulting alignments are symmetrized. The set of one-word paraphrase candidates C_L is then defined as all word pairs connected by at least one alignment link. Multi-word candidates C_M are extracted using standard phrase extraction procedures for phrase-based MT, applying the usual phrase consistency criterion [Och et al., 1999]. This approach ensures that multi-word paraphrases represent phrases that are consistently aligned as units, filtering out many unrealistic or noisy combinations that would otherwise arise from a purely greedy selection.

(2) Paraphrasing

After extracting the candidate sets C_L and C_M, we filter them using paraphrase tables. For the set of one-word candidates C_L, we keep only pairs appearing in the Czech WordNet or in the *filtered* METEOR tables (see Section 3.1.2). If a lemma appears in multiple synonymous pairs we give preference to those found in the WordNet or even better in the intersection of paraphrases from the WordNet and the *filtered* METEOR, as these are more likely to represent reliable lexical paraphrases.

Similarly, multi-word candidates C_M are filtered to include only those phrase pairs that occur in the Czech METEOR tables. An example of this filtering is shown in Figure 3.4.

We evaluate three paraphrasing strategies, which differ in the order in which one-word and multi-word substitutions are applied.

One-word only: We proceed word by word from the beginning of the reference sentence to its end. If the lemma of the current reference word appears as the first member of a paraphrase pair from C_L, it is replaced by the corresponding word of the hypothesis (i.e., the second element of that pair). Otherwise, we keep the original word from the reference sentence. This

[4] We also experiment with augmenting the data using additional Czech–Czech synthetic parallel corpora (generated with MT), but this does not improve paraphrasing quality. The simpler strategy with pairs of hypotheses therefore remains the preferred approach, as it requires no additional data.

$\mathbf{C_L}$	{(už, samotný), (už, místo), (poloha, samotný), (poloha, místo)}
$\mathbf{C_M}$	{(už, samotné místo), ..., (klasická, je klasické)}
$\mathbf{C_L} \cap$ WordNet	{}
$\mathbf{C_L} \cap$ *filtered* METEOR	{(poloha, místo)}
$\mathbf{C_M} \cap$ METEOR tables	{}

Figure 3.4: Filtering candidates from Figure 3.3. Even though there were 84 candidates in C_M there was no intersection with the Czech METEOR tables.

variant uses only one-word substitutions and does not consider any multi-word paraphrases.

One-word first We begin by applying the *One-word only* method. After all one-word substitutions, we apply multi-word paraphrases. Multi-word substitutions are attempted in order of decreasing length, from the longest to the shortest candidate phrases. This ordering is important because *filtered METEOR* frequently contains both multi-word paraphrases and their subsequences; applying shorter paraphrases first could block a valid longer substitution. We do not replace any word that was already changed by the *One-word only* method.

Multi-word first: We begin by applying multi-word substitutions, again proceeding from the longest to the shortest phrase pairs in C_M. Only sequences that have not yet been modified are eligible for replacement. After all possible multi-word substitutions have been applied, we perform a second pass using the *One-word only* method to substitute any remaining unmatched words.

(3) Depfix

Depfix [Rosa et al., 2012] is an automatic post-editing system originally developed to improve the quality of phrase-based English–to–Czech machine translation outputs. It consists of a set of linguistically-motivated rules combined with a statistical component, designed to correct various types of errors, particularly those related to grammar. Using a range of natural language processing tools, Depfix analyzes the input sentences and attempts to fix common machine translation issues. For the purposes of this work, we adapt Depfix to suit the specific characteristics of our paraphrasing setting.

During preliminary analysis, we observe a similarity between errors in the outputs of our paraphrasing algorithms and those commonly encountered in phrase-based machine translation systems. In particular, errors in morphological agreement occur frequently. This observation suggests that Depfix might also be well suited for correcting substitution-related grammatical issues. Unlike conventional grammar-checking tools found in word processors, Depfix targets

error types characteristic of machine-generated text rather than errors typical of human writing.

Furthermore, Depfix benefits from access to the English source sentence, which it uses to guide some of its corrections. This represents an advantage in our scenario: other grammar correction tools typically lack access to the source sentence and therefore cannot use it to improve their performance. For these reasons, we apply Depfix as a post-editing step to reduce grammatical errors in our paraphrased outputs.

However, it is worth noting that certain error types, commonly encountered in phrase-based machine translation, such as inaccuracies in preserving the correct verb tense, do not frequently appear in the paraphrasing process and may even introduce errors. To account for this, we evaluate two configurations of Depfix:

full: The original Depfix system with all 33 fixing blocks, as described in Rosa [2013].

limited: A manually guided analysis of 50 randomly selected outputs reveals that while Depfix often corrects genuine errors, it may also introduce new ones. Most of these errors correspond to changes crucial when translating from English (e.g., adding missing negation) but redundant for paraphrasing. To address this, Depfix is adapted for fixing paraphrasing errors by disabling 10 fixing blocks[5], identified as the main sources of incorrect modifications.

3.2.2 Automatic Filtering of the Meteor Tables

In Section 3.1.2, we describe the noise in the Czech Meteor tables and introduce a heuristic filtering method applied to one-word pairs only, i.e., *filtered Meteor*. In this section, we extend this idea and explore several strategies for automatically assigning a score to each paraphrase pair from the Czech Meteor tables. We then iteratively remove low-scoring paraphrases and measure how this reduction affects the correlation between human judgments and BLEU computed on references created using the filtered paraphrase tables. We experiment with three scoring schemes: pivot scores from the Meteor paraphrase tables, lexical translation probabilities, and a random baseline.

random selection We use random selection as a baseline in order to compare quality of our filtering strategies to simply choosing paraphrases by chance. The paraphrases are randomly shuffled and we then keep the first 10, 20,... percent of them.

pivot scores The second straightforward approach is to use the pivot scores already provided in the Meteor paraphrase tables, which represents the probability that two phrases are equivalent because they translate to the same foreign phrase as discussed in more detail in Section 3.1.2.

[5]The following fixing blocks are disabled in limited Depfix: Fixing reflexive tantum, Fixing morphological number of nouns, Translation of "by", Translation of "of", Translation of present continuous, Subject categories projection, Missing reflexive verbs, Subject personal pronouns dropping, Tense translation, Negation translation

lexical pivoting Finally, we propose an alternative scoring method based on pivoting and lexical translation scores. While the standard pivot score treats phrases as atomic units, it can be misled by co-occurrences in noisy data. To mitigate this, we introduce a fine-grained metric that verifies the semantic equivalence at the word level:

$$\text{lex_p}(\mathbf{s}, \mathbf{t}) = \sum_{s \in \mathbf{s}} \sum_{t \in \mathbf{t}} \sum_{pivot} \text{lex}(s|pivot)\text{lex}(pivot|t),$$

In this equation, lex denotes lexical translation probabilities computed using maximum likelihood estimation from single-best word alignments on the Czech-English parallel corpus CzEng 1.0 [Bojar et al., 2012]. The sum aggregates over all *pivot* words that align to both a word s in the source phrase $\mathbf{s}$ and a word t in the target phrase $\mathbf{t}$.

We refer to this score as *lexical pivoting*. The intuition behind this score is to examine phrase pairs at the word level in the phrases if constituent words also share translation links. A high lex_p score provides strong evidence that the internal words of $\mathbf{s}$ and $\mathbf{t}$ align to the same concepts in the pivot language, thus confirming the validity of the phrase pair. Conversely, a low score suggests that while the phrases might co-occur, their internal semantic content does not match.

It is important to note that while this method is designed to capture compositional alignment in multi-word phrases, it generalizes well to single-word paraphrases. In the case of single words, the outer summations over phrase positions collapse, and the metric functions as a pivot-based measure of distributional similarity. A high score indicates that two individual words share a significant distribution of common translations in the pivot language, thereby validating their synonymy.

We evaluate the effectiveness of these three filtering techniques on the WMT12 data. Due to different nature of our data source, the evaluation was performed in two steps:

1. **Single-Word Paraphrases.** Initially, we attempt to filter using the *One-word only*. In this case, our paraphrase table is the already cleaned *filtered* METEOR table.

2. **Multi-Word Paraphrases.** In the second step, we scale the approach to the noisier the Czech METEOR tables. We apply the best performing method from the first step to multi-word phrases to determine if it can successfully separate valid paraphrases from noise.

Figure 3.5 shows the performance of the different filtering techniques for one-word paraphrases. Filtering according to the pivot scores from the METEOR paraphrase tables performs worse than random selection. In contrast, lexical pivoting maintains a high correlation even when up to 80% of the paraphrases are discarded. However, none of the filtering methods improves by a relevant margin over the baseline correlation of 0.802 achieved by *One-word only* paraphrasing with the full *filtered* METEOR table.

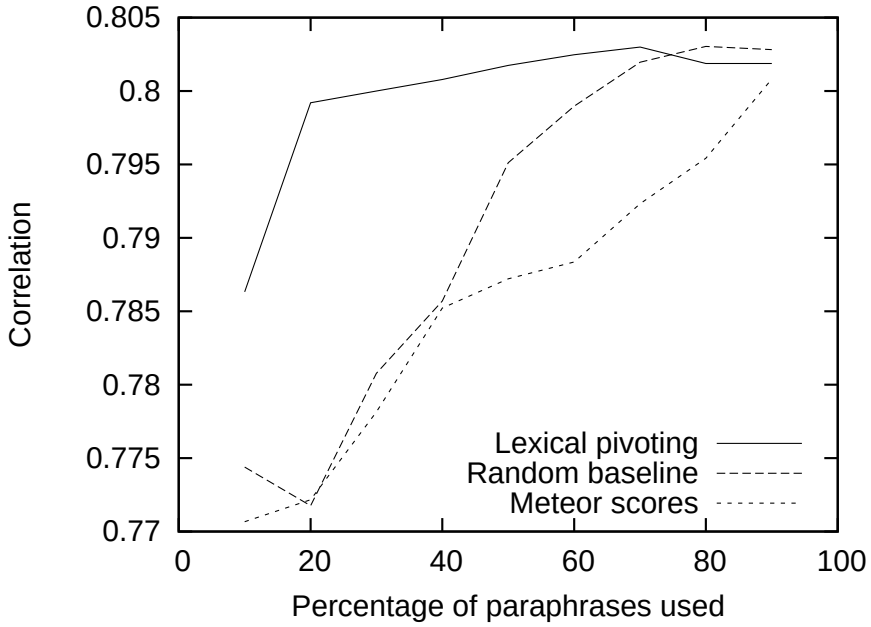


Figure 3.5: Correlation between human judgments and BLEU scores computed on paraphrased references produced by the *One-word only* strategy, using the top 0,10,20,...,100% paraphrases from *filtered* METEOR selected by automatic filtering techniques: random baseline, METEOR score, and lexical pivoting.

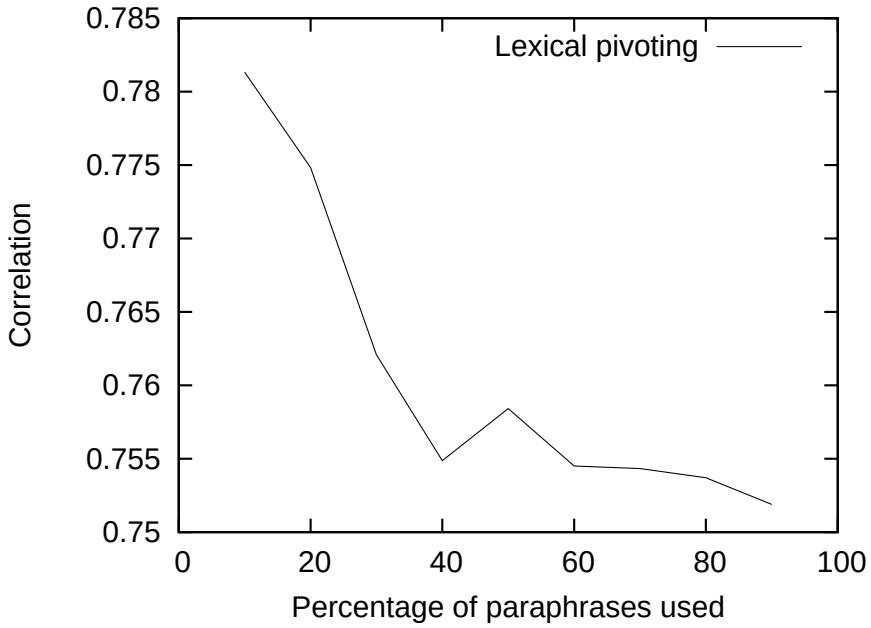


Figure 3.6: Automatic filtering of the multi-word paraphrases from the Czech METEOR tables for the *Multi-word-first* scenario on WMT12 data.

We also evaluate the best-performing technique, lexical pivoting, in the *Multi-word first* scenario by applying it to filter multi-word paraphrases from the Czech METEOR tables. Figure 3.6 shows considerable improvement in correlation with a decreasing number of used paraphrases. However, the highest correlation is achieved with only 10% of paraphrases used. Nevertheless, it still does not outperform the *One-word only*. In this case, the filtering simply mitigates the dam-

WMT12

Method	Greedy selection			Word alignment		
Depfix	No	Full	Limited	No	Full	Limited
One-word only	0.80	**0.83**	**0.83**	0.79	0.81	0.81
One-word first	0.79	0.82	0.82	0.77	0.79	0.80
Multi-word first	0.77	0.81	0.80	0.76	0.78	0.78
Baseline correlation: **0.75**						

WMT13

Method	Greedy selection			Word alignment		
Depfix	No	Full	Limited	No	Full	Limited
One-word only	0.86	**0.89**	0.88	0.86	0.88	0.87
One-word first	0.85	0.88	0.88	0.83	0.87	0.86
Multi-word first	0.84	0.87	0.86	0.83	0.87	0.86
Baseline correlation: **0.84**						

Table 3.3: Pearson's correlation of BLEU and the silver standard.

age done by the multi-word paraphrases. Achieving higher scores would require a more fine-grained understanding of what constitutes a good multi-word paraphrase.

Overall, automatic filtering proves to be inconclusive, and therefore we do not use it further in our experiments. Instead, we continue working exclusively with the *filtered* METEOR tables and the Czech WordNet in the *One-word only* and *One-word first* strategies, and with all multi-word pairs from the Czech METEOR tables in the *Multi-word first* scenario.

3.2.3 Results

Table 3.3 summarizes the results of our simple substitution paraphrasing method. The baseline correlations obtained with the original reference sentences are 0.75 and 0.84 for WMT12 and WMT13, respectively. All evaluated approaches surpass this baseline, with the simplest method, *One-word only*, achieving the strongest performance overall. An example of the *One-word only* method is shown in Figure 3.7.

The differences between correlation coefficients are tested using the method for comparing correlated correlations [Meng et al., 1992]. BLEU shows a statistically significant improvement when evaluated using the new *One-word only* references, with 99% confidence.

Multi-word paraphrases are very noisy, and while they do bring the system outputs closer to the reference (the average BLEU score of the systems increases), they often propose non-equivalent translations or violate the correctness of the sentence, thus blurring the differences between systems.

When paraphrasing is restricted by word alignment, all methods perform worse. As shown in Table 3.4, the number of applied paraphrases drops con-

Source	*The location alone is classic.*
Hypothesis	*Samotné místo je klasické.* Actual place is classic The place alone is classic.
Reference	*Už poloha je klasická.* Already position is classic. The position itself is classic.
New reference	*Už místo je klasická.* Already place is classic *The place itself is classic.
After Depfix	*Už místo je klasické.* Already place is classic The place itself is classic.

Figure 3.7: Example of the *One-word only* method. The hypothesis is grammatically correct and has very similar meaning to the reference sentence. The new reference replaces a paraphrase pair from *filtered* METEOR and becomes lexically closer to the hypothesis, but it has a noun-adjective disagreement. Depfix resolves the error and the final reference is correct and much more similar to the hypothesis.

siderably. While manual inspection suggests a higher proportion of correct paraphrases, the total number of substitutions is reduced too much and overall, our technique is harmed by this restriction.

On the other hand, applying Depfix is consistently beneficial, supporting our assumption that the grammatical correctness of reference sentences plays an important role in reliable MT evaluation. The differences between the *full* and *limited* Depfix configurations are small and not statistically significant; therefore, we do not claim a consistent advantage of either variant.

Results on the WMT12 and WMT13 data are consistent. Paraphrasing increases the accuracy of the evaluation using the BLEU metric, although the differences in the WMT13 data are not as prominent due to a much higher baseline. This is also reflected in the smaller amount of substitutions (see Table 3.4).

3.2.4 Conclusion

Our findings confirm the positive impact of paraphrasing reference sentences on the performance of the BLEU metric. Among the examined strategies, the *One-word only* greedy substitution method achieves the best results. We also demonstrate the usefulness of Depfix in ensuring grammatical correctness of paraphrased references. Finally, we explore several techniques for reducing noise in the METEOR paraphrase tables. We answer the following research questions:

Does simple substitution of paraphrases improve the correlation between BLEU and human judgments? Yes. Using the additional references

WMT12

Method	Greedy selection		Word alignment	
	Words	Phrases	Words	Phrases
One-word only	1.59	–	0.86	–
One-word first	1.59	0.23	0.86	0.22
Multi-word first	1.38	0.31	0.81	0.27

WMT13

Method	Greedy selection		Word alignment	
	Words	Phrases	Words	Phrases
One-word only	1.33	–	0.76	–
One-word first	1.33	0.20	0.76	0.20
Multi-word first	1.04	0.68	0.74	0.24

Table 3.4: Average number of replaced words/phrases per sentence for each method on data from WMT12 and WMT13.

produced by *One-word only* plus Depfix increases system-level Pearson correlation from 0.75 to 0.83 on WMT12 and from 0.84 to 0.89 on WMT13 (Table 3.3). The improvement over the baseline is statistically significant.

Are multi-word paraphrases beneficial? Generally no. Multi-word substitutions are noisy: while they can raise raw BLEU by pulling hypotheses closer lexically, they frequently introduce non-equivalences or grammatical issues and blur system differences. Even with filtering, *Multi-word first* does not outperform *One-word only* or *One-word first* on either WMT12 or WMT13.

Does constraining substitutions to aligned word/phrase pairs help? No. Alignment constraints reduce the number of applicable substitutions too aggressively and act as a bottleneck, preventing a sufficient volume of lexical variations from being generated, which limits the improvement in BLEU correlation.

Does post-editing with Depfix help, and which configuration is better? Yes. Depfix consistently improves correlation, confirming the importance of the grammatical well-formedness of the generated references. The *full* configuration is typically as good as or better than *limited*; *limited* does not yield consistent improvements over the out-of-the-box system, suggesting that the standard *full* Depfix model is robust enough for this task without task-specific constraints.

Can we automatically filter noisy paraphrase tables without hurting performance? Partly. Lexical pivoting scores allow heavy pruning (discarding up to ∼80%) while retaining high correlation for one-word substitutions, but they do not surpass the best unfiltered *One-word only* result. Pivot scores from the METEOR paraphrase tables perform worse than random scores in our evaluation.

metric	reference	**WMT12**	**WMT13**
BLEU	Original	0.751	0.835
	Paraphrased	0.834	**0.891**
Meteor-next	Original	0.833	0.817
	Paraphrased	**0.927**	**0.891**
1 - TER	Original	0.274	0.760
	Paraphrased	0.283	0.781

Table 3.5: Pearson's correlation of different metrics with the silver standard. The Paraphrased references are created using the best performing method from the previous section – *one-word only* with *full Depfix*.

3.3 Parmesan

In the preceding section, we demonstrate that targeted paraphrasing of reference sentences substantially improves the correlation between automatic evaluation metrics and human judgment when using simple string-based metrics such as BLEU. This naturally raises two follow-up research questions:

- **Does this improvement generalize to other commonly used evaluation metrics, in particular Meteor-next, which already incorporates paraphrase matching internally?**
- **Could the noise present in the Meteor paraphrase tables actually harm the performance of the Meteor-next metric itself?**

In this Section[6], we investigate both questions and introduce Parmesan, a variant of Meteor-next that moves paraphrase matching from the metric itself to a targeted paraphrasing step applied to reference sentences prior to evaluation.

As a first step, we compare the effect of paraphrased references on three widely used metrics: BLEU, Meteor-next, and TER. The results presented in Table 3.5 show that paraphrased references consistently improve correlation with human judgment not only for BLEU and TER, but also for Meteor-next.

The improvement is somewhat unexpected, since Meteor-next internally integrates the same paraphrase table (i.e., the Meteor paraphrase tables), which is used to generate new references (in the form of the *filtered Meteor* tables). This suggests that the way paraphrases are incorporated into Meteor-next may be suboptimal, and that noise in the Meteor paraphrase tables could outweigh their intended benefit.

We therefore formulate the following hypothesis: while paraphrases are useful for bringing reference sentences closer to valid system outputs, using noisy paraphrase tables directly inside the evaluation metric may degrade its ability to discriminate between systems.

To test this hypothesis, we evaluate the performance of the Meteor-next metric under several configurations that progressively reduce or entirely remove paraphrase matching. This configurations are summarized in Table 3.6.

[6]This section is based on Barančíková [2014].

METEOR-NEXT setting	size	paraphrase table description
Original	684k	the METEOR paraphrase tables
One word	181k	**Original** without multi-word pairs
Same POS	122k	**One-word** with the same part-of-speech.
Different Lemma	71k	**Same POS** in which the pairs have different lemmas
Same Lemma	51k	**Same POS** in which the pairs have the same lemma
No paraphrases	0	no paraphrase tables, i.e., exact match only

Table 3.6: Different paraphrase tables for the METEOR-NEXT metric and their size (number of paraphrase pairs) The **Original** configuration corresponds to the original METEOR-NEXT metric, implying no filtering for the paraphrase match. Conversely, **No paraphrases** correspond to a total reduction, i.e., an empty set, so the METEOR-NEXT metric has to rely solely on exact match.

paraphrases	**WMT12**	**WMT13**	**WMT14**
WordNet	0.26	0.22	0.24
filtered METEOR	1.53	1.29	1.39
together	1.59	1.34	1.44

Table 3.7: Average number of one-word paraphrases per sentence found in WordNet, *filtered METEOR* tables and their union over all systems.

Note that the **Original** size reported here does not correspond to the 756k paraphrase pairs mentioned earlier in Section 3.1.2.The earlier figure refers to the original Czech the METEOR paraphrase tables as introduced inDenkowski and Lavie [2010b], whereas the METEOR metric implementation used in our experiments (version 1.3; Denkowski and Lavie, 2011) employs a internally filtered version of the paraphrase tables containing approximately 684k pairs.

Based on our observations, we propose the PARMESAN[7] metric, our submission to the WMT14 Metrics Shared Task to evaluate English-to-Czech translation.

Specifically, PARMESAN operates in two stages:

1. Reference sentences are paraphrased using the *one-word only* method, followed by the application of *full Depfix*.

2. The resulting paraphrased references are evaluated using METEOR-NEXT with paraphrase matching disabled, relying solely on exact matching.

In other words, PARMESAN uses paraphrases from the METEOR paraphrase tables only in the form of *filtered METEOR* for generating a paraphrased reference sentence for each machine translation output prior to evaluating using METEOR-NEXT. The PARMESAN score is computed using a strictly exact-match version of METEOR-NEXT on the newly generated references.

3.3.1 Results on WMT12 and WMT13 data

We first report results on WMT12 and WMT13 in Table 3.8 for the silver standard and Table 3.9 for the gold standard. We evaluate the METEOR-NEXT metric using

[7]PARMESAN stands for PARaphrasing for MEteor SANs paraphrases.

WMT12

METEOR-NEXT conf.	Original	Paraphrased	Depfixed
Original	0.833	0.905	0.927
One-word	0.836	0.908	0.930
Same POS	0.840	0.911	0.931
Same Lemma	0.838	0.911	0.932
Different Lemma	0.863	0.931	0.950
No paraphrases	0.861	0.931	**0.951**

WMT13

METEOR-NEXT conf.	Original	Paraphrased.	Depfixed
Original	0.817	0.865	0.891
One-word	0.820	0.867	0.892
Same POS	0.823	0.869	0.893
Same Lemma	0.821	0.868	0.892
Different Lemma	0.850	0.895	**0.915**
No paraphrases	0.848	0.895	**0.915**

Table 3.8: Pearson's correlation of METEOR-NEXT and the silver standard.

all paraphrase table configuration from Table 3.6 and three reference sentences:

Original: original references provided by WMT for particular year,

Paraphrased: paraphrased references using the *One-word only* method using *filtered METEOR* and WordNet tables, average number of paraphrases is described in Table 3.7,

Depfixed: application of *full Depfix* on **Paraphrased** references.

PARMESAN scores are emphasized by enclosing them within a box, while METEOR-NEXT is underscored. The best scores are represented using bold font.

The results in both tables exhibit clear consistency. There is a notable positive impact of the prior paraphrasing of reference sentences and the application of Depfix. The strongest results are consistently obtained when paraphrases are used exclusively to generate paraphrased references, while the evaluation itself relies on exact matching.

The original METEOR-NEXT metric (i.e. original references + the **Original** METEOR-NEXT setting) shows the weakest performance for both WMT12 and WMT13. Filtering the METEOR paraphrase tables leads to improved performance in METEOR-NEXT, regardless of the reference sentence used.

We use methods for comparing correlation coefficients Meng et al. [1992] to statistically determine the significance of differences in correlation coefficients. The tests indicate that PARMESAN outperforms METEOR-NEXT with 99% certainty.

On the original reference sentences, **Different Lemma** setting yield the best results. That is because it is essentially a limited version of the *filtered METEOR* tables used for creating new references.

WMT13 reference			
METEOR-NEXT setting	Original	Paraphrased	Depfixed
Original	0.856	0.894	0.918
One-word	0.859	0.896	0.918
Same POS	0.862	0.898	0.919
Same Lemma	0.860	0.897	0.919
Different Lemma	0.885	0.918	**0.933**
No paraphrases	0.883	0.917	**0.933**

Table 3.9: Pearson's correlation between METEOR-NEXT and the gold standard – *Expected Wins* Bojar et al. [2013a]. Gold standard results are not presented for WMT12 since the gold standard and silver standard are identical for that year of the evaluation campaign.

The main reason for the poorer performance of the metric when employing the Meteor Paraphrase tables is noise. Significant amounts of the noise are pairs of different word forms of a single lemma, which may award even completely non-grammatical sentences. This is reflected in the low correlation of the **Same Lemma** setting.

Even worse is the fact that the metric may award even parts of the hypothesis left untranslated, as the METEOR paraphrase tables contain English words and their Czech translations as pairs of paraphrases (See Section 3.1.2).For these reasons, the differences among the systems are more blurred and the metric performs worse than without using the paraphrases.

3.3.2 Results on WMT14 data

However, the WMT14 results presented in Table 3.10, show distinct patterns compared to the preceding two years. PARMESAN still outperforms METEOR-NEXT on the silver standard, but only by a marginal difference (0.005). However, when using the gold standard, METEOR-NEXT performs better by precisely the same margin. In particular, Pearson's correlation between the two standards remains exceptionally high at 0.997.

However, in contrast to previous years, targeted paraphrasing of reference sentences is detrimental to the performance of METEOR-NEXT. Applying Depfix on paraphrased references is beneficial but does not outperform the original references. However, the variations in correlation between the original references and the new ones are minimal (up to 0.012).

Similarly, as in WMT12 and WMT13, reducing the paraphrase tables in the METEOR-NEXT evaluation metric remains beneficial for both original and paraphrased reference sentences – the best results are achieved with the **No paraphrases** configuration on the original references, i.e., no paraphrases are used.

The correlations with human judgment have an exceptionally high score of 0.973, suggesting that any paraphrasing adversely affects the performance of the metric. This observation corresponds with the official results of WMT14 metrics task [Macháček and Bojar, 2014], where simple string matching metrics outperform more sophisticated ones in evaluation from English to Czech. Notable is

silver standard

METEOR-NEXT setting	Original	Paraphrased	Depfixed
Original	0.963	0.957	0.968
One-word	0.967	0.958	0.965
Same POS	0.965	0.959	0.969
Same Lemma	0.968	0.959	0.969
Different Lemma	0.970	0.965	0.968
No paraphrases	**0.973**	0.965	0.968

gold standard

METEOR-NEXT setting	Original	Paraphrased	Depfixed
Original	0.967	0.958	0.966
One-word	0.968	0.959	0.966
Same POS	0.969	0.959	0.966
Same Lemma	0.972	0.960	0.967
Different Lemma	0.972	0.963	0.962
No paraphrases	**0.974**	0.963	0.962

Table 3.10: Pearson's correlation of Meteor and the silver, respectively, the gold standard – *TrueSkill* Bojar et al. [2014]. Note that as opposed to official WMT14 results, the version 1.4 of METEOR-NEXT is still used in this table.

that METEOR-NEXT is surpassed by TER, NIST [Doddington, 2002] or CDER [Leusch et al., 2006].

3.3.3 Additional Analyses

Using unfiltered Meteor paraphrases. We also experiment with paraphrasing reference sentence using the METEOR paraphrase tables (instead of *filtered METEOR* and WordNet) for a comparison. We use again *One word only* method with one-word paraphrases from the METEOR paraphrase tables. Even though the paraphrase table is much larger than our *filtered METEOR* tables, the amount of substituted words is much smaller (see Table 3.11) due to not being lemmatized (compare with Table 3.7.

The **Original** setting in Table 3.11 closely corresponds to the **One-word** setting in Table 3.8 evaluated on the original reference sentences. While the **No paraphrases** setting in Table 3.11 outperforms all correlations with original references, it fails to outperform the new reference sentences constructed using *filtered METEOR* and WordNet.

Effect of source-language composition. There is an important distinction between the data from previous years and WMT14 in terms of the composition of the English source data used for translation into Czech. In WMT12 and WMT13, the source data are evenly derived from all competing languages, meaning only a fifth or sixth of the data, respectively, is originally written in English and the remaining $\frac{4}{5}$,respectively, $\frac{5}{6}$ are translations from the other languages. On the

	WMT12	WMT13
number of substitutions	0.75	0.61
Original	0.837	0.818
No paraphrases	0.869	0.852

Table 3.11: The number of paraphrase substitution per sentence using the original METEOR paraphrase tables only and Pearson's correlation with the silver standard using those paraphrases.

WMT12

Reference	Original	Paraphrased	Depfixed
Original	0.781	0.872	0.897
One-word	0.779	0.872	0.897
Same POS	0.782	0.874	0.897
Same Lemma	0.772	0.874	0.897
Different Lemma	0.807	0.898	**0.923**
No paraphrases	0.798	0.899	**0.923**

WMT13

Reference	Original	Paraphrased	Depfixed
Original	0.805	0.843	0.874
One-word	0.810	0.846	0.877
Same POS	0.813	0.849	0.878
Same Lemma	0.813	0.848	0.877
Different Lemma	0.842	0.879	0.877
No paraphrases	0.840	0.877	**0.902**

Table 3.12: Pearson's correlation of Meteor and the silver standard on sentences originally Czech or English only. In this case, the interpretation of human judgment was computed only on those sentences as well.

other hand, the English source data for translation into Czech in WMT14 were created exclusively from English and Czech sentences.

Translating sentences from an original source language to English and subsequentlu to Czech, might be a contributing factor to the beneficial effect of prior paraphrasing of reference sentences. In light of this, we conduct an experiment by restricting the WMT12 and WMT13 data to sentences originally written in either Czech or English. In this restricted setting, PARMESAN still significantly outperforms METEOR-NEXT, and the results (see Table 3.12) follow similar patterns as observed on the full datasets. In other words, the presence of an additional language in the translation pipeline does not appear to explain the differing outcomes observed for the WMT14 data.

3.3.4 Conclusion

In summary, we highlight the adverse impact of noise in the METEOR paraphrase tables on the performance of the METEOR-NEXT metric. We demonstrate that substantial reduction in the paraphrase tables can be very beneficial for targeted paraphrasing of reference sentences. The METEOR-NEXT scores computed without paraphrase support, when applied to these new reference sentences, exhibit a significantly improved correlation with human judgment compared to the original METEOR-NEXT scores, particularly on the WMT12 and WMT13 data. However, the WMT14 data do not confirm this improvement, and the observed enhancement was marginal. Furthermore, Parmesan's performance was even worse than METEOR-NEXT on the gold standard. We answer the following research questions:

Does this improvement generalize to other commonly used evaluation metrics, in particular Meteor-next, which already incorporates paraphrase matching internally? Yes, but not universally. On WMT12 and WMT13, targeted paraphrasing of reference sentences significantly improves the correlation of METEOR-NEXT with human judgment, however the results was not confirmed by WMT14.

Could the noise present in the Meteor paraphrase tables actually harm the performance of the Meteor-next metric itself? Yes, our results provide strong evidence that noise in the METEOR paraphrase tables harms the performance of the METEOR-NEXT metric. Across multiple experiments, we observe that entirely removing paraphrase matching within METEOR-NEXT consistently leads to the highest correlations with human judgment. The original METEOR-NEXT configuration, which relies on the full paraphrase tables has the weakest performance on WMT12 and WMT13.

3.4 Paraphrasing using phrase-based machine translation

In this and the following sections,[8]s, we investigate targeted paraphrasing using machine translation techniques. Since machine translation and paraphrasing both aim to preserve meaning—differing primarily in whether the output language changes—it is natural to consider adapting translation systems for monolingual paraphrasing.

Machine translation benefits from a well-established set of tools and resources, in contrast to the relatively limited availability of dedicated paraphrasing systems. This motivates our research question: how can existing machine translation frameworks be adapted to enable targeted paraphrasing within a single language? We address this question by exploring modifications required to generate Czech targeted paraphrases.

This section focuses on phrase-based machine translation using Moses, while the following section examines the rule-based system Treex. Moses, developed by Koehn et al. [2007], had been recognized as a leading open-source Statistical Machine Translation (SMT) framework for more than a decade. For an extended period, Moses played a pivotal role in machine translation – until the rise of neural machine translation (NMT) caused a significant shift in the field. The introduction of the Transformer [Vaswani et al., 2017] architecture and pre-trained models have led to substantial improvements in translation quality and made NMT the prevailing approach for automatic translation.

Despite NMT rendering SMT obsolete in recent years, we continue to utilize Moses. The reason is not only technical, reflecting the fact that our research started prior to the widespread adoption of NMT, situating our foundational methodologies and experiments within the context of SMT's prevailing technologies at the time. More importantly, SMT offers considerable advantages in terms of explainability and transparency, it provides an ability to control the translation process, where the influence of specific features and rules on the translation process can be directly observed and adjusted. This characteristic is particularly beneficial for tasks that require the targeted manipulation of translation outputs, such as paraphrasing. Additionally, while NMT excels in producing fluent and coherent translations, SMT frameworks such as Moses allow for more granular experimentation with linguistic features and translation hypotheses.

This chapter first introduces the necessary components of phrase-based SMT, then describes our adaptations for monolingual paraphrasing, and finally evaluates their effectiveness on WMT12 Callison-Burch et al. [2012] data. We address the following research question:

- **Can phrase-based SMT be adapted for targeted monolingual paraphrasing?**

[8]This section is based on the article [Barančíková and Tamchyna, 2014], which is a joint work with Aleš Tamchyna. Aleš was in charge of training and tuning Moses. My responsibilities included data preparation, paraphrasing, and evaluation of experiments.

3.4.1 Moses overview

The Moses system generates translations using statistical models. The translation process involves the following phases: the creation of language and translation models, parameter tuning, and the decoding process. A simplified representation of the Moses translation engine is shown in Figure 3.8.

The initial step in statistical machine translation involves creating language and translation models. The language model is responsible for ensuring the syntactical correctness and grammatical integrity of the translated sentence. A translation model (also referred to as a phrase table) provides all possible translations for a given word or phrase.

Following the construction of the models, the next phase consists of parameter tuning. During this step, the models weights are learned to optimize translation quality. Various optimization techniques, such as minimum error rate training (MERT) or maximum likelihood estimation, are applied to fine-tune the statistical models.

During decoding, the final phase, a beam search algorithm is employed to efficiently find the best translation among potentially numerous candidates generated from a source sentence. More specifically, an input sentence is divided into smaller units, words, and phrases, and they are translated independently. The algorithm keeps only a limited number of hypotheses (partial translations), focusing only on the most promising ones based on their cumulative scores. These scores are computed as a combination of different model scores and their corresponding weights, which are learned during the training process.

In our paraphrasing setup, Moses is repurposed to translate Czech into Czech. When trained exclusively on monolingual data, Moses would generate Czech paraphrases. However, without constraints, monolingual SMT produces arbitrary paraphrases.

This observation leads to another research question:

- **Can task-specific features enable effective targeted paraphrasing?**

To answer this question, we have developed and integrated into Moses a novel feature specifically designed to mitigate the randomness of paraphrase variations (Section 3.4.4). Let us examine the components of Moses in more detail and present our specific modifications first.

3.4.2 Language Model

The language model (LM) is essential to ensure that translations are not only grammatically correct but also fluent in the target language. It plays an important role in resolving ambiguities and ensuring the contextual integrity of the translated text.

The process of constructing a language model does not require modifications, since it is based solely on monolingual data. We created our language model using the SRILM toolkit [Stolcke, 2002] utilizing data from the Czech component of the Czech-English parallel corpus CzEng [Bojar et al., 2012].

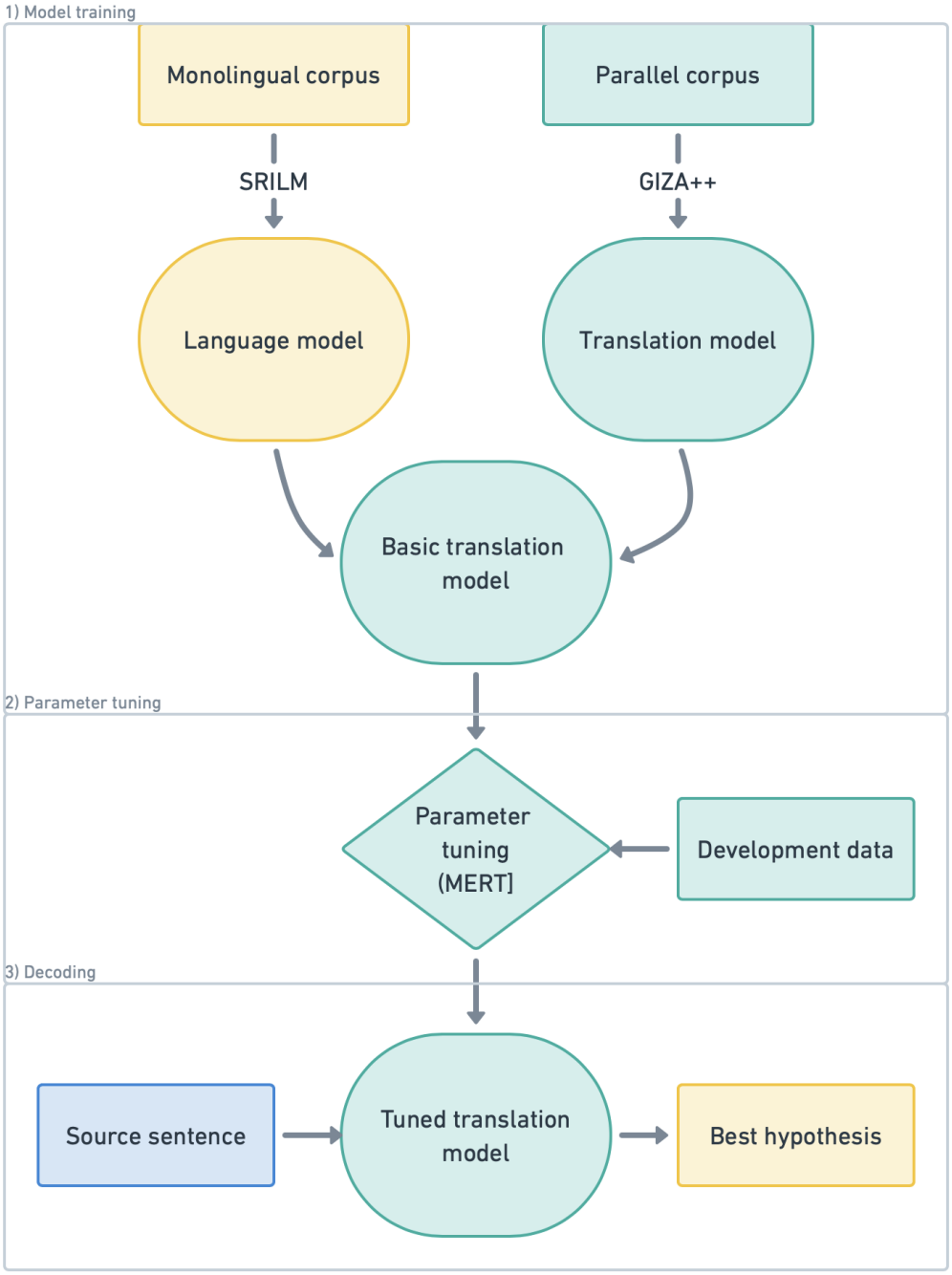


Figure 3.8: The schema of the Moses translation system, employing a color-coded scheme to denote the languages associated with various models and datasets. The blue color is used to signify the source language, whereas yellow designates the target language. Green represents parallel data or models, i.e, those incorporating both the target and source languages.

3.4.3 Translation models

The translation model is responsible for generating possible translations of a source language sentence into a target language. This model identifies the most probable translations of individual words or phrases and then combines them to form complete sentences in the target language.

Fundamentally, a translation model is structured as an extensive table. Each entry in this table consists of a phrase, its corresponding translation, several feature scores (such as translation probability and lexical weight) and optionally also an alignment between the phrases and their frequencies from a training corpus.

These feature scores, along with the alignment and the frequencies, are automatically learned by analyzing large parallel data sets using tools such as GIZA++ [Och and Ney, 2000]. It computes statistical probability, focusing on how words and phrases in the source language correspond to those in the target language. The effectiveness of the phrase table depends on the quality and volume of the bilingual corpora used for training.

In the pursuit of paraphrasing using Moses, the crucial change is to employ monolingual data instead of using data in two languages. Unfortunately, there are no extensive Czech-Czech parallel datasets available to train the translation model. In view of this, we have to adapt a different approach to acquire the translation model. We turn to the available Czech paraphrase tables, which are inherently parallel, though not in the traditional sense of aligning two distinct languages. From these tables, we can derive synthetic translation models that substitute direct Czech-Czech parallel data. This approach raises another research question:

- **Can synthetic paraphrase tables replace monolingual parallel data?**

To address this question, we introduce two complementary phrase tables designed to compensate for the lack of Czech–Czech parallel data.

Enhanced Meteor tables

The *Enhanced* Meteor tables are derived from the original Czech paraphrase Meteor table [Denkowski and Lavie, 2010b]. Each paraphrase pair has a pivoting score that we repurpose as a feature in our phrase table. However, our analysis in Section 3.5 shows that this score performs worse than random selection, so we anticipate its minimal impact during the tuning process.

To compensate for this limitation, we introduce additional feature scores based on distributional semantics [Miller and Charles, 1991]. This approach is based on the hypothesis that words or phrases that occur in similar contexts tend to have similar meanings. This principle is often summarized by the aphorism "You shall know a word by the company it keeps" [Firth, 1957].

Distributional similarity has previously been used to re-rank automatically extracted paraphrases, notably by Chan et al. [2011] and Ganitkevitch and Callison-Burch [2014], who applied it to paraphrases obtained from parallel bilingual corpora. Distributional semantics is efficient in eliminating incorrect paraphrases generated by lexical ambiguity, as such words rarely occur in similar contexts.

Given that the METEOR paraphrase tables was also constructed automatically from parallel corpora (as detailed in Section 3.1.2), it is reasonable to incorporate distributional similarity scores as additional features in our translation model.

To compute these scores, we extract all contexts in which paraphrases from the METEOR paraphrase tables appear in the Czech National Corpus [Křen et al., 2010]. Contexts are defined as words within a fixed window around the target phrase. The window size parameter specifies how many words are considered on each side; for example, a window size of 1 includes only immediately adjacent words, whereas a window size of 3 includes three words to the left and right.

For each paraphrase pair, we compute the context similarity using cosine similarity as an auxiliary signal over context-frequency vectors. This yields a graded estimate of contextual relatedness, allowing us to distinguish true paraphrases from spurious pairs more reliably. A worked example of the cosine similarity computation on a toy corpus is provided in Appendix 5.1.

Using this procedure, we add six distributional semantic scores to each paraphrase pair in the METEOR paraphrase tables. These scores vary according to (i) the context window size (1–3 words) and (ii) whether word order within the window is preserved. Together with the original pivot score, the resulting *Enhanced METEOR* phrase table contains the following seven features:

1. **pivot score:** score adopted from the original the METEOR paraphrase tables,
2. **DS_W1_WO** distributional semantic score with window size 1 using word order
3. **DS_W2_WO** distributional semantic score with window size 2 using word order
4. **DS_W3_WO** distributional semantic score with window size 3 using word order
5. **DS_W1_NWO** distributional semantic score with window size 1 not considering word order
6. **DS_W2_NWO** distributional semantic score with window size 2 not considering word order
7. **DS_W3_NWO** distributional semantic score with window size 3 not considering word order

One-word paraphrase table

As a preliminary step to create one-word paraphrase table, we create a candidate set of words by compiling all words from the Czech part of the CzEng corpus [Bojar et al., 2012]. We set a frequency threshold to at least five occurrences in the corpus in order to filter out rare words and potential typographical errors. The resulting set of candidates containing 445,782 words was further expanded to incorporate all the words found in the hypotheses and the reference sentences from WMT12, totaling 447,540 words.

This set of candidate words constitutes the basis for creating another phrase table. Subsequently, a morphological analysis is performed on these words using the Morče tool [Spoustová et al., 2007] to obtain lemmas and POS tags.

The one-word paraphrase table is created in the following way. For every word x from the candidate set, each pair of words that satisfies at least one of the following criteria is added to the one-word paraphrase table:

- x, x, because not every word should be paraphrased;
- x, y, provided that the lemma of x is the same as the lemma of y, because some words might have different morphology in the paraphrased sentences;
- x, y, if the lemmas of x and y are paraphrases according to Czech WordNet PDT 1.9;
- x, y, if the lemmas of x and y are paraphrases according to the *filtered* METEOR.

This phrase table contains more than 1,121,988 pairs of words. For each pair of words x, y, we add the following five feature scores:

- **identity:** $e(= e^1)$ if $x = y$, $1(= e^0)$ otherwise,
- **same lemmas:** e if x and y have the same lemma, $1(e^0)$ otherwise,
- **WordNet:** e if lemmas of x and y are paraphrases according to the Czech WordNet, $1(e^0)$ otherwise,
- ***filtered* Meteor:** e if lemmas of x and y are paraphrases according to *filtered* METEOR, $1(e^0)$ otherwise,
- **POS similarity:** $e^{\frac{1}{a+1}}$, where a represents the minimal Hamming distance between POS tags of the two words, i.e., the number of positions at which the individual POS tags differ. This probability should reflect how morphologically distant the paraphrases are, offering insight into the extent to which the pair of words varies in grammatical function and structure.

We represent binary feature values using the exponential function with the natural base, assigning values of e^1 and $= e^0$. This representation is convenient because Moses internally operates in log space. After logarithmic transformation, these values become 1 and 0, respectively, allowing binary features to be seamlessly integrated into the log-linear model.

Importantly, these values are not interpreted as probabilities. Instead, they serve as feature activations whose influence is controlled by the learned model weights. This approach avoids numerical issues associated with zero values while preserving the intended relative contribution of individual features during decoding.

An example of our phrase table is presented in Figure 3.9.

Together, these tables aim to balance semantic coverage and lexical control.

```
okno ||| okno ||| 2.71828182846 2.71828182846 1.0 1.0 2.71828182846 |||  |||
okno ||| okna ||| 1.0 2.71828182846 1.0 1.0 1.35914091423 |||  |||
okno ||| oknech ||| 1.0 2.71828182846 1.0 1.0 0.90609394282 |||  |||
okno ||| okny ||| 1.0 2.71828182846 1.0 1.0 0.90609394282 |||  |||
okno ||| okně ||| 1.0 2.71828182846 1.0 1.0 1.35914091423 |||  |||
okno ||| oken ||| 1.0 2.71828182846 1.0 1.0 0.90609394282 |||  |||
okno ||| skla ||| 1.0 1.0 2.71828182846 1.0 1.35914091423 |||  |||
okno ||| sklo ||| 1.0 1.0 2.71828182846 1.0 2.71828182846 |||  |||
okno ||| obrazovka ||| 1.0 1.0 1.0 2.71828182846 1.35914091423 |||  |||
```

Figure 3.9: In the One-word paraphrase table, each entry within the phrase table divided into five sections, separated by the delimiter |||. The first block contains a source phrase, and the second block presents a target phrase (its corresponding translation, or a paraphrase, in our context). The third block contains all feature scores associated with the phrase pair. The fourth and fifth blocks, which are left empty in our case, are intended for the alignment of the phrases (not necessary in our case as this is one-word phrase table) and counts on how often phrase pair appeared in the training data, respectively. The last two columns are not applicable in our case as we do not have training data.

3.4.4 CoveredReferenceFeature

We develop and integrate into the Moses translation engine a novel feature designed to guide the machine translation decoder to generate outputs that exhibit a higher degree of similarity to the evaluated translation hypotheses.

This feature named **CoveredReferenceFeature** evaluates the overlap between each candidate sentence produced during the paraphrase generation process and a specified translation hypothesis. The goal of this feature is to prioritize candidate translations that share more words with the hypothesis, thereby facilitating targeted paraphrasing.

The overlap is computed as the count of words in the candidate translation that are also found in the evaluated hypothesis. The mechanism for measuring overlap is computationally efficient in accommodating the constraints of the decoding process. This simplicity is important to maintain the speed and tractability of the search algorithm during the machine translation process.

The integration of this feature into Moses' beam search algorithm, used in the phrase-based decoding phase, requires one to keep track of feature state. The feature state refers to the set of words from the reference translation that have been covered by the hypothesis generated so far. This state tracking is essential for accurate hypothesis recombination, as it allows the beam search algorithm to maintain a comprehensive understanding of which parts of the reference translation have been addressed.

Furthermore, we implement an estimator that predicts the potential of a given phrase to cover additional words from the reference translation. It is defined as the number of reference translation words covered by the given phrase.

The code for **CoveredReferenceFeature** was written by Aleš Tamchyna and it is included in the Moses GitHub repository.[9]

[9] `https://github.com/moses-smt/mosesdecoder/`

3.4.5 Additional Features

In addition to the already mentioned language model, see Section 3.4.2, the features of the translation models (Section 3.4.3) and **CoveredReferenceFeature** (Section 3.4.4), there are three additional three features computed:

- **Distortion:** This feature measures the cost associated with reordering phrases from their positions in the source sentence to their positions in the translated sentence. A higher distortion cost penalizes more significant rearrangements, encouraging the translation to preserve the original order of the source sentence as much as possible.

- **WordPenalty:** This feature controls the length of the output translation. Typically, it penalizes translations for each word produced, encouraging the model to generate more concise translations. This feature helps to balance the length of the translation, preventing too long or too brief translations.

- **PhrasePenalty:** Similar to the **WordPenalty**, the **PhrasePenalty** feature is concerned with the number of phrases in the translation. This penalty is applied for each phrase used in the translation, influencing the model to prefer translations with an optimal number of phrases. A low **PhrasePenalty** (close to 0) means little to no discouragement for using multiple phrases, potentially leading to more detailed, possibly fragmented translations.

3.4.6 Parameter tuning using MERT

We use the minimum error rate training (MERT) [Och, 2003] to find the optimal weights for our models to minimize the error rate in a development dataset. In the original setting, these validation datasets consist of parallel data, source sentences, and their professional translations.

MERT learns the weights by iteratively translating each source sentence from the validation dataset, calculating an error metric (such as BLEU) by comparing it to the reference translation, and adjusting the model weights to minimize average error across all candidates until the error rate stops improving significantly.

In case of MERT, we cannot replace parallel data as simply as we did for the translation models. We briefly experimented with Many Czech References dataset [Bojar et al., 2013c], which contains human-created paraphrases of 50 sentences. Despite the high quality of these paraphrases, the dataset is very limited in its size and variability.

Consequently, we decided to use reference sentences and the highest rated hypothesis as parallel data from WMT13 Bojar et al. [2013a] as our validation dataset. This decision was driven by the absence of more suitable datasets. We believe that directly estimating the necessary, non-trivial model weights from the outputs generated by a single system presents a viable strategy, without incurring a significant risk of overfitting.

Results of the parameter tuning are presented in Table 3.13. The **CoveredReferenceFeature** not surprisingly obtains the highest weight as it provides an oracle guide towards the hypothesis. Similarly, **Distortion** gets really low

Feature	Weight
Language Model	0.016
Covered Reference Feature	0.513
Distortion	0.026
Word Penalty	0.291
Phrase Penalty	0.006
***Enhanced* Meteor**	
Pivot Score	0.010
DS_W1_WO	-0.021
DS_W2_WO	0.009
DS_W3_WO	0.006
DS_W1_NWO	-0.004
DS_W2_NWO	0.011
DS_W3_NWO	0.002
One-word Paraphrase Table	
Identity	0.025
Same Lemmas	-0.003
WordNet	0.021
FM	0.003
POS Similarity	0.031

Table 3.13: Weights learned in the optimization process using MERT.

weight, which is not surprising suggesting not drastic syntactic differences between hypotheses and reference translations.

On the other hand, the weight of **Language Model** is really low, which is surprising as language model weights typically have a substantial influence on ensuring the fluency of the output text. Similarly, the **WordPenalty** is typically used with a negative weight to discourage overly long translations by penalizing each additional word. The positive WordPenalty weight is counter-intuitive from a traditional translation perspective but consistent with the optimization objective in this setting. Since BLEU favors higher n-gram overlap, generating longer paraphrases increases the likelihood of matching reference n-grams, even at the cost of grammaticality or semantic precision.

This behavior can be explained by the optimization objective of MERT. Since tuning directly minimizes BLEU error with respect to the hypothesis, lexical overlap becomes disproportionately important. As a result, the **CoveredReferenceFeature** dominates the optimization process, while the language model — which primarily enforces fluency rather than lexical similarity — receives a very low weight.

The diagram of Moses translation engine adjusted for paraphrasing is presented in Figure 3.10.

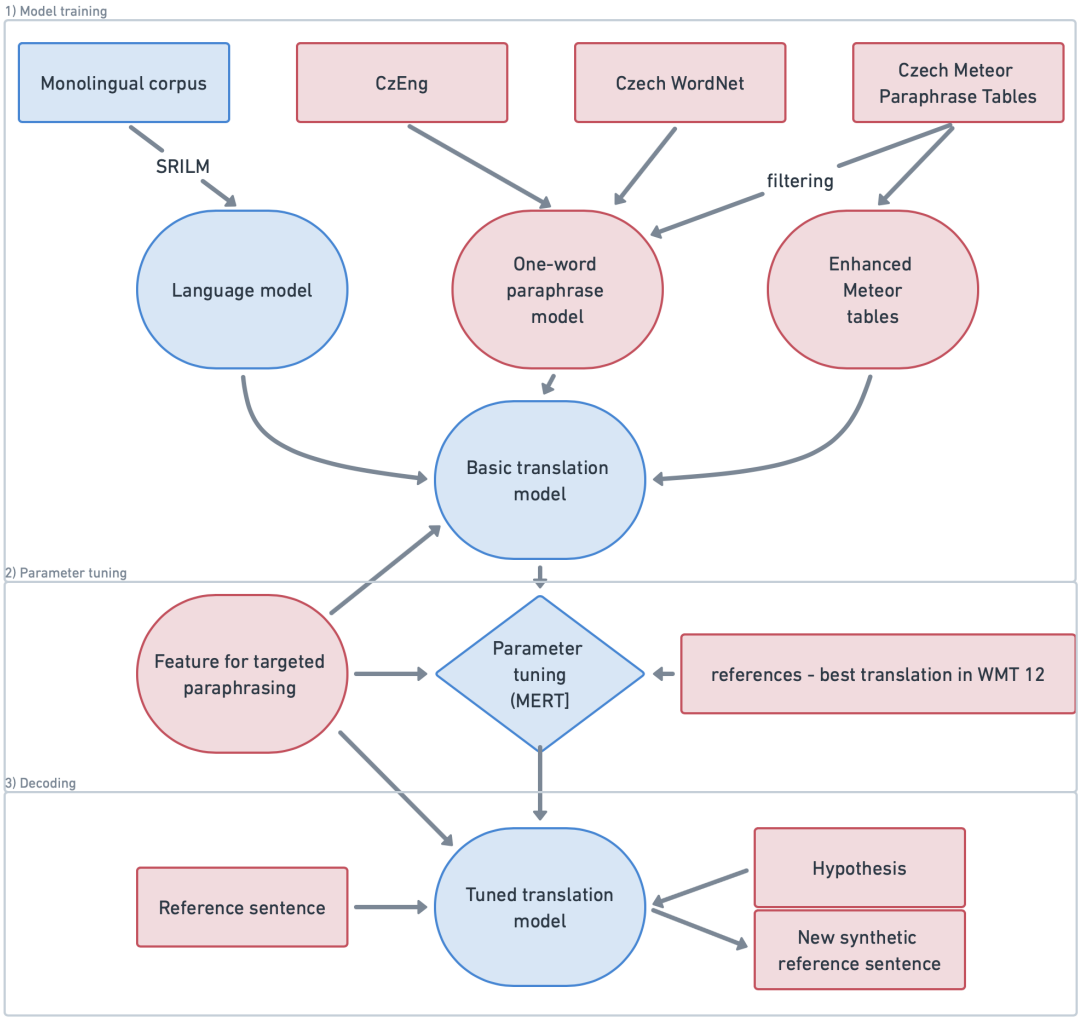


Figure 3.10: The schema of the Moses translation system, as adapted for the specific requirements of targeted paraphrasing and the constraints posed by the unavailability of parallel monolingual data. Due to the fact that only Czech data was used, we employ dual-color coding to distinguish new and original parts of the engine. Components retained from the original configuration of the Moses system, as depicted in Figure 3.8, are blue. Modifications introduced to address the challenges of targeted paraphrasing are marked in red.

3.4.7 Results of paraphrasing via Moses

Default Settings

The results are presented in Table 3.14 as the Pearson's correlation coefficient between BLEU and the silver standard of human judgement on the following four configurations:

- **Original:** original reference sentences are used, no paraphrasing
- **Paraphrased:** paraphrased reference sentences generated by Moses, using weights learned via MERT.

configuration	correlation	average BLEU
Original	**0.75**	12.8
Paraphrased	0.50	15.8
LM+0.2	0.24	9.1
LM+0.4	0.22	6.7

Table 3.14: Description of basic settings and the results - Pearson's correlation of BLEU and the human judgment, the average BLEU scores.

- **LM+0.2:** paraphrased reference sentences generated by Moses, using weights learned via MERT with an incremental adjustment of 0.2 to the weight of the language model.
- **LM+0.4:** paraphrased reference sentences generated by Moses, using weights learned via MERT with an incremental adjustment of 0.4 to the weight of the language model.

Figure 3.11 provides a visual overview of our findings. The graphs demonstrate that while paraphrasing via Moses, with MERT-learned weights consistently increases BLEU scores, it fails to improve the correlation with human judgment. In fact it is substantially degraded. Increasing the weights of the language model intended to improve grammatical accuracy instead contributed to unpredictability and decreased coherence between BLEU scores and human judgment.

Importantly, none of the paraphrasing configurations surpasses the baseline correlation obtained using the original, non-paraphrased reference sentences. This contrasts with the simpler paraphrasing approach discussed in Section 3.2, where targeted lexical variation yielded consistent gains.

The observed degradation can be attributed to a small number of interacting factors. First, MERT optimization against BLEU disproportionately rewards lexical overlap, causing the **CoveredReferenceFeature** to dominate the model. As a consequence, other components—most notably the language model—receive negligible weights.Specifically, the paraphrases generated by the **Paraphrased** setting, despite achieving a significantly higher BLEU score, were mostly ungrammatical, which negatively affected the correlation of our metric.

Second, manually increasing the weight of the language model paradoxically introduces new complications. This adjustment produces grammatically correct references that, regrettably, tend to have little or no relevance to the original source sentence. This observation indicates that our efforts to mitigate noise from the paraphrase tables were by no means sufficient, leaving a considerable amount of incorrect phrase pairs within our phrase tables, which the language model cannot reliably compensate for.

Finally, the design of the targeting feature itself contributes to the problem. Operating purely at the word level and ignoring word order or syntactic structure, it permits translations with scrambled structure as long as lexical overlap is maintained. As a result, the feature acts as a strong oracle for word coverage but provides no guaranties with respect to fluency or syntactic adequacy.

These effects are further reflected in the learned feature weights. All features of both translation models—the One-word paraphrase table and *Enhanced* METEOR —receive very small weights, with the highest assigned to POS similarity. This

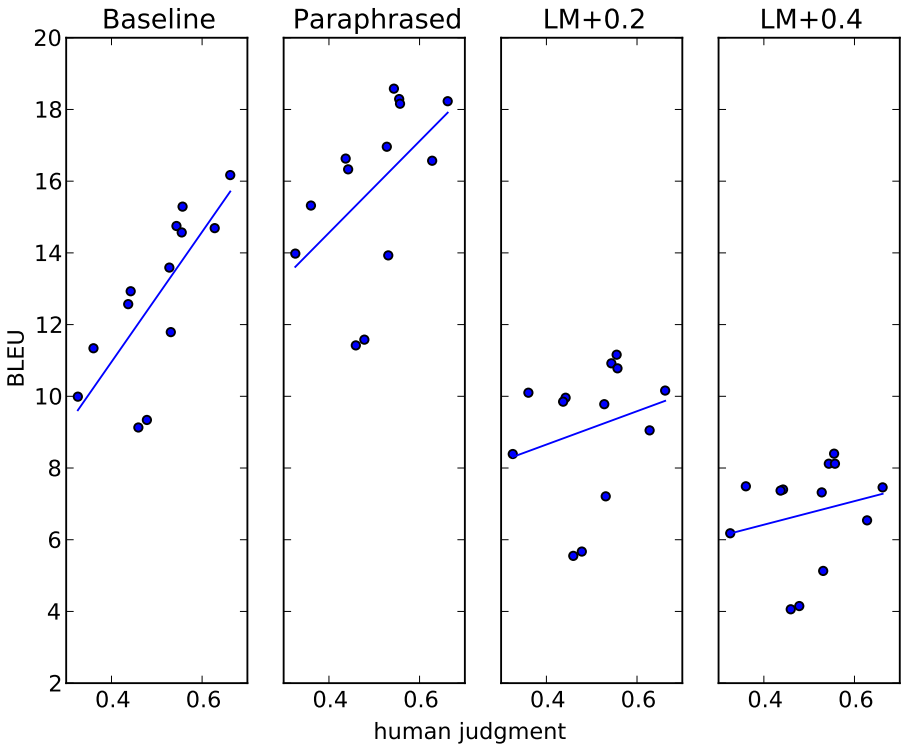


Figure 3.11: The visualization of BLEU and human judgment across the four basic settings – **Original**, **Paraphrased**, **LM+0.2**, and **LM+0.4** —- includes linear regression lines to better demonstrate the linear correlation between these two metrics. This illustration is adopted from Barančíková and Tamchyna [2014]

indicates that neither the pivot score nor the distributional semantic features are effective in distinguishing valid paraphrases from noise.

Figure 3.12 illustrates these failure modes. While the hypothesis is fluent and semantically appropriate, the paraphrased reference generated under the default setting contains grammatical errors. Increasing the language model weight yields fluent outputs, but at the cost of semantic drift, resulting in references that no longer correspond to the source sentence.

Monotony

Given the unsatisfactory results, we experimented with three additional configurations:

- **Lexical:** This setting is based on the basic **Paraphrased** setting, the only difference is utilizing only one translation model – the One-word paraphrase table, focusing on lexical substitutions without considering the *Enhanced* METEOR table.
- **Lexical & LM+0.2:** This setting combines the lexical approach with an increment of 0.2 in the language model weight, aiming to enhance grammatical accuracy while maintaining a focus on simple lexical paraphrases.

Source sentence	*Paclík claims he would dare to manage the association.*
Original reference	Paclík tvrdí, že by si na vedení asociace troufl. *Paclík claims he would dare to lead the association.*
Hypothesis	Paclík tvrdí, že by se odvážil k řízení komory. *Paclík claims he would find the courage to control the chamber.*
Paraphrased reference	*Paclík tvrdí, že by se na řízení organizace troufl. **Paclík claims he would dare to control the organization.*
LM+0.2 reference	#Paclík tvrdí, že by si troufl na řízení ekonomiky. #Paclík claims he would dare to control the economy.
LM+0.4 reference	# Říká se, že Paclík si troufl na řídící rady. # They say that Paclík ventured to governing boards.

Figure 3.12: An illustrative example of the targeted paraphrasing. The hypothesis is grammatically correct and has very similar meaning to the source sentence. The paraphrased reference more closely matches the hypothesis in terms of wording; however, there is a grammatical error. Specifically, the reflexive verb *troufnout* (to dare) require pairing with the reflexive pronoun *si* (oneself [in dative]), but the paraphrase sentence incorrectly employs reflexive pronoun *se* (oneself [in accusative]) instead. The reference sentences generated with increased weights of the language model are both grammatically correct, but they both significantly deviated from the meaning of the original reference sentence (marked with #). In the**LM+0.4** setting, the reference also differs a lot in wording from both the hypothesis and the reference sentence.

- **Lexical & Monotone:** This configuration applies lexical paraphrasing within a monotone translation framework, i.e., explicitly disabling reordering to address issues related to scrambled word order.

The results of these new settings are presented in Table 3.15. To reduce noise in the translation process, we opted to exclude the *Enhanced* METEOR tables from our experiments. This decision proved to be reasonable as the correlation between BLEU scores and human judgment in the **Lexical** showed an improvement of 0.06 compared to the **Paraphrased** setting. The idea to explore monotone translation was motivated by the frequent occurrence of disordered word sequences and misplaced punctuation in the **Paraphrased** setting. The **Lexical & Monotone** setting yielded the best results among all paraphrased hypotheses tested.

Despite these modifications leading to an improved correlation with human judgment, they did not surpass the baseline results. We observed a noticeable improvement in the BLEU score, suggesting a greater overlap between the newly generated references and the hypotheses. Interestingly, this increase was even higher when monotony was added – indicating that disabling reordering helped to cover longer n-grams that were previously shuffled.

This suggests that while targeting specific areas of improvement such as lexical choice and sentence structure can yield positive changes, achieving an overall translation quality that exceeds the baseline using Moses remains a challenge. These findings underscore the complexity of phrase-base machine translation

setting	correlation	avg. BLEU
Original	0.75	12.8
Paraphrased	0.50	15.8
LM+0.2	0.24	9.1
LM+0.4	0.22	6.7
Lexical	0.56	15.1
Lexical & LM+0.2	0.33	9.5
Lexical & Monotone	**0.61**	18.1

Table 3.15: Results for the additional settings, showing Pearson's correlation between BLEU and the silver standard, as well as the average BLEU scores. Results from **??** are highlighted in gray for easier comparison.

and the nuanced balance required between various components to produce paraphrases that are both accurate and grammatical.

In conclusion, we answered the following research questions:

Can a phrase-based SMT system be adapted for targeted monolingual paraphrasing? Yes, we demonstrate that Moses, traditionally used for machine translation, can be technically adapted for targeted monolingual by employing monolingual resources and task-specific features. However, our adaptation of Moses enables the generation of paraphrases that are lexically similar to given hypotheses, although grammatical and semantic fidelity are not consistently preserved.

Can synthetic paraphrase tables replace monolingual parallel data? Partially, synthetic paraphrase tables provide partial coverage in the absence of monolingual parallel corpora but introduce substantial noise. Consequently, they cannot fully substitute for large-scale, high-quality monolingual parallel data in paraphrasing tasks.

Can task-specific features enable effective targeted paraphrasing? Yes, we introduce a novel feature to guide the decoder toward lexically targeted paraphrases. The CoveredReferenceFeature feature, implemented within Moses, effectively enforces lexical overlap between generated paraphrases and selected hypotheses. However, its word-level design and strong influence during decoding limit its ability to preserve grammatical correctness and semantic adequacy.

3.5 Paraphrasing at a Deep Syntactic Layer

In this chapter,[10] we investigate targeted paraphrasing of reference sentences at a deep syntactic layer. We employ the NLP framework Treex and extend it with modules specifically designed for targeted paraphrasing and word-order changes. Our goal is to examine whether linguistically informed paraphrasing of reference sentences can improve the correspondence between automatic evaluation scores and human judgment.

In the previous section, we explored targeted paraphrasing using the SMT system Moses, adapted for monolingual phrase-based translation. The results of that approach were inconclusive, suggesting that Moses may not be well suited for targeted paraphrasing in scenarios where large, high-quality paraphrase tables are unavailable. In particular, the need to construct extensive translation tables, define targeting features, and tune complex weight configurations motivated the exploration of an alternative approach that allows for more conservative and linguistically controlled sentence modifications.

Reflecting on these challenges, we decided to employ a rule-based system, which would allow for partial sentence modifications, thus enabling the generation of more conservative paraphrases. In particular, we chose to adapt Treex [Popel and Žabokrtský, 2010], a highly modular NLP software system. Treex incorporates the TectoMT [Žabokrtský et al., 2008] rule-based machine translation system, which operates on the deep syntactic layer. As such, Treex is already equipped with a pipeline capable of transforming a sentence from its surface form at the word level to the deep semantic layer and back, making it an ideal platform for our targeted paraphrasing tasks.

Utilizing Treex, we analyze both the hypothesis and its corresponding reference sentence to the deep semantic (t-)layer, where only content words are represented as nodes in dependency graphs and function words are encoded as nodes attribute. At this layer, we identify lemmas unique to only one of the sentences. Based on the paraphrase tables, selected lemmas from the hypothesis are substituted to the reference sentence, producing a modified t-tree. The process concludes with synthesizing the modified reference back to the surface word layer, where appropriate word forms are generated automatically.

This targeted approach addresses several problematic issues that have previously been encountered when using Moses. By directly comparing two sentences, the potential for introducing noise is substantially reduced. Furthermore, Treex's highly developed NLP pipeline significantly lower the risk of generating ungrammatical sentences. By only modifying only specific parts of sentences directly related to paraphrased word, while keeping the remaining structure intact, we obtain conservative paraphrases that preserve meaning and context more reliably than using phrase-based methods.

We concentrate to answer the following research questions:

- **Can targeted paraphrasing at a deep syntactic layer improve the correlation between automatic MT evaluation metrics and human judgment?**

[10] This chapter is based on the article [Barančíková and Rosa, 2015], which is a joint work with Rudolf Rosa. Rudolf wrote the two new Treex blocks. My responsibilities included data preparation, paraphrasing and evaluation of experiments.

- **How does a rule-based, linguistically motivated paraphrasing approach compare to our previous methods?**
- **Does explicit word-order reordering of paraphrased references further improve MT evaluation?**

3.5.1 Treex

Treex is an open-source NLP system and is available on GitHub,[11] including the two blocks that we contributed.[12]

Treex implements a stratificational approach to language, adopted from the Functional Generative Description theory [Sgall, 1967] and its later extension by the Prague Dependency Treebank (PDT) [Bejček et al., 2013]. For a more detailed description of PDT, as well as a comprehensive exploration of the various layers of the language system and their associated linguistic phenomena, see Hajičová [2020], Hajič et al. [2016].

The PDT schema represents sentences at four layers:

- **w-layer (word layer):** A sentence is represented without any linguistic annotation, consisting of the sequence of words/tokens as they appear in the text only.
- **m-layer (morphological layer):** At this level, each token is lemmatized to its dictionary form and assigned with a tag indicating its part of speech and other morphological categories, as relevant in the given context. The input words is also normalized (e.g., incorrect tokenization or spelling errors are corrected, figures are normalized).
- **a-layer (analytical layer/shallow-syntax layer):** The analytical layer represents a sentence as a surface syntactic dependency tree, nodes representing individual tokens and edges representing syntactic relation between pairs of tokens.
- **t-layer (deep-syntax/tectogrammatical layer):** At this layer, sentence is represented as a deep-syntactic dependency tree-based graph, where content words (i.e. semantically full lexical units) only have their own nodes. Each node in this tree (t-node) consists of a t-lemma and a set of attributes – a *formeme* indicating the original syntactic form, and *grammatemes* representing meaning counterpart of morphological features (where relevant for sentence meaning).

In our experiments with rule based paraphrase generation, we utilize the analysis and generation pipeline from the TectoTM system. We transfer both a hypothesis and its corresponding reference sentence to the t-layer, where we introduce a module specifically designed for paraphrasing t-lemmas. Following the paraphrasing process, we perform synthesis of the altered sentence back to a-layer, where a reordering module is applied to adjust word order as necessary before further synthesis to the w-layer completes the whole process.

[11] `https://github.com/ufal/treex`

[12] Since 2017, Treex is not developed any more, as several better alternatives are available (such as Udapi [Popel et al., 2017] or UDPipe [Straka, 2018]).

3.5.2 Analysis from w-layer to t-layer

The analysis from the w-layer the to a-layer within the Treex framework includes several essential NLP tasks. Initially, tokenization segments the text into words or tokens. Following this, part-of-speech (POS) tagging and lemmatization are performed using MorphoDiTa [Straková et al., 2014]. Additionally, dependency parsing is performed using the MSTParser [McDonald et al., 2005] adapted by Novák and Žabokrtský [2007], trained on PDT. It constructs the dependency relationships between tokens, forming a structured representation of the sentence's syntactic organization.

In the next step, a surface-syntax a-tree is converted into a deep-syntax t-tree – auxiliary words are removed, with their function now represented using t-node attributes (*formemes* and *grammatemes*) of content words they belong to (e.g., two a-nodes of the analytical verb form *spal jsem* (I slept) are merged into a single t-node *spát* (sleep) with the tense grammateme set to the past tense. Similarly, the phrase *v květnu* (in May) is collapsed into *květen* (May) with the formeme indicating the prepositional group *v+X* (in+X).

The choice to use the t-layer for paraphrasing is strategic, primarily because it represents words in their lemmatized form, stripped of syntactical complexity. Moreover, functional words, which are typically not targets for paraphrasing and tend to introduce considerable noise into paraphrase tables, are absent at this layer. This absence significantly reduces the potential for generating erroneous paraphrases, making the t-layer an ideal stage for implementing paraphrase transformations. This layer's focus on content over syntactic form simplifies the identification of paraphrasable elements, enhancing the quality and relevance of the generated paraphrases.

3.5.3 Paraphrasing at t-layer

The paraphrasing module T2T::ParaphraseSimple developed by Rudolf Rosa is freely available at GitHub.[13]

The T2T::ParaphraseSimple block is based on the selective replacement of a t-lemma of a reference t-nodes (R) based on specific conditions that should ensure coherence with the context of both the hypothesis and the reference sentence. The conditions for changing a t-lemma of a reference t-node R from A to B are as follows:

1. **B is present in the hypothesis:** There exists a hypothesis t-node with lemma B. We are not trying to introduce any new words to the new reference sentences, just words that already appears in the hypothesis.

2. **A is absent in the hypothesis:** There is no hypothesis t-node with lemma A. If there is lemma A already present, paraphrasing is redundant.

3. **B is unique in the reference**: There is no other reference t-node with lemma B, ensuring that the paraphrase introduces no redundancy within the reference sentence.

[13] `https://github.com/ufal/treex/blob/master/lib/Treex/Block/T2T/ParaphraseSimple.pm`

Source	*The Internet has caused a boom in these speculations.*
Hypothesis	Internet vyvolal boom v těchto spekulacích . *Internet caused boom in these speculations .* *The Internet has caused a boom in these speculations.*
Reference	Rozkvět těchto spekulací způsobil internet . *Boom these speculations caused internet .* *A boom of these speculation was caused by the internet.*

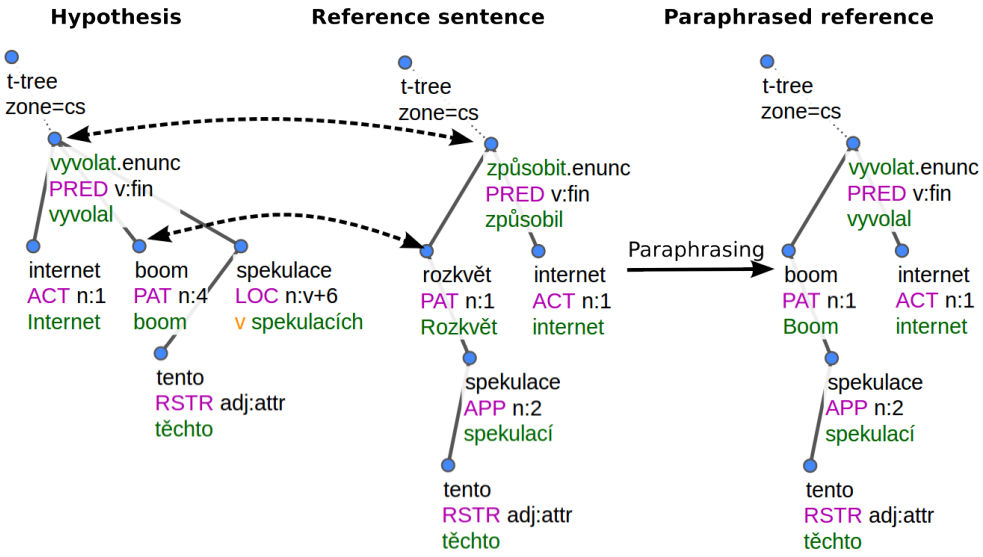


Figure 3.13: Example of the paraphrasing. The hypothesis is grammatically correct and has the same meaning as the reference sentence. We analyse both sentences to t-layer, where we create a new reference sentence by substituting synonyms from hypothesis to the reference. In this step a new reference sentence *Boom těchto spekulací vyvolal internet.* (A boom of these speculation was caused by the internet.) would be generated. This new reference has more similar choice of word to the hypothesis, however there is still no overlap in term of longer phrases. To address this, the subsequent step is word reordering to more closely mirror the structure of the hypothesis.

4. **A and B are paraphrases**: Lemmas A and B are recognized as paraphrases according to the provided paraphrase tables, confirming their meaning equivalence or close relationship.

The other attributes of the t-node are preserved, reflecting the assumption that deep syntactic properties are largely independent of the specific lexical realization. However, in practice, there is at least one case where this is not true: t-nodes corresponding to nouns are marked for grammatical gender, which is very often a grammatical property of the given lemma with no effect on the meaning (for example, the English noun "*a house*" could correspond to the Czech masculine noun "*dům*" or the feminine noun "*budova*", both of which carry different grammatical genders but denote the same object. We really believe this should be marked on the lemma only, but this is not the case in Treex.

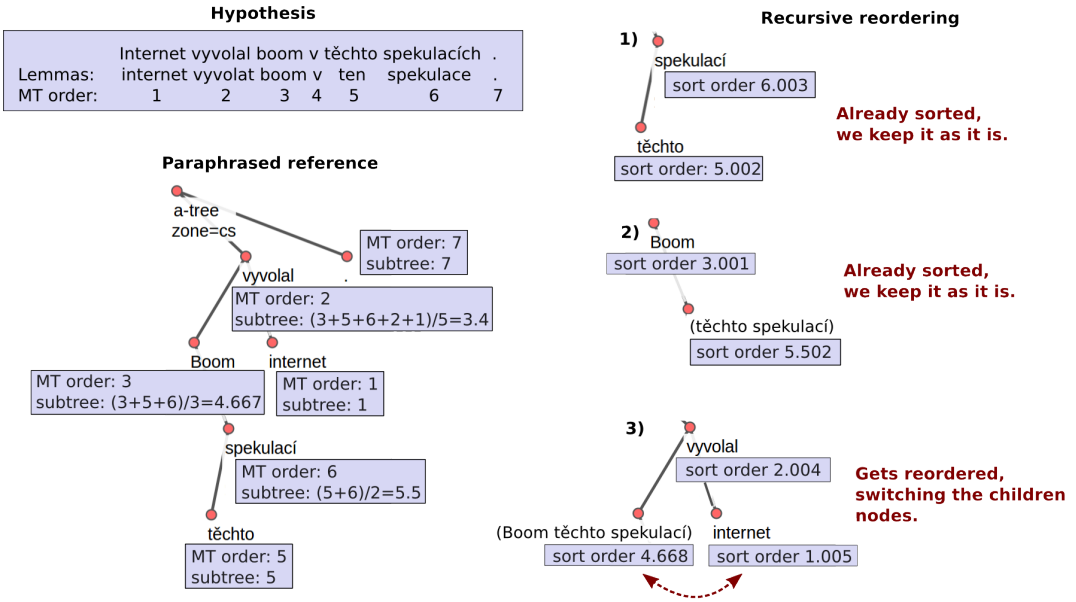


Figure 3.14: Continuation of Figure 3.13, reordering of the paraphrased reference sentence. This visualisation originally comes from Figure 3 in Barančíková and Rosa [2015].

To address this inconvenience, when paraphrasing nouns, the module deletes the value of the corresponding gender grammateme of the noun's t-node. This action prevents incorrect gender assumptions from being carried over to the new lemma. The subsequent synthesis pipeline within Treex then generates the correct value of the morphological gender feature. This feature is critical for ensuring that the noun correctly agrees with its dependents, such as adjectives and verbs, in terms of morphological properties.

Example of the method is presented in Figure 3.13.

3.5.4 Synthesis from t-layer to a-layer

In this phase, a-nodes corresponding to auxiliary words and punctuation are generated, morphological feature values on a-nodes are initialized and set to enforce morphological agreement among the nodes. Correct inflectional forms based on lemma and POS, and morphological features are generated using MorphoDiTa.

3.5.5 Tree-based reordering at the a-layer

During the synthesis, the newly generated reference sentence is reordered at the analytical layer using the reordering block A2A::ReorderByLemmas developed by Rudolf Rosa. This block is freely available at GitHub.[14]

The idea behind the block is to make the word order of the paraphrased reference sentence more similar to that of the hypothesis, while respecting the syntactic constraints of the dependency tree in order to avoid ungrammatical constructions.

[14] `https://github.com/ufal/treex/blob/master/lib/Treex/Block/A2A/ReorderByLemmas.pm`

The algorithm operates on the analytical dependency tree and reorders modifier subtrees of a given head node so that their linear order approximates the order observed in the hypothesis. The procedure consists of the following steps (illustrated in Figure 3.14):

1. **Alignment:** Lemmas in the newly generated reference sentence that also occur in the hypothesis are aligned. Reference words that do not appear in the hypothesis are considered unaligned, because reordering them would have little effect on word-based MT evaluation metrics. For simplicity, repeated words are also considered unaligned, since resolving ambiguity among multiple possible alignments is non-trivial. In practice, such repetitions are relatively rare.

2. **MT order:** Each aligned reference a-node is assigned an *MT order*, defined as the linear position of its aligned counterpart in the hypothesis sentence. Unaligned nodes have an undefined MT order.

3. **Subtree MT order:** For each reference a-node (R), *subtree MT order* is computed as the average MT order of all a-nodes within the subtree rooted at R, including R itself. If no aligned nodes are present in the subtree, the subtree MT order remains undefined.

4. **Sorting order:** For a head node H, the sorting order is defined as its MT order. For each dependent node D_i, the *sorting order* is defined as the subtree MT order of D_i. If the sorting order of a node is undefined, it is inherited from the nearest preceding node in linear order, which biases the algorithm toward keeping neighboring nodes together in the absence of evidence for reordering. To break ties, a small fraction (1/1000) of the original word order index is added, ensuring that the original order is preferred when no clear reordering signal is present.

5. **Recursive reordering:** The algorithm processes the tree bottom-up. For each head node H, its dependent subtrees D_i are reordered according to their sorting order. If a dependent node D_i is the root of a subtree, the entire subtree is moved as a unit, preserving its internal word order. Reordering is performed only when the current linear order violates the sorting order, ensuring minimal and controlled changes.

Non-projective edges are not treated specially: if they participate in reordering, they are projectivized; otherwise, their original order is preserved. Importantly, no new non-projective edges are introduced, as subtrees are always moved as whole units.

Each node participates in at most two reorderings—once as a head and once as a dependent. As a result, reorderings are independent of processing order, although a bottom-up traversal is used for implementation convenience.

The reordering process is illustrated on Figure 3.14.

3.5.6 Synthesis from a-layer to w-layer

The word forms are already generated at the a-layer, so there is little to be done. Superfluous tokens are deleted (e.g. duplicated commas), prepositions are

WMT12

Reference	Original	Paraphrased	Reordered
BLEU	0.751	0.783	0.804
Meteor-next	0.833	0.864	0.868
Meteor-next, exact match	0.861	0.900	**0.903**

WMT13

Reference	Original	Paraphrased	Reordered
BLEU	0.834	0.850	0.878
Meteor-next	0.817	0.871	0.870
Meteor-next, exact match	0.848	**0.893**	**0.893**

Table 3.16: Pearson correlation of a metric and the silver standard of human judgment on original references, paraphrased references before reordering and paraphrased reordered references. We evaluate it using BLEU and METEOR-NEXT, which we used in two settings - with paraphrases and exact match only (i.e. no paraphrase support).

vocalized, the first letter in a sentence is capitalized, and the tokens are concatenated (a set of rules is used to decide which tokens should be space-delimited and which should not).The example in Figure 3.14 results in the following sentence: *Internet vyvolal boom těchto spekulací* (The Internet has caused a boom of these speculations.), which has the same meaning as the original reference sentence, is grammatically correct and, most importantly, is much more similar in wording to the hypothesis.

3.5.7 Results

Our experiments were performed using datasets from the English-to-Czech translation task of the WMT12 Callison-Burch et al. [2012] and WMT13 Bojar et al. [2013a] competitions. For paraphrasing purposes, we employed two Czech paraphrase tables – *filtered METEOR* (see Section 3.1.2) and the Czech WordNet 1.9 PDT Pala and Smrž [2004], both of which contain lemmatized paraphrases.

Our method of paraphrasing is independent of the evaluation metric used. We employ three different metrics – BLEU, METEOR-NEXT and METEOR-NEXT without paraphrase support. Using paraphrases in a metric on already paraphrased sentences seems redundant and furthermore, we showed that employing METEOR-NEXT without paraphrase support can be beneficial as discussed in Section 3.3.

The results are presented in Table 3.16 as Pearson correlation of each metric with human judgment. In both datasets, paraphrasing clearly helps to reflect the human perception better. Notably, even the METEOR-NEXT metric, which includes paraphrase components, shows improved performance when utilizing references crafted from its own paraphrase tables. We attribute this improvement

to the reduction of noise within the paraphrase table, which otherwise blurs the difference between hypotheses from different MT systems.

Unlike the earlier substitution-based methods, the Treex-based approach modifies sentence structure only when supported by syntactic evidence, which limits both its gains and its error profile. The sentence reordering proves clearly beneficial when assessed using the BLEU metric, which penalizes any changes in word order to the reference sentence. Conversely, METEOR-NEXT is more forgiving of word order changes, and thus reordering has no significant impact on its scores.

In general, the rule-based paraphrasing approach operating in the deep syntactic layer demonstrates a consistent improvement in correlation with human judgment in datasets and evaluation metrics. Although the absolute gains are smaller than those achieved by the earlier simple substitution method, (See Section 3.2, the current approach offers a linguistically grounded alternative that enables controlled paraphrasing and explicit adjustment of word-order within a single unified framework.

We answered the following research questions:

Can targeted paraphrasing at a deep syntactic layer improve the correlation between automatic MT evaluation metrics and human judgment? Yes. Experiments on WMT12 and WMT13 datasets show that paraphrasing reference sentences at the deep syntactic (t-) layer consistently improves the correlation of automatic metrics with human judgment. This holds across different evaluation metrics, including BLEU and METEOR-NEXT.

How does a rule-based, linguistically motivated paraphrasing approach compare to our previous methods? In contrast to our previous experiment with SMT-based paraphrasing, the rule-based Treex approach leads to consistent improvements in correlation with human judgment. However, the earlier, simple substitution-based methods still achieve larger absolute gains in correlation. However, it offers greater linguistic control, avoids metric-specific heuristics, and enables syntactically constrained transformations, including word-order adaptation.

Does explicit word-order reordering of paraphrased references further improve MT evaluation? Yes, partially. Tree-based reordering at the analytical layer leads to additional improvements for word-order–sensitive metrics such as BLEU. In contrast, metrics that are more tolerant to word-order variation, such as METEOR, show little or no additional benefit from reordering.

4. Sentence Embedding Evaluation

There is a significant time gap between this chapter and the previous one, during which the field of natural language processing has evolved significantly. Complex neural architectures, precomputed word and sentence representations, and large language models (LLMs) have transformed NLP research and its application. Models such as BERT [Devlin et al., 2019], RoBERTa [Liu et al., 2019a], or GPT-3 [Brown et al., 2020] have become foundational, setting new benchmarks in practically every NLP task. With newer, larger and more efficient models appearing frequently, State-of-the-Art (SOTA) results are constantly being surpassed.

Sentence embeddings have become ubiquitous in NLP, with novel representation methods appearing almost every other week. However, with such an overwhelming number of methods available to compute sentence vector representations, studying their general properties has become increasingly difficult. Furthermore, it remains an open research question how to evaluate sentence embeddings as representations, rather than as components of downstream systems. In particular, it is unclear to what extent modern embeddings preserve fine-grained semantic structure in a stable, interpretable, and linguistically meaningful way.

Machine translation evaluation serves as a great example of this evolution. Traditionally, MT evaluation relied heavily on string-based metrics such as BLEU or TER. These metrics compare machine-generated translations to human references by measuring n-gram overlaps. However, they struggle to capture the full range of possible valid translations because they rely on the exact matching of words and phrases. To overcome this limitation, techniques like paraphrasing [Denkowski and Lavie, 2014, Kauchak and Barzilay, 2006, Bawden et al., 2020] were introduced to better align semantically similar sentences that differed in surface form, thus improving the evaluation's ability to account for linguistic diversity.

Despite the historical utility of paraphrasing, there is currently strong pressure to stop using string-matching metrics [Mathur et al., 2020, Freitag et al., 2022] in favor of semantic metrics that correlate better with human judgment. String-based approaches are nowadays being replaced by more advanced techniques based on precomputed continuous-space representations, commonly referred to as embeddings. Embeddings are high-dimensional vectors of real numbers that are learned from large-scale datasets. They effectively capture the semantic and syntactic nuances of sentences, offering a more sophisticated understanding of meaning.

Modern metrics, which we introduced in more detail in Section 2.2, such as COMET [Rei et al., 2020], BERTscore [Zhang et al., 2020], or BLEURT [Sellam et al., 2020b], assess translations based on the vector similarity between a hypothesis and a reference translation. A high similarity indicates that the hypothesis conveys a meaning close to the reference, providing more reliable evaluations even when their surface forms differ significantly.

This shift represents a major advancement, as it allows for more flexible and accurate evaluations, particularly for sentences with multiple valid translations

or significant syntactic variations. Furthermore, the rise of massively multilingual models such as XLM-R [Conneau et al., 2020], LaBSE [Feng et al., 2022], and M5 [Wang et al., 2024] allows these metrics to be applied across languages without extensive adaptations.

As embedding-based metrics become prevalent, the need for explicit paraphrasing to account for variations in sentence structure or word choice has diminished[1]. However, the field has recently seen another shift: the use of LLMs as evaluators. Kocmi and Federmann [2023b] demonstrated that LLMs can function as SOTA evaluators for machine translation in reference-free assessment. For reference-based evaluation, Kocmi and Federmann [2023a] introduced GEMBA-MQM, a high-performing metric based on GPT-4 [OpenAI, 2023].

However, using LLMs has significant disadvantages. They are computationally intensive, slow, and often reside behind paid, black-box APIs, limiting transparency and control over the evaluation process. Most importantly, a key feature of LLMs – their non-deterministic nature – becomes a liability in this context, making them unsuitable for evaluation processes that require reproducibility and stability [Shen et al., 2023]. In scenarios requiring efficiency, privacy, and determinism, encoder-based embedding metrics, which are the focus of this chapter, are much more suitable.

Although sentence embeddings provide a powerful tool for evaluating sentence similarity, they are not without limitations. An ideal sentence embedding, which perfectly captures the full meaning of a sentence, would eliminate any need for paraphrasing entirely. However, in practice, this ideal is difficult to achieve. The process of embedding a sentence into a fixed-length vector inherently involves lossy compression and generalization, and as a result, embeddings may not always fully capture the richness of the sentence's meaning. If an embedding fails to encapsulate essential semantics, such as negation or causality, metrics derived from it can be misleading. Furthermore, the selection of a specific embedding model for evaluation becomes essential – different models may produce embeddings with varying degrees of fidelity to the original sentence, and the choice of model can significantly affect the outcome of the evaluation.

This raises critical questions: How well does a particular embedding preserve subtle semantic distinctions? Is it sensitive to grammatical structure? How does it handle small perturbations? Choosing a model best suited for modern metrics is a significant challenge. To address this, we focus on a systematic evaluation of sentence embeddings as representations, rather than on downstream task performance. Our goal is to analyze how different embedding models preserve semantic, syntactic, and structural information, and to identify which properties make embeddings robust and reliable for evaluation purposes.

This chapter is structured as follows. First, in Section 4.1 we review existing approaches to intrinsic and extrinsic evaluation of sentence embeddings. In Section 4.3, we introduce an experiment analyzing linear properties of sentence embedding spaces. We then present the Costra dataset for Czech sentence embedding evaluation in Section 4.4, followed by a comparison of intrinsic and extrinsic evaluation methodologies in Section 4.5.

[1]While sentence embeddings have reduced reliance on paraphrasing, the metric PRISM [Thompson and Post, 2020] demonstrates that integrating both approaches can still produce superior results by capturing both semantic similarity and translation diversity

4.1 Related Work on Intrinsic and Extrinsic Evaluation of Sentence Embeddings

When Kiros et al. [2015] introduced Skip-Thought vectors, one of the first effective methods for encoding sentences into a fixed-length vector from unlabeled text, they evaluated their quality on eight supervised tasks such as paraphrase detection or sentiment polarity classification. This *extrinsic* evaluation, or 'transfer tasks', i.e., using sentence embedding as input features for downstream tasks, became the de facto standard for the evaluation and comparison of sentence embeddings. Furthermore, this method corresponds with the primary objective of the user, to determine which embedding performs the best in their specific application such as sentiment analysis, information retrieval, natural language inference, and machine translation.

Despite its widespread adoption, extrinsic evaluation has limitations. Wieting et al. [2015] demonstrated that even simple bag-of-words (BOW) approaches often achieve competitive results in transfer tasks, questioning the necessity of complex models for certain applications. Extrinsic evaluations may also lack transparency in explaining why certain embeddings perform better, making it difficult to understand the underlying properties that contribute to their success or failure in various tasks. Moreover, they might not reflect the embedding's ability to capture nuanced linguistic features beyond the specific context of the evaluated tasks. Eger et al. [2019] highlight additional issues in existing evaluation techniques, including biases that favor larger embeddings.

Alternatively, *intrinsic* evaluation should not just test classification accuracy, but the structural correspondence between meaning and geometry, quality of sentence representations in capturing the sentence's meaning, and various semantic nuances such as formality, sentiment, and syntactic structure. This involves analyzing the embeddings to see how effectively they preserve not just the core semantic content but also more subtle aspects of language that contribute to the overall communicative intent.

For **word** embeddings, there are numerous datasets for intrinsic evaluation. The word analogy task (e.g., "king" is to "queen" as "man" is to "woman") [Mikolov et al., 2013b,c, Gladkova et al., 2016] and the word semantic similarity task [Faruqui and Dyer, 2014, Hill et al., 2015, Gerz et al., 2016] are among the most popular methods. The attributes of word embeddings have been studied in depth from various aspects, from sentiment polarity [Chen et al., 2013] to the ability to predict syntactic properties (such as POS, gender, case or number) [Köhn, 2015] to data efficiency [Jastrzkebski et al., 2017]. Bakarov [2018] provide a typology of 16 intrinsic evaluation methods. In fact, there are so many studies on the properties and improvements of one of the most popular contextualized word embeddings, BERT, that it motivated a metastudy to summarize all current knowledge of BERT [Rogers et al., 2020]. Li et al. [2020] discuss calibration methods such as BERT-flow to improve sentence embedding quality.

For sentence embeddings, this is a different story – intrinsic evaluation is more challenging due to sentence complexity. Early studies by Adi et al. [2016] and Ettinger et al. [2016] introduced 'probing tasks' for intrinsic evaluation of sentence embedding, focusing on surface-level linguistic features such as sentence length, verb tense, and word order. Building on this foundation, Conneau et al. [2018]

expanded the scope of these tasks, introducing a broader set of linguistic probes to systematically assess the extent to which sentence embeddings capture various linguistic properties. However, Linzen et al. [2016] and Bacon and Regier [2018] have raised concerns that high performance in these tasks may be due to flaws in the test design, such as the use of grammatically simple sentences.

Conneau et al. [2018] was extended to SentEval [Conneau and Kiela, 2018], a toolkit designed for both intrinsic and extrinsic evaluation of sentence embeddings. SentEval contains 10 probing tasks and 17 transfer tasks, which mainly focus on single sentence classification and semantic textual similarity (STS). STS measures how well a system can determine the semantic similarity between two sentences, i.e., directly evaluating the ability of the embeddings to capture semantic meaning. Perone et al. [2018] applied SentEval to many sentence embedding techniques, showing that no method had consistently good performance on all tasks, highlighting the challenges of developing universally robust sentence embeddings.

However, Zhelezniak et al. [2019] and Wang et al. [2022a] warned against using STS as the only indicator of the quality of sentence embeddings, because strong performance in STS tasks may not necessarily correlate with the effectiveness of sentence representations in a wider range of downstream applications [Neelakantan et al., 2022]. Hewitt and Liang [2019] questioned whether high performance on probing tasks truly reflects linguistic understanding or if models are exploiting statistical artifacts. They emphasized the need for control tasks to differentiate genuine linguistic competence from dataset biases.

Muennighoff et al. [2023] introduced the Massive Text Embedding Benchmark (MTEB)[2]. It started with 8 tasks on 58 datasets for more than a hundred languages and currently[3] it contains more than 250 datasets and 10 tasks. MTEB is open-source and offers a public leaderboard where researchers can compare model performances on various tasks and contribute new results. Recently, Enevoldsen et al. [2025] presented the Massive Multilingual Text Embedding Benchmark (MMTEB), which comprises more than 500 distinct tasks in 10 task categories, covering more than 250 languages.

In the context of evaluating sentence embeddings, particularly for languages such as Czech, the introduction of MMTEB represents a significant advancement. It allows for more thorough intrinsic and extrinsic evaluations, ensuring that embeddings are not only effective in capturing semantic meaning but also versatile across various applications. By participating in benchmarks such as MTEB or MMTEB, researchers can better understand the strengths and limitations of their models, contributing to the development of more accurate and reliable sentence embeddings. However, such benchmarks primarily provide performance rankings and offer limited insight into why certain embeddings behave well or poorly, motivating complementary analyses that directly probe representation structure.

In the context of the Czech language, there are a few intrinsic evaluation datasets – Svoboda and Brychcín [2018] have translated and subsequently annotated the SentEval toolkit into Czech, creating the first Czech STS dataset. Other Czech STS benchmarks include Konopík et al. [2017] and Sido et al. [2021b]. Costra [Barančíková and Bojar, 2020, Barančíková and Bojar, 2020] is a dataset

[2] `https://huggingface.co/spaces/mteb/leaderboard`

[3] November 17 2024

designated for evaluating the quality of Czech sentence embeddings spaces. It examines the proficiency of sentence embeddings in capturing intricate linguistic phenomena such as style, formality, or simplicity. We introduce Costra in more detail in Section 4.4.

4.2 Sentence Embeddings

In this chapter, we experiment with a diverse range of models, moving from early recurrent architectures to modern instruction-tuned LLMs. This selection allows us to trace how the evolution of sentence representations—specifically a transition from unsupervised pre-training to contrastive fine-tuning—affects the linear geometric properties of their embedding spaces.

Rather than aiming for exhaustive coverage, we select representative models that illustrate key architectural and training paradigms. The goal is not to identify a single "best" embedding, but to analyze how different design choices shape the geometry and other properties of sentence representations. All selected models and their basic characteristics are summarized in Table 4.1 at the end of this section.

Although in this work we focus primarily on Czech embedding models, Section 4.3 is computed using English data. Therefore, we use Czech, English, and multilingual models. In order to increase clarity in model languages and their sizes[4], we introduce the following symbols:

Embedding Type approx. parameters	*small* (14–30M)	*base* (110M)	*large* (335M+)	*proprietary* unknown
Czech Monolingual	△	□	⬠	-
English Monolingual	▲	■	⬟	-
Multilingual	▲	■	⬟	●

Legacy Architectures

The following models provide historical baselines from the era before Transformer Vaswani et al. [2017]. We include these models to provide a historical baseline and to evaluate how early attempts handle linear relationships. They often rely on simpler compositionality, which shows how deep contextualization impacts vector arithmetic.

- `InferSent` ⬟ [5] [Conneau et al., 2017] is one of the first embedding models that uses supervised learning to compute sentence representations. Specifically, it was trained to predict inference labels in the SNLI dataset [Bowman et al., 2015]. The authors tested 7 different architectures, finding that a BiLSTM (Bidirectional Long Short-Term Memory) encoder with max pooling achieved the best results. We use a version trained using GloVe word embeddings [Pennington et al., 2014].

[4] Following the naming convention of Clark et al. [2020], we categorize models based on the number of their parameters

[5] `https://github.com/facebookresearch/InferSent`

- Similarly to `InferSent` ⬟ , `USE` ⬟ (Universal Sentence Encoder)Cer et al. [2018] combines unsupervised learning augmented with training on supervised data from SNLI. It comes in several versions, we use a larger transformer-network[6] designed for higher accuracy.

- `ELMo` ■ [7] (Embeddings from Language Models) [Che et al., 2018] derives contextualized representations from a biLSTM trained on a language modeling objective over a large text corpus. We compute sentence embeddings by averaging all token representations from the final layer.

- `LASER` ■ [8] (Language-Agnostic SEntence Representations) [Artetxe and Schwenk, 2018] is a five-layer BiLSTM trained on machine translation from over 90 languages to English or Spanish, with random source language selection per batch. It uses max-pooling over the last states to produce a 1024-dimensional vector, optimized for cross-lingual alignment.

Modern Baselines

These models represent the transition to attention-based architectures. While BERT-based models serve as the standard foundation, their off-the-shelf representations often exhibit high *anisotropy* (occupying a narrow cone in the vector space), which limits their effectiveness for semantic similarity without specific fine-tuning [Ethayarajh, 2019].

- `BERT-Large` ⬟ [9] (Bidirectional Encoder Representations from Transformers) [Devlin et al., 2019] is a deep unsupervised representation pre-trained on Masked Language Modeling (MLM) and next-sentence prediction. Although the original approach extracted the [CLS] token (a special token inserted at the beginning of every sentence), Reimers and Gurevych [2019] demonstrated that mean pooling consistently outperforms [CLS] for sentence-level tasks. In our experiments, we use mean pooling to reduce the inherent bias of raw BERT embeddings.

- `SBERT` ▲ [10] (Sentence-BERT) [Reimers and Gurevych, 2019] addresses the limitation of BERT by using a siamese network architecture, where two identical BERT networks share weights to process sentence pairs. It is optimized such that the cosine similarity between vectors directly reflects semantic similarity. We adopt the `all-MiniLM-L6-v2` variant, which balances competitive performance with high inference speed.

Contrastive and Generative Approaches

- `SimCSE` ■ [11] (Simple Contrastive Learning of Sentence Embeddings) [Gao et al., 2021] has a simple yet effective unsupervised training objective. It

[6] `https://tfhub.dev/google/universal-sentence-encoder-large/3`
[7] `https://github.com/HIT-SCIR/ELMoForManyLangs`
[8] `https://github.com/facebookresearch/LASER`
[9] `https://huggingface.co/google-bert/bert-large-cased`
[10] `https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2`
[11] `https://github.com/princeton-nlp/SimCSE`

uses dropout as a minimal data augmentation technique: the same sentence is passed through the encoder twice with different dropout masks, creating a "positive pair" that the model learns to pull together in vector space, while pushing away representations of other sentences. This approach is specifically designed to address the "anisotropy" of BERT, pushing apart unrelated sentences to make the embedding space more uniform.

Czech Encoders

A primary focus of this work is the evaluation of Czech-specific models. We include three *base*-size Transformer architectures, each using masked language modeling as their primary pretraining objective:

- CZERT-b-cased[12] (`Czert` □ , Sido et al., 2021a)
- FERNET-C5[13] (`FERNET` □ , Lehečka and Švec, 2021)
- `RobeCzech` □ [14] [Straka et al., 2021]

We also adopt two efficient Czech *small* encoders from Bednář et al. [2024]:

- Seznam/retromae-small-cs[15] (`RetroMAE` △), a BERT-small model trained with the RetroMAE objective on a custom Czech corpus from the Seznam.cz search engine
- Seznam/simcse-dist-mpnet-czeng-cs-en[16] (`SimCSE-dist-cs` △), a distilled BERT-small model derived from `sentence-transformers/all-mpnet-base-v2`, trained on the parallel Czech–English dataset CzEng [Bojar et al., 2012] and fine-tuned with the SimCSE objective [Gao et al., 2021]. `SimCSE-dist-cs` △ was the best performing model in zero-shot evaluation among recently published *small* -size Czech sentence embedding models that are efficient in both size and speed Bednář et al. [2024].

Multilingual Models

Although the experiment in Section 4.3 utilizes English data (SNLI and MultiNLI), we include massively multilingual models to serve as a bridge to Section 4.4, where we evaluate Czech representations. It is essentially an open research question whether the cross-lingual alignment objectives used to train these models distort the vector space in a way that affects the linearity of monolingual (English) relationships.

- `LaBSE` ■ [17] (Language-agnostic BERT Sentence Embedding) [Feng et al., 2022] is a dual-encoder model that generates language-agnostic representations for more than 100 languages with remarkable cross-lingual alignment, since its training objective was machine translation into a shared semantic space.

[12] `https://huggingface.co/UWB-AIR/Czert-B-base-cased`
[13] `https://huggingface.co/fav-kky/FERNET-C5`
[14] `https://huggingface.co/ufal/robeczech-base`
[15] `https://huggingface.co/Seznam/retromae-small-cs`
[16] `https://huggingface.co/Seznam/simcse-dist-mpnet-czeng-cs-en`
[17] `https://huggingface.co/sentence-transformers/LaBSE`

- `BGE-M3` ⬟ [18] The BAAI General Embedding [Chen et al., 2024] are currently among top-performers on benchmarks like MTEB. It supports over 100 languages with hybrid training for dense, sparse, and multi-vector retrieval. We use its dense output, trained via retro-fitting on diverse datasets including hard negatives for improved discrimination.

- `XLM-RoBERTa-large` (`XLM-R` ⬟) [19] [Conneau et al., 2019] represents a significant milestone in multilingual encoders, scaling the RoBERTa architecture to a massive scale. Unlike previous models trained on constrained Wikipedia data, XLM-R was pretrained on 2.5TB of filtered CommonCrawl data. Geometrically, it is unique because it does not use explicit language identifiers, learning a shared multilingual manifold purely through MLM.

- `multilingual-e5-large` (`mE5` ⬟ , [Wang et al., 2024]), a model pretrained using a contrastive learning approach on a diverse range of tasks, including natural language inference and question answering across multiple languages.

Instruction-Tuned Encoders

This category represents the current state-of-the-art for encoder-only models (2023–2025). These models move beyond simple self-supervised pre-training, utilizing massive-scale contrastive learning and instruction tuning to align the embedding space with human semantic expectations.

- The `E5-Large-v2` ⬠ [20] and `E5-Small-v2` ▲ [21] Wang et al. [2022b] models represent a paradigm shift by leveraging a massive, weakly-supervised dataset (CCPairs) for contrastive training. Geometrically, these models are unique because they require task-specific instruction prefixes (e.g., *"query: "* or *"passage: "*). **e5-small-v2** is roughly 10x smaller than (335M parameters), which allow us to observe how model scale influences the linear relationships.

- `ModernBERT` ■ [22] [Warner et al., 2025] represents the 2024 modernization of the BERT architecture. It incorporates *Rotary Positional Embeddings (RoPE)* and unpadding to support long-context windows (up to 8k tokens) efficiently. Unlike the original BERT, which used absolute position embeddings, RoPE introduces a relative geometric bias that allows the model to better capture long-range dependencies, potentially affecting the global structure of the sentence-level embedding space.

LLM-based and Proprietary Embeddings

Recent trends involve adapting massive Large Language Models (LLMs) into embedding models. This shift represents a move from Encoder-only architectures

[18] `https://huggingface.co/BAAI/bge-m3`
[19] `https://huggingface.co/facebook/xlm-roberta-large`
[20] `https://huggingface.co/intfloat/e5-large-v2`
[21] `https://huggingface.co/intfloat/e5-small-v2`
[22] `https://huggingface.co/nomic-ai/modernbert-embed-base`

(like BERT) to Decoder-only architectures, using the whole world knowledge and reasoning capabilities inherent in models with billions of parameters. Unlike BERT (which uses the [CLS] token or mean pooling), these decoder-only models usually extract the representation from the last token in the sequence.

- `Qwen3-4B` ⬟ [23] [Zhang et al., 2025] represents open-weights LLM embedding models. Based on a 4-billion parameter decoder, it is fine-tuned with contrastive loss and *Matryoshka Representation Learning* (MRL). MRL is specifically designed to make the vector space *nested*: the model is trained so that the first few dimensions (e.g., 256) contain the most critical semantic information, while larger dimensions provide finer-grained detail.

- `TE3-small` ● and `TE3-large` ● [24] gpt4 from OpenAI are the commercial standards for closed-source embeddings. Like Qwen3, these models utilize MRL, allowing users to truncate vectors (down to 256 dimensions) with minimal performance degradation. While their exact training data is proprietary, their inclusion allows us to compare properties of highly optimized "black-box" models against open-source alternatives.

- `Gemini-001` ● [25] [Google DeepMind, 2023] from Google utilizes a similar LLM-based backbone. It is optimized for a wide variety of tasks including clustering and classification. By including Gemini alongside OpenAI and Qwen, we can evaluate whether the massive scaling of decoder-only models creates a geometric structure of their semantic spaces.

Control Baselines

We introduce two control baselines designed to isolate the effects of the model architecture from the information acquired during pre-training.

- Untrained `Random BERT` ■ and `Random BERT` ■ : To isolate the contributions of the model architecture and tokenization, we employ two random versions of BERT where all weights are randomly initialized, using Xavier normal initialization: (i) `Random BERT` ■ for English texts, this model utilizes the `bert-base-uncased` architecture and its corresponding English tokenizer. (ii) `Random BERT` ■ for Czech texts, this model utilizes the `bert-base-multilingual-cased` architecture and its multilingual tokenizer. The only component inherited from the pretrained model is the tokenizer. In other words, the model processes input according to the tokenization patterns learned from data but does not have any pretrained language representations. This setup isolates and evaluates the contribution of the tokenizer alone, establishing a lower performance bound and offering a meaningful point of comparison to evaluate the benefits of pretraining.

- `Random`: As a second baseline, we used completely random vectors assigned to each sentence. This provides a minimal benchmark, helping to contextualize the performance of trained embedding models.

[23] `https://huggingface.co/Qwen/Qwen3-Embedding-4B`
[24] `https://platform.openai.com/docs/guides/embeddings`
[25] `https://ai.google.dev/gemini-api/docs/models`

Category	Model	Language	Arch.	Params	Dim
Legacy	`InferSent` ⬟	English	BiLSTM	≈200M	4096
	`ELMo` ■	English	BiLSTM	94M	1024
	`USE` ⬟	English	Encoder	147M	512
	`LASER` ■	Multilingual	BiLSTM	≈200M	1024
Baselines	`BERT-Large` ⬟	English	Encoder	336M	1024
	`SBERT` ▲	English	Encoder	≈23M	384
Czech	`Czert` □	Czech	Encoder	109M	768
	`FERNET` □	Czech	Encoder	≈200M	768
	`RobeCzech` □	Czech	Encoder	125M	768
	`RetroMAE` △	Czech	Encoder	≈14M	256
	`SimCSE-dist-cs` △	Czech	Encoder	≈23M	384
Contrastive	`SimCSE` ■	English	Encoder	110M	768
Multilingual	`LaBSE` ■	Multilingual	Encoder	471M	768
	`BGE-M3` ⬟	Multilingual	Encoder	567M	1024
	`XLM-R` ⬟	Multilingual	Encoder	560M	1024
	`mE5` ⬟	Multilingual	Encoder	560M	1024
Instruction	`E5-Small-v2` ▲	English	Encoder	≈33M	384
	`E5-Large-v2` ⬟	English	Encoder	≈335M	1024
	`ModernBERT` ■	English	Encoder	149M	768
LLM	`Qwen3-4B` ⬟	Multilingual	Decoder	4.0B	2560
	`TE3-small` ●	Multilingual	Prop.	Prop.	1536
	`TE3-large` ●	Multilingual	Prop.	Prop.	3072
	`Gemini-001` ●	Multilingual	Prop.	Prop.	3072
Controls	`Random BERT` ■	English	Encoder	110M	768
	`Random BERT` ■	Multilingual	Encoder	110M	768
	`Random`	N/A	N/A	N/A	256

Table 4.1: Detailed comparison of embedding models across architectures and linguistic focus. Size labels and symbols denote parameter scales and language: `gray` (English), `white` (Czech), `black` (Multilingual/Proprietary). For models that support truncation via MRL, dimensions are the default/maximum output size.

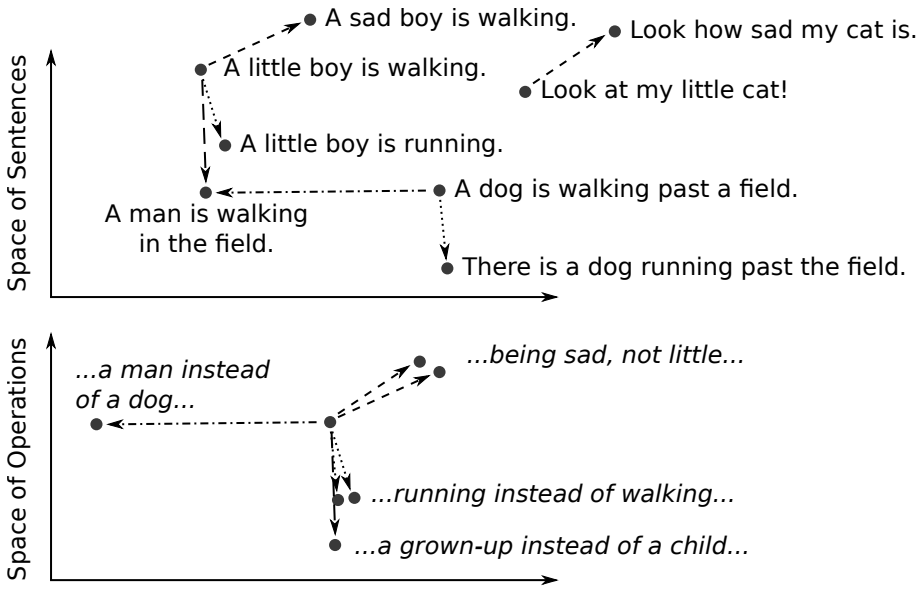


Figure 4.1: An illustration reprinted from Barančíková and Bojar [2019]. It represents a continuous multi-dimensional vector space representing individual sentences, a 'space of sentences' (upper plot) where each sentence is represented as a dot. Pairs of related sentences are connected with arrows; dashing indicates various relation types. The lower plot illustrates a possible 'space of operations' (here a vector difference shown, so all arrows are simply moved to start at a common origin). The hypothesis is that consistent semantic changes (e.g. all vector transformations extracted from sentence pairs differing in the speed of travel "running instead of walking") should be represented close to each other in the space of operations, i.e., form a more or less compact *cluster*.

4.3 In Search for Linear Relations in Sentence Embedding Spaces

In this section,[26] we present an initial exploration of continuous-space vector representations of sentences. A notable observation by Mikolov et al. [2013a] demonstrated that meaningful relationships between words have immediate geometric counterparts in the continuous vector space. Mikolov et al. [2013a] defined a series of tests to measure how well word embeddings capture semantic and syntactic information. These tests include, for example, declination of adjectives ("easy"→"easier"→"easiest"), changing the tense of a verb ("walking"→"walk") or matching a capital to a corresponding country ("Athens"→"Greece") or currency ("Angola"→"kwanza").

Our objective is to determine whether *whole sentence* embeddings exhibit the same linear behavior, looking for an analogous behavior to that observed with word embeddings. This idea is illustrated in Figure 4.1. Specifically, we examine whether semantic shifts between sentences can be modeled as transformation

[26]This section is based on the article [Barančíková and Bojar, 2019], a joint work with Ondřej Bojar. The methodology and experimental framework is adapted from the original study, however, all experiment with the exception of Figure 3.11 are recomputed using updated scripts and updated set of sentence embedding model. As a result, reported numbers may differ from those published in the original article.

vectors.

Following Mikolov et al. [2013a], we deliberately do *not include* this aspect in the training objective of the sentence representations, but instead search for properties that are learned in an unsupervised way, as a side-effect of the original training objective, data and setup. This approach has the potential to *explain* good or bad performance of the representation types examined in various tasks.

We acquire pairs of sentences that differ by small alterations (such as a change of noun, the addition of an adjective, or punctuation) from natural language inference datasets. We hypothesize that if P and H represent two sentences differing only by a single linguistic edit (e.g., `X man Y` $\rightarrow$ `X woman Y`), the operation between their representation in vector space $V_{trans} = V_H - V_P$ should isolate that specific edit in a consistent direction across the vector space.

In particular, we proceed in the following steps:

1. **collect sentence pairs:** each pair consists of a premise and a hypothesis that differ by a specific small alteration (the modification pattern). The process of collecting such pairs of sentences is described in detail in Section 4.3.1.

2. **compute sentence embeddings:** each sentence is embedded in a vector space using various sentence embedding models. We use a representative subset of sentence embeddings that are introduced in Section 4.2,

3. **compute vector operations:** apply arithmetic operations to the embeddings of this pair of sentences to capture the transformation between them, i.e., their *transformation vectors*. The operations used are presented in Section 4.3.3.

4. **cluster transformation vector and evaluate clustering:** For evaluation, we first use the k-Means algorithm for clustering to group transformation vectors based on their position in the space of operations. In order to see how well the clusters correspond to the actual modification patterns, we evaluate them using cluster evaluation metrics. Clustering and evaluation metrics are described in more detail in Section 4.3.4.

In this chapter, we investigate whether sentence embedding spaces exhibit linear semantic structure analogous to that observed in word embeddings, focusing on the following research questions:

- **Can semantic and syntactic transformations between sentences be captured through simple arithmetic operations in continuous vector spaces?**
- **To what extent do model architecture and scale influence the linearity of the sentence embedding space?**
- **Is the linearity of a model the result of a learned semantic understanding or an inherent inductive bias of its architecture?**
- **Which linguistic phenomena are most (and least) consistently represented as linear transformations in sentence embeddings?**

Step	Premise and hypothesis
0.	P: a man with a tattoo behind his **ear is playing a guitar.**
	H: a woman with a tattoo behind her **ear is playing a guitar.**
1.	P_1: a man **with a tattoo behind** his X
	H_1: a woman **with a tattoo behind** her X
2.	P_2: a man Y his X
	H_2: a woman Y her X
3.	P_3: a man X his Y
	H_3: a woman X her Y

Figure 4.2: Example of our pattern extraction method. In step 0, the longest common substring of tokens (*ear is playing a guitar .*) between the original premise P and original hypothesis H is identified and replaced with the variable X in step 1. In step 2, *with a tattoo behind* is identified as the longest common substring and substituted with the variable Y. As the variables are not listed alphabetically in the premise, they are switched in step 3, which also yields the final pattern `a man X his Y` → `a woman X her Y`.

4.3.1 Sentence Patterns

Because manual creation of sentence variations is costly, we reuse existing English data from the Stanford Natural Language Inference Corpus (SNLI) [Bowman et al., 2015] and Multi-Genre Natural Language Inference (MultiNLI) [Williams et al., 2018b]. Both collections consist of pairs of sentences, a premise and a hypothesis, and their relationship (entailment, contradiction, or neutral). The two datasets together contain almost one million unique sentence pairs. All sentences were lowercased and tokenized using the NLTK library [Loper and Bird, 2002].

From the available data, we select only a subset where the difference between sentences in a pair can be described with a simple pattern. We adopt the following approach: given a premise P and a corresponding hypothesis H, we identify the longest common substring consisting of whole words and replace it with a variable X. This step is repeated once more, allowing our sentence patterns to contain up to two variables X and Y. In the last step, we make sure that the pattern is in canonical form by renaming the variables to ensure that the variable X always precedes Y within the premise P. The process is illustrated in Figure 4.2.

The ten most common patterns for each NLI relation are shown in Figure 4.3. Many of the obtained patterns clearly match the sentence pair label. For example, the entailment patterns are mainly generalizations and change of the word order, while the contradiction patterns show typical oppositions like man–woman, dog–cat. Neutral pairs often represent various refinements of the content, likely due to the original SNLI instruction that a hypothesis "might be true" given the premise in a neutral relation.

We keep only patterns within at least 20 different sentence pairs in order to have large and variable sets of sentence pairs in subsequent experiments. We also excluded the most common pattern—identity—as it does not alter the sentence. Surprisingly, identity is not just among entailment pairs (693 cases), but also in neutral (41 cases) and contradictory (22) pairs, showing label noise or human error in the original datasets.

Entailment

Pattern	Count
X → X	693
X man Y → X person Y	224
X . → X	207
X woman Y → X person Y	118
X boy Y → X person Y	65
X Y → Y , X .	61
X men Y → X people Y	56
two X → X	56
X girl Y → X person Y	55
X , Y → Y X .	53

Contradiction

Pattern	Count
X man Y → X woman Y	413
X woman Y → X man Y	196
X men → X women Y	111
X boy Y → X girl Y	109
X dog Y → X cat Y	98
X girl Y → X boy Y	97
X women Y → X men Y	64
X Y, → X not Y	56
two X, → three X	46
X child Y → X man Y	44

Neutral

Pattern	Count
X Y → X sad Y	701
X Y → X big Y	119
X Y → X fat Y	69
X young Y → X sad Y	68
X people Y → X men Y	60
X → sad X	51
X → X	41
X person Y → X man Y	34
X Y → X red Y	30
X Y → X busy Y	28

Figure 4.3: Top 10 patterns extracted from sentence pairs labeled as entailment, contradictions, and neutrals, respectively. Note the X → X pattern indicating no change in the sentence string at all.

In total, we collected 4,200 unique sentence pairs across 60 patterns. Approximately 10% of this data originates from MultiNLI, with the majority derived from SNLI.

4.3.2 Sentence Embeddings

In this chapter, we evaluate the linear properties in the diverse set of sentence embedding models introduced in detail in Section 4.2. These include legacy architectures (e.g., `InferSent` ⬟ , `ELMo` ■ , `USE` ⬟), modern Transformer-based models (e.g., `BERT-Large` ⬟ , `SBERT` ▲ , `ModernBERT` ■ , `SimCSE` ■), multilingual variants (e.g., `LaBSE` ■ , `BGE-M3` ⬟ , `LASER` ■), and recent instruction-tuned encoders (`E5-Large-v2` ⬟ , `E5-Small-v2` ▲), LLM-based decoder (`Qwen3-4B` ⬟), proprietary models such as `TE3-small` ● , `TE3-large` ● , `Gemini-001` ● . As control models, we use `Random` and `Random BERT` ■ . For complete specifications, dimensions, training objectives, see Section 4.2 and Table 4.1.

Although Table 4.1 lists 26 embedding models, only a subset of 18 models is used in the experiments presented in this chapter. This selection is motivated by the experimental setup: the analysis of linear properties of embedding spaces is conducted exclusively on English data, and Czech-specific models are therefore excluded. In addition, we do not include `XLM-R` ⬟ in the experiments, as its raw sentence representations exhibit strong anisotropy and perform poorly in semantic similarity settings as we show further in Section 4.5.

4.3.3 Choosing Vector Operations

Mikolov et al. [2013a] used a simple vector difference as the operation that relates two word embeddings. For sentences embeddings, we experiment a little and consider four simple operations: addition, subtraction, multiplication and division, all applied element-wise to a pair of embedding vectors. More operations could be also considered as long as they are reversible, so that we can isolate the vector change for a particular sentence alternation and apply it to the embedding of any other sentence. Hopefully, we would then land in the area where the correspondingly altered sentence is embedded.

The underlying idea of our analysis was already sketched in Figure 4.1. From every sentence pair in our dataset, we extract the modification pattern, i.e., the specific change made from one sentence to the other (e.g., changing a noun, adding an adjective, or negation). The arithmetic operation needed to move from the embedding of the first sentence to the embedding of the second sentence (in the continuous space of sentences) can be represented as a point in what we call the *space of operations*.

4.3.4 Evaluation Metrics

Considering all sentence pairs that share the same modification pattern, we obtain many points in the space of operations. If the sentence embedding space reflects the particular modification pattern in an accessible way, all the corresponding points in the space of operations will be close together, forming a cluster. To evaluate the clustering of sentence transformation vectors, we test patterns with

a metric suite designed to capture the information-theoretic, pairwise, and geometric properties of the resulting clusters:

- **Adjusted Mutual Information (AMI)** [Vinh et al., 2010] measures the information shared in cluster assignments; specifically, it calculates how much uncertainty about ground-truth labels is reduced by knowing the cluster assignments. It typically ranges between 0 and 1, where 1 indicates that the clusters perfectly match the ground truth, while 0 represents a result not better than what would be expected by random label assignment. The correction for chance ensures that random label assignments do not artificially inflate the success score, a common problem in experiments involving a large number of clusters Vinh et al. [2010], such as the 60 distinct sentence patterns used in this experiment.

- **Adjusted Rand Index (ARI)** [Hubert and Arabie, 1985] measures pairwise accuracy – it evaluates every possible pair of data points and determines whether their co-occurrence in a cluster matches their co-occurrence in the ground truth. The score ranges from -1 to +1, with 1 being the perfect match score and values around 0 meaning random label assignment. Negative numbers indicate worse agreement than a random result due to systematic disagreement. By reporting both AMI and ARI, we can distinguish between models that correctly identify general "neighborhoods" of transformations and those that achieve high-fidelity membership accuracy across all 60 patterns.

- **Silhouette Coefficient** [Rousseeuw, 1987] is employed to analyze the internal geometric separation of the vectors within the embedding space. Using a cosine distance metric, the Silhouette score measures the ratio of intra-cluster cohesion to inter-cluster separation, providing a label-independent validation of the cluster structure. High Silhouette score confirms that the linguistic patterns are mapped into isolated "islands" rather than a continuous, overlapping distribution. Given that sentence embeddings typically reside on a high-dimensional hypersphere, the use of a cosine-based Silhouette score is more semantically appropriate than Euclidean distance for assessing the density and separation of the transformation vectors.

- **Intra-Cluster Cosine Similarity** is used to validate the consistency of the vector arithmetic. This metric calculates the mean directional alignment of all transformation vectors (e.g. `X child Y` $\rightarrow$ `X man Y`) within a specific pattern. Mikolov et al. [2013c] shows that a high consistency score demonstrates that a linguistic transformation maintains a parallel trajectory across diverse lexical contexts All transformation vectors are L2-normalized prior to cosine similarity computation

4.3.5 Comparative Analysis of Vector Operations

We use these metrics to evaluate how well an unsupervised clustering algorithm can recover known sentence modification patterns in different sentence embedding spaces. Sentence embeddings are L2-normalized prior to computing transformation vectors, ensuring that subtraction captures directional semantic change

operation	AMI	ARI	Silhouette	Cosine mean ± std
×	0.15	0.03	-0.08	0.65 ± 0.03
+	0.19	0.04	-0.06	0.51 ± 0.06
−	0.82	0.48	0.29	0.63 ± 0.25
/	-0.00	-0.00	-0.03	0.03 ± 0.04

Table 4.2: Averaged performance of cluster evaluation metrics over all evaluated models.

rather than magnitude differences. Transformation vectors were then clustered using k-Means with the number of clusters fixed to the number of ground-truth patterns (i.e., 60). We compare the resulting cluster assignments to the gold labels using AMI, ARI, and the Silhouette score. To reduce variance due to random cluster initialization, we report the best clustering obtained over 100 random restarts. High agreement between predicted clusters and ground-truth patterns indicates that the corresponding linguistic transformations are encoded in a consistent and separable manner within the sentence embedding space.

To identify the optimal mathematical representation of linguistic transformations, we evaluated four primary vector operations: addition (+), subtraction (-), multiplication (×) and division (/). Table 4.2 presents the performance of these operations, averaged across all sentence embedding models (excluding random BERT and random).

The experimental results show that the best cluster distribution is achieved by far using the most intuitive operation, vector subtraction. Subtraction achieved an AMI of 0.82 and an ARI of 0.48, significantly higher than the scores produced by other operations. Most significantly, subtraction is the only operation that resulted in a positive Silhouette score (0.29), indicating that it produces actual cluster separation rather than a disordered distribution of points.

From a theoretical perspective, the success of subtraction supports the *transformation vector* hypothesis [Mikolov et al., 2013c, Vylomova et al., 2015]. If sentence embeddings are structured linearly, the difference vector $V_{diff} = V_{target} - V_{source}$ approximates the removal of shared semantic content and isolates the specific linguistic shift. The high AMI score confirms that this resulting vector retains highly specific information about the transformation, while the positive Silhouette score proves that these shifts occupy distinct regions in the vector space.

In contrast, addition and multiplication fail to produce a meaningful clustering structure. Although addition (+) yields a relatively high intra-cluster cosine similarity (0.51), its negative Silhouette score (0.06) and low ARI (0.04) reveal that it does not create distinct boundaries. The addition combines the semantics of both sentences into a "centroid" representing the sum of all parts; it doubles the context rather than isolating the change. Consequently, the resulting vectors for different patterns overlap in a single semantic mass, making them indistinguishable from clustering algorithms.

Multiplication (×) and division (/) are even less effective. Multiplication is sensitive to the magnitude of specific dimensions and does not align with the directional, cosine-based logic on which modern transformer embeddings are optimized. Division fails to preserve directional information and produces results

indistinguishable from random clustering, as evidenced by an AMI of -0.00, which is functionally equivalent to random chance.

The results suggest that it would make a little sense to explore addition, multiplication, or division in more detail. These operations fail to uncover the latent structural regularities of the pattern in embedding spaces, their transformation vectors being essentially random noise. However, results for subtraction suggest that the transformation between related sentences is encoded as a consistent directional shift. Therefore, in the rest of the chapter, we focus entirely on subtractions. In the next section, we compare subtraction results in more detail from different architectures—from legacy LSTMs to modern Transformers.

4.3.6 Result

Across all trained sentence embedding models, vector subtraction produces compact and well-separated clusters that align closely with the corresponding sentence transformation patterns. As shown in Table 4.3, semantically trained models achieve high AMI and ARI, indicating strong agreement between clustered transformation vectors and ground-truth labels.

This behavior is robust in various model families, including BiLSTM-based encoders (`InferSent` ⬟), encoder-only Transformers (`SBERT` ▲ , `BGE-M3` ⬟), multilingual models (`LaBSE` ■) and proprietary embedding services (`TE3-small` ●). The absence of a clear performance gap between model families indicates that the ability to encode sentence-level transformations as linear directions is a robust and widely shared property of modern sentence embedding spaces.

Interestingly, our experiment shows that the model scale is not a reliable predictor of the linear semantic structure. Although the scaling laws of Transformer-based model accurately predict improvements in linguistic modeling and downstream task performance s [Kaplan et al., 2020], this does not transfer into a more organized geometric space for vector arithmetic. As shown in Table 4.3, `SBERT` ▲ with 23M parameters performs comparably to substantially larger decoder-only architectures (e.g.`Gemini-001` ●). This finding suggests that semantic alignment is driven less by model size than by the training objective. In particular, contrastive supervision appears to impose a form of geometric pressure that organizes linguistic transformations into consistent directional shifts—an emergent property that does not arise naturally from large-scale next-token prediction alone.

Interestingly, among the best performing models are not just modern methods, but also `InferSent` ⬟ , a legacy architecture, which yielded an AMI of 0.87 and a Silhouette score of 0.35. However, we interpret this not as a sign of superior architecture, but as task-specific overfitting. `InferSent` ⬟ was trained specifically on the SNLI dataset from which our patterns are derived. Furthermore, Dasgupta et al. [2018] investigated the compositionality in sentence embedding spaces and found that `InferSent` ⬟ relies strongly on shallow word-level heuristics rather than exhibiting genuine compositional understanding. Supporting this conclusion, McCoy et al. [2019], Gururangan et al. [2018] found that `InferSent` ⬟ (and early transformers to some extent) exploit superficial statistical regularities in SNLI rather than learning robust entailment. Consequently, if transformations at the word-vector level are minimal, then the difference between two sentence

name	AMI	ARI	Silhouette	Cos. mean ± std	RandCos
InferSent ⬟	**0.87**	0.54	0.35	0.65 ± 0.22	0.54
TE3-large ●	0.86	0.56	0.26	0.54 ± 0.22	0.14
TE3-small ●	0.86	0.56	0.31	0.61 ± 0.24	0.15
SBERT ▲	0.84	0.49	0.36	0.72 ± 0.28	0.05
Gemini-001 ●	0.84	0.52	0.27	0.60 ± 0.26	0.59
LaBSE ■	0.84	**0.59**	0.32	0.76 ± 0.22	0.48
E5-Large-v2 ⬟	0.83	0.52	0.28	0.65 ± 0.27	0.72
Qwen3-4B ⬟	0.83	0.44	0.31	0.64 ± 0.28	0.26
SimCSE ■	0.83	0.50	0.27	0.70 ± 0.24	0.36
E5-Small-v2 ▲	0.83	0.45	0.38	0.76 ± 0.26	0.78
BGE-M3 ⬟	0.82	0.41	0.36	0.72 ± 0.27	0.62
ELMo ■	0.82	0.48	0.33	0.61 ± 0.21	0.52
ModernBERT ■	0.82	0.52	0.33	0.64 ± 0.27	0.49
LASER ■	0.76	0.45	0.23	0.45 ± 0.24	0.60
BERT-Large ⬟	0.72	0.38	0.12	0.41 ± 0.17	0.50
Random BERT ■	0.71	0.45	**0.46**	0.62 ± 0.34	0.95
USE ⬟	0.70	0.24	0.13	0.66 ± 0.28	0.17
Random	0.00	0.00	-0.03	0.00 ± 0.00	-0.00

Table 4.3: Result of different models based on the subtraction. In the RandCos column, we state Cosine similarity between two random sentence vectors from our dataset averaged over one thousand runs.

embeddings is strongly influenced by the embedding of the underlying transformation pattern. This observation provides a plausible explanation for why such patterns are consistently clustered together in the embedding space. In summary, our results are not evidence that InferSent ⬟ is a superior general-purpose sentence encoder, but rather strong evidence that our evaluation protocol detects task-specific overfitting.

In our control experiments, Random does not exhibit any meaningful structure, while Random BERT ■ unexpectedly generates moderately well-separated clusters and notably the highest Silhouette coefficient. However, this result does not represent genuine semantic understanding. The high cosine similarity of the random sentence embeddings (0.95) suggests very high anisotropy. However, the model still encodes small, consistent *micro-shifts* in response to systematic linguistic modifications in the input sentences. These micro-shifts arise from a combination of factors: the tokenizer, trained on large corpora, introduces subtle language-derived structure by consistently mapping similar linguistic patterns to similar token sequences (or to systematic differences in those sequences); meanwhile, the Transformer architecture—even with random weights—reacts to these structured input variations through its attention and feed-forward mechanisms, amplifying small differences into detectable linear patterns. Therefore, we can consider Random BERT ■ to be an upper bound on the contribution of architecture and tokenizer alone.

Compared to this architectural baseline, semantically trained models exhibit both stronger alignment with ground-truth transformation labels and more sta-

ble directional consistency. These results demonstrate that the linear structure observed in trained sentence embeddings is not an artifact of high dimensionality or model architecture, but a learned property induced by semantic training objectives.

4.3.7 Visualisation

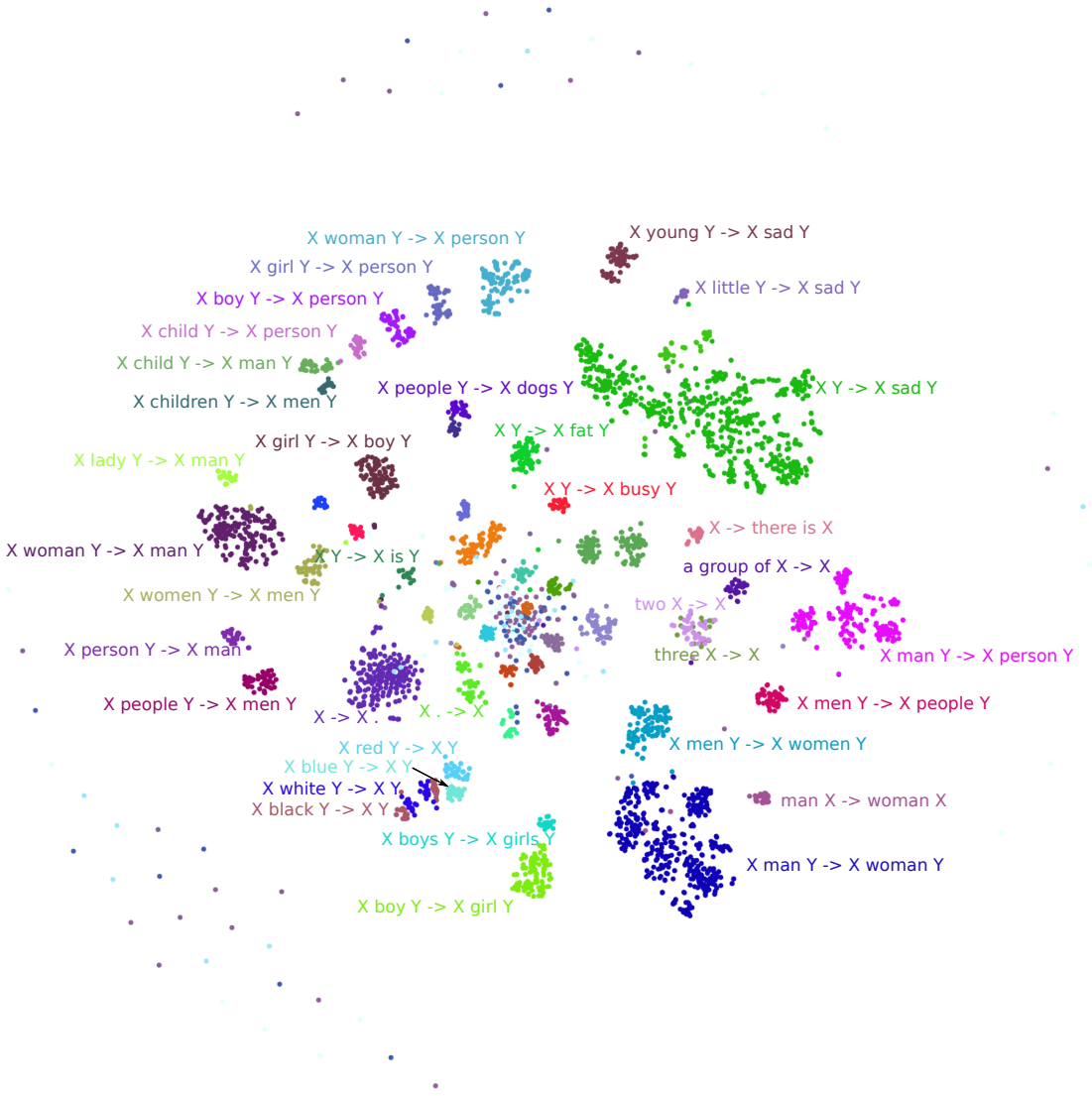


Figure 4.4: t-SNE representation of pattern reprinted from Barančíková and Bojar [2019]. The points in the operation space are obtained by subtracting the ELMo embedding of the hypothesis from the ELMo embedding of the premise. Best viewed in color. Colors correspond to the sentence patterns. All 60 patterns is included.

We visualize the subtraction vectors via t-SNE [van der Maaten and Hinton, 2008] in Figure 4.4 using `ELMo` ■ as a case study, as this embedding model was not trained using the SNLI dataset. The visualization confirms that many patterns are grouped into compact clusters without any supervised knowledge of the labels with the exception of a 'chaos cloud' in the middle and several outliers. There

are also several patterns that seem inseparable, e.g. `two` $X \rightarrow X$ and `three` $X \rightarrow X$, or X `white` $Y \rightarrow X\ Y$ and X `black` $Y \rightarrow X\ Y$,

We identified the patterns responsible for the noisy center and outliers by computing the *weighted inertia* for each pattern. For a given pattern, inertia is computed as the mean squared cosine distance between each transformation vector and the centroid of that pattern in the space of operations. Formally, for a pattern with vectors $\{v_i\}_{i=1}^{n}$ and centroid μ, inertia is defined as:

$$\frac{1}{n}\sum_{i=1}^{n}(1-\cos(v_i,\mu))^2.$$

Lower inertia indicates compact, coherent clusters, while higher inertia corresponds to scattered or inconsistent transformations.

In clustering, *inertia* typically denotes the sum of squared distances from each data point to its nearest cluster centroid. Dividing this sum by the number of samples in a cluster yields the mean squared distance (or variance). In other words, it indicates how tightly points are grouped around the centroid.

Our analysis reveals that the patterns with the highest inertia involve alterations in word order and the insertion or removal of punctuation. Specifically, these patterns include:

`X is Y .` $\rightarrow$ `Y is X`
`X Y .` $\rightarrow$ `Y X .`
`X` $\rightarrow$ `X .`
`X , Y .` $\rightarrow$ `Y X .`
`X , Y .` $\rightarrow$ `Y , X .`
`X Y .` $\rightarrow$ `Y , X .`
`X .` $\rightarrow$ `X`

The failure of subtraction to capture word-order changes (`X is Y .` $\rightarrow$ `Y is X`) suggests that the ELMo model still behaves more like weighted bags of words than true structural encoders in this specific geometric context.

To investigate whether the space of operations can be interpreted also automatically, i.e., to determine if sentence relations are generalizable, we remove the previously identified noisy patterns selected above and apply fully unsupervised clustering: we do not even disclose the expected number of patterns, i.e. clusters.

To identify the optimal number of clusters, we employ two metrics to find the optimal number of clusters: Davies-Bouldin's: Davies-Bouldin's index [Davies and Bouldin, 1979] and Silhouette Coefficient [Rousseeuw, 1987]. They are both designed to measure compactness and separation of the clusters, i.e. they award dense clusters that are far from each other. Both Davies-Bouldin index and Silhouette Coefficient agree that the best separation is achieved at 9 clusters. Consequently, we perform k-Means clustering with $k = 9$ and obtain the results illustrated in Figure 4.5.

By manually inspecting the automatically identified clusters, we observe that many exhibit meaningful patterns. For instance, Cluster 1 captures 90% (altogether 264 out of 292) sentence pairs that generalize women, boys, or girls to person. The corresponding generalization for men occurs in Cluster 5 (218 out of 227 pairs) for the singular case, and in Cluster 7 for the plural case. Specifically,

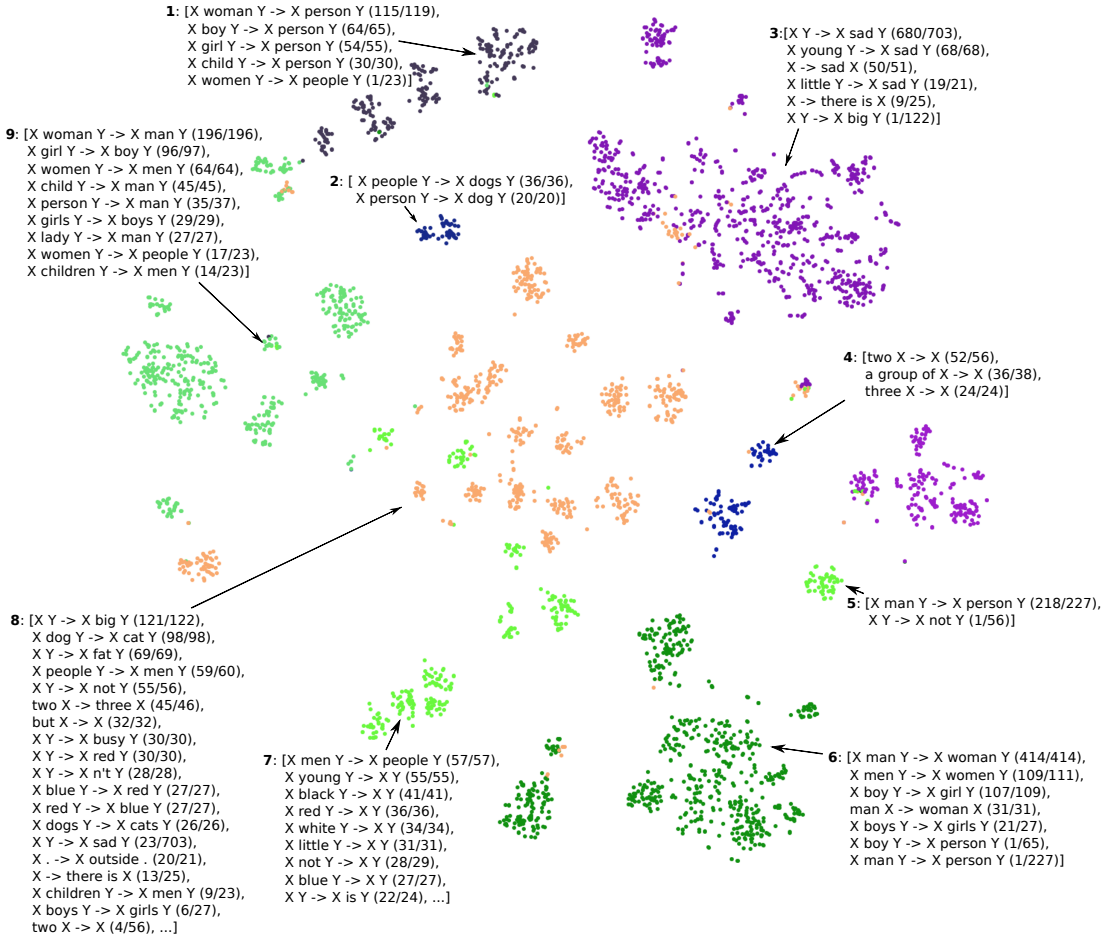


Figure 4.5: t-SNE representation of patterns as in Figure 4.4 with colors coding now fully automatic clusters. Each cluster is labelled with the set of patterns extracted from sentence pairs assigned to the cluster. The numbers in parentheses indicate how many sentence pairs belong to the given pattern within this cluster and overall, resp. For instance the line `two X` → `X`(52/56) says that of the 56 sentence pairs differing in the prefix "two", 52 were automatically clustered together based on the subtraction of their ELMo embeddings. This illustration is reprinted from Barančíková and Bojar [2019].

Cluster 7 contains all 57 plural pairs of the form `X men Y` → `X people Y`, along with various oppositions.

Cluster 2 contains all sentence pairs in which a person is replaced with a dog, whereas Cluster 3 is primarily connected with sentence pairs that introduce a negative mood. Cluster 4 unites patterns that represent omitting a numeral or a group of entities. Cluster 6 covers gender oppositions in one direction, and Cluster 9 adds the other direction with some noise related to transformations such as child/man or person/man. Cluster 8 does not exhibit a clearly defined semantic theme and can be interpreted as a residual cluster.

4.3.8 Pattern Analysis

Based on the inertia results of the ELMo ■ model, we continue to evaluate the stability of linguistic patterns across different sentence embedding spaces. By measuring how tightly a model clusters vectors for a specific linguistic transformation (e.g., gender swaps or negation), we can examine the internal consistency of the model and its sensitivity to context.

We exclude the Random baseline from this analysis because its inertia values depend solely on random initialization and therefore do not provide an informative baseline for comparing semantically motivated transformation patterns.

For clarity, we visualize the results for a representative subset of patterns that spans the full range of observed inertia values. In particular Figure 4.6 presents the four most compact and the four most dispersed patterns among all evaluated models, and the four patterns that show the highest cross-model disagreement.

- **BEST:** This category comprises patterns with the lowest average inertia (dark blue), indicating a high degree of semantic coherence across models. These transformations induce stable and compact clusters in the embedding space, suggesting that the underlying semantic relations are robustly encoded irrespective of architectural differences.

- **DISPUTED:** Patterns in this group exhibit substantial variance in inertia across models, revealing pronounced architectural sensitivities. Notably, sentence-order transformations (e.g., `X is Y .` → `Y is X`) dominate this category. Transformer-based models, such as E5-Small-v2 ▲ and SimCSE-dist-cs △ , consistently maintain low inertia, whereas sequential or fixed-encoder architecture (including ELMo ■ , LASER ■ , and) markedly higher dispersion. This result is consistent with previous work that demonstrates that self-attention–based architectures are less sensitive to linear word order than recurrent models, allowing more stable representations under syntactic reordering Vaswani et al. [2017], while LSTM-based encoders are known to be highly sensitive to word order perturbations [Conneau et al., 2018, Dasgupta et al., 2018].

- **WORST:** This category contains patterns with consistently high inertia values (red), which indicates loose or noisy representations. These patterns mainly involve transformations involving negation (e.g., `X Y` → `X not Y`), which systematically disrupt semantic clustering. This finding corresponds to previous evidence that Transformer-based sentence embeddings often place negated sentences close to their affirmative counterparts and perform poorly in negation-sensitive instances Ettinger [2020], Hossain et al. [2022]. In contrast to the **DISPUTED** category, modern Transformer-based models show the highest inertia, while legacy models such as ELMo ■ , LASER ■ , and InferSent ⬟ demonstrate relatively greater consistency.

4.3.9 Conclusion

In this chapter, we investigate the linear geometric properties of sentence embedding spaces across a wide range of architectures, ranging from legacy BiLSTMs

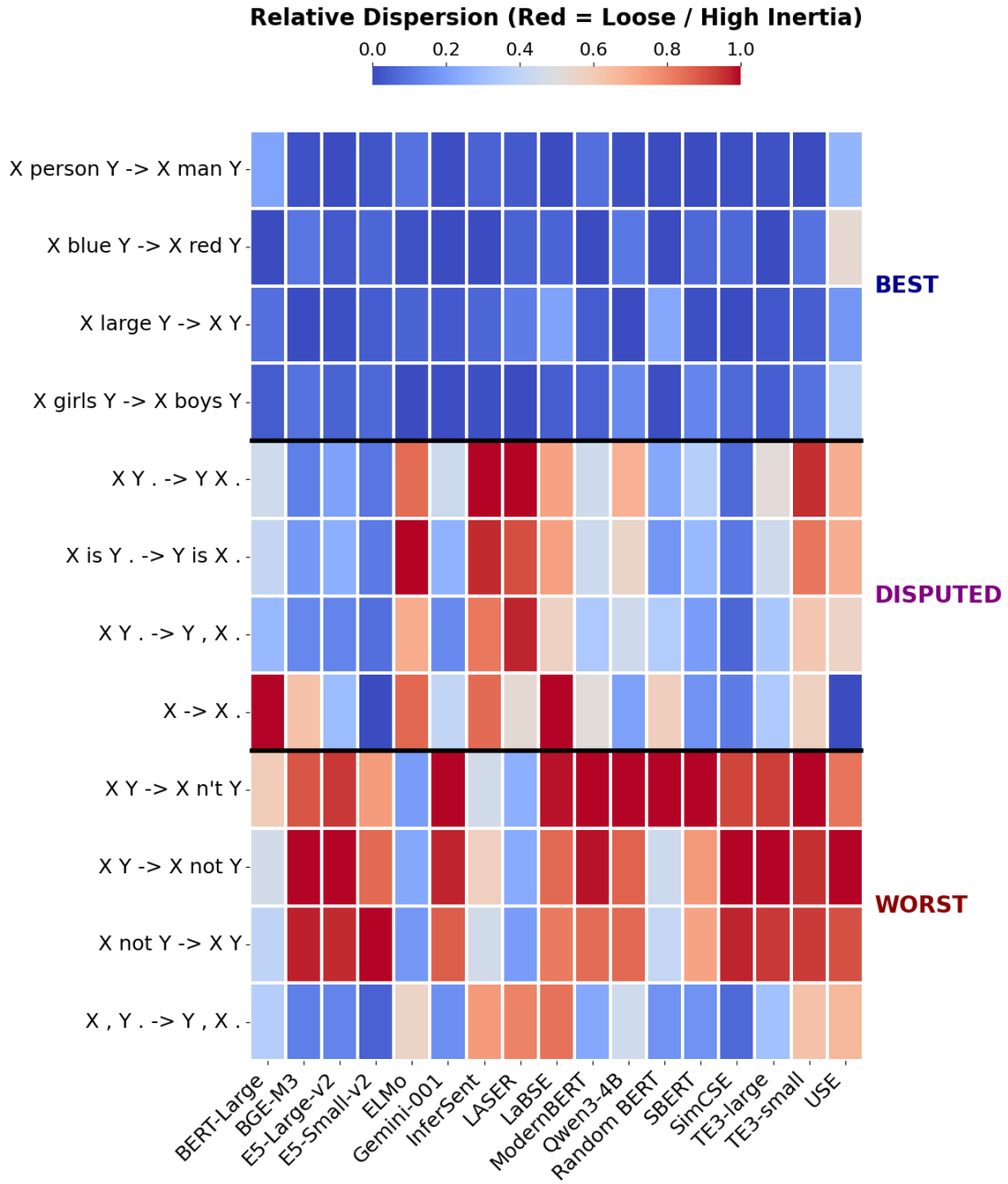


Figure 4.6: Example of inertia values for selected patterns across different embedding models. Red color represent patterns with high inertia, which correspond to incoherent, loosely scattered transformation vector in the operation space. Blue indicates compact clusters.

to modern decoder-only LLMs. We analyze 4,200 pairs of sentences across 60 distinct transformation patterns. Our primary objective is to find whether simple arithmetic operations within the vector space correspond to meaningful semantic edits at the string level.

Our first results of sixty sentence-edit patterns support this hypothesis – simple linguistic transformations correspond to a consistent directional shift in the embedding spaces. Among the operations tested, vector subtraction significantly

outperforms addition, multiplication, and division, producing well-separated clusters that closely align with known sentence modification patterns.

Importantly, this observed linearity is not an artifact of high dimensionality or the Transformer architecture. Although `Random BERT` ■ shows geometric regularities, it fails to produce stable semantically aligned transformation vectors. This indicates that the observed linear structure is learned from semantic supervision. Furthermore, the scale and parameter count of the model do not reliably predict the quality of the linear semantic structure. Smaller contrastively trained models often match or outperform substantially larger proprietary and LLM-based embeddings. This suggests that training objective and representation alignment play a more decisive role than the size of the model.

Unlike the rest of this thesis, the present experiment was conducted exclusively on English data due to the scarcity of similar resources for the Czech language. To address this gap, the following chapter introduces Costra: a novel dataset specifically designed for the thorough testing of Czech sentence embeddings.

We answered the following research questions:

Can semantic and syntactic transformations between sentences be captured through simple arithmetic operations in continuous vector spaces? Yes, but only through specific operations. Our results demonstrate that vector subtraction is highly effective in isolating linguistic transformations. By subtracting a source sentence embedding from a target (modified) sentence embedding, a transformation vector that represents the specific change (e.g., gender swap, negation, or generalization) is created.

To what extent do model architecture and scale influence the linearity of the sentence embedding space? Surprisingly, model scale (parameter count) is not a reliable predictor of linear semantic structure.

Is the linearity of a model a result of learned semantic understanding or an inherent inductive bias of its architecture? It is a combination of both, but genuine semantic consistency requires training.

Which linguistic phenomena are most (and least) consistently represented as linear transformations in sentence embeddings? Lexical substitutions and generalizations are consistent, while structural and punctuational changes are noisy.

4.4 Costra

As we mentioned in Section 4.1, resources for evaluating Czech sentence embeddings are scarce. We consider the lack of an intrinsic evaluation dataset an important issue. To address this, this chapter[27] introduce Costra, a dataset of COmplex Sentence TRAnsformations, together with a lightweight evaluation toolkit. The dataset is intended for the study of sentence-level embeddings beyond simple word alternations or standard paraphrasing. Although the current release contains only Czech sentences, the construction protocol is language-agnostic and readily extensible to other languages.

The goal of Costra is *not* to rank sentence embedding models according to overall usefulness or the performance of the downstream task. Instead, Costra is designed as a *diagnostic intrinsic benchmark* that investigates whether sentence embedding spaces exhibit an interpretable and consistent *semantic structure*. In particular, we focus on whether specific semantic, stylistic, or temporal transformations correspond to coherent geometric relations in the embedding space. This perspective treats sentence embeddings as representational objects whose internal organization can be examined independently of task-specific fine-tuning.

In an attempt to bring together more traditional representations of sentence meanings and vector representations, Bojar et al. [2019] introduce several aspects or desirable properties of sentence embeddings. One of them, "relatability", highlights the correspondence of meaningful differences between sentences on the one hand and geometrical relations between their respective embeddings in the highly dimensional continuous vector space on the other. If such an alignment holds, geometric operations in the embedding space could correspond to particular changes in the meaning of a sentence, similarly to that shown in Figure 4.1.

The purpose of Costra is to support the exploration of semantic and syntactic relations between sentences in continuous space. Our dataset is the prerequisite for one of the possible ways of exploring the relatability of sentence meaning:[28] We envision that the continuous space of sentence representations induced by an ideal embedding method would exhibit topological similarity to the graph of sentence variations. For example, one could argue that a subset of sentences could be organized along a linear scale reflecting the formality of the language used. Another set of sentences could form a partially ordered set of gradually less and less concrete statements. Yet another set, intersecting both of the previous ones in multiple sentences, could be partially or linearly ordered according to the strength of the speaker's confidence in the claim. Such structures would provide interpretable axes within the embedding space.

Our long-term objective is to search for a sentence embedding method that exhibits this behavior: where the topology of the embedding space corresponds to meaningful operations or changes in the set of sentences of a language (or more languages at once). We prefer this behavior to *emerge*, as happened for word vector operations, but regardless of whether the behavior is emergent or

[27] This chapter is based on the papers Barančíková and Bojar [2020] and Barančíková and Bojar [2020].

[28] The term "relatability" is used to indicate that we search for specific types of relations among sentences. The common term "relatedness", in our opinion, suggests some vagueness on the relation type. We do not build a dataset of sentences related in just some way, we seek for a set of clear-cut, "orthogonal" relations.

trained, we need a dataset of sentences illustrating these patterns. A large dataset could serve for training; a small one would provide a test set. In either case, these sentences could provide a "skeleton" to the continuous space of sentence embeddings.[29]

In this chapter, we create a dataset for semantic evaluation of Czech sentence embeddings, focusing on the following research questions:

- **To what extent can modern sentence embeddings distinguish between paraphrases and sentences with high lexical overlap but different meanings?**
- **How effectively do embedding spaces capture the ordinal progression of stylistic, temporal, and semantic transformations?**
- **Do specialized Czech-language models provide a significant advantage over multilingual or general-purpose commercial models in complex semantic tasks?**
- **What is the relationship between model complexity and performance on the Costra benchmark?**

4.4.1 Related Work

We place Costra within intrinsic evaluations of sentence embeddings that test whether embedding spaces reflect expected semantic relations among controlled sentence variants. A common strategy is to compare the relative distances among systematically modified versions of a base sentence. For example, Zhu et al. [2018] automatically generates variants to probe semantic sensitivity:

(1) Original sentence: A rooster pecked grain.

(2) Synonym substitution: A cock pecked grain.

(3) Not-Negation: A rooster didn't peck grain.

(4) Quantifier-Negation: There was no rooster pecking grain.

They then evaluate whether the embedding space respects expected relationships (e.g., synonyms are closer than edits that alter sentence meaning). Benchmarks such as SentEval [Conneau and Kiela, 2018] and SICK [Marelli et al., 2014] use similar evaluation principles.

While useful for basic cases, these methods focus on small or categorical edits, and they may fail to show how embeddings capture gradual semantic or stylistic change. Our aim is to move beyond relatively simple transformations [Zhu et al., 2018, Ribeiro et al., 2020, Dhole et al., 2023, Chiang et al., 2023] toward more challenging and non-trivial comparisons: changes in style or register, minimal edits that drastically alter meaning, and word-order perturbations that affect interpretation.

The PAWS dataset [Zhang et al., 2019] and its multilingual extension PAWS-X [Yang et al., 2019] address the tendency of models to over-rely on surface

[29]The Czech word form"skeleton" is "kostra".

Change	Example of change	%
seed sentence	Several hunters slept on a clearing.	-
change of aspect	Hunters have fallen asleep on a clearing.	4
different meaning	On a clearing, several hunters were dancing.	15
less formal	Several deer stalkers kipped down on a clearing.	6
possibility	Several hunters can sleep on a clearing.	4
ban	Hunters are forbidden to sleep on a clearing.	4
exaggeration	Crowds of hunters slept on a clearing.	7
concretization	Several hunters dozed off after lunch on the Upper clearing.	15
generalization	There were several men in a forest.	9
change of locality	Several hunters slept in a cinema.	3
change of gender	Several huntresses slept on a clearing.	2
total		65

Table 4.4: Examples of transformations given to annotators for the given seed sentence. The third column shows how many of all the transformation suggestions collected in the first round closely mimic the particular example. The number is approximate as annotators typically call one transformation by several names, e.g. *less formally*, *formality diminished*, *decrease of formality*, *not formal expressions*, *non-formal*, *less formal*, *formality decreased*, ... The examples were translated to English for presentation purposes only.

overlap. They provide paraphrase/non-paraphrase pairs with very high lexical overlap, enabling tests of whether embeddings capture semantics beyond surface word matching.

To our knowledge, no comparable resource exists for Czech that jointly targets stylistic, temporal, and semantic axes with ordinal structure. To help fill this gap, we introduce COSTRA, a dataset of complex sentence transformations covering fifteen modification types, designed to evaluate the semantic properties of sentence embeddings beyond simple word substitutions.

4.4.2 Annotation

We acquired the data in four rounds of annotation. In the first round, we acquired suggestions for original and uncommon sentence changes. In the second round, we collected sentence alternations inspired by ideas from the first round. These two rounds can be broadly described as *collecting ideas* and *collecting data*, respectively. In the third round, we focused on increasing the density of the sentence space by adding more related sentences. In order to project the sentence transformation to a linear scale, we decided to collect *interpolations* and *extrapolations*. At this stage, the dataset contains various transformations, but their mutual relationships remain unknown. To address this, in the fourth and final round, we collected pairwise comparisons of sentences from the second and third rounds.

Change and instructions
paraphrase 1: Reformulate the sentence using different words.
paraphrase 2: Reformulate the sentence using other different words.
different meaning: Shuffle words in the sentence in order to get different meaning.
opposite meaning: Reformulate the sentence to get a sentence with an opposite meaning.
nonsense: Shuffle words in the sentence to make grammatical sentence with no sense. (e.g. A hen pecked a grain. → A grain pecked a hen.)
minimal change: Significantly change the meaning of the sentence using only a minimal alternation.
generalization: Make the sentence more general.
gossip: Rewrite the sentence in a gossip style – strongly exaggerated meaning on the sentence.
formal: Rewrite the sentence in a more formal style.
nonstandard: Rewrite the sentence in nonstandard, colloquial style.
simple sentence: Rewrite the sentence in a simplistic style, so even a person with a limited vocabulary could understand it.
possibility: Change the modality of the sentence into a possibility.
ban: Change the modality of the sentence into a ban.
future: Move the sentence into the future.
past: Move the sentence into the past.

Table 4.5: Sentences transformations requested in the second round of annotation with the instructions to the annotators. The annotators were given no examples (with the exception of *nonsense*) not to be influenced as much as in the first round.

4.4.3 Round 1: Collecting Ideas

We manually selected 15 newspaper headlines. Eleven annotators were asked to produce up to 20 modifications to each headline and to label each modification with a short[30] name. Annotators were given an example sentence together with several possible alternations and their labels (see Table 4.4).

Unfortunately, the examples highly influenced annotator decisions: modifications suggested in examples represent nearly two-thirds of all transformation labels collected in the first round. Other frequent transformations included changes in word order and transformations to interrogative or imperative form.

Other suggested interesting alterations involve a change into a fairy-tale style,

[30]This requirement was not always respected. Annotators sometimes created very complex descriptions, such as the *specification of information about the society affected by the presence of an alien.*

heavy use of diminutives or vulgarisms, and a "dadaistic" role swap (swapping participant roles so that the resulting sentence remains grammatical but becomes non-sensical in the real world). Of these suggestions, we selected only the role-swap variant for the current study (see *nonsense* in Table 4.5).

In total, we collected 984 sentences described by 269 unique labels. We use them as inspiration for the second round of annotation.

Transformations and their translations
seed: *Je to prostě blbost, zrušte ten projekt, prosím, a nemrhejte státními penězi.* (It's just crap, please cancel the project, and don't squander state money.)
different meaning: *Zrušení projektu je blbost a nesmyslné mrhání státními penězi.* (Cancellation of the project is stupid and pointless waste of state money.)
formal sentence: *Ten projekt je od základu špatný, zrušte jej prosím a neplýtvejte státními prostředky.* (The project is fundamentally bad, please cancel it and don't waste state resources.)
future: *Bude to určitě blbost, projekt bude nejspíš zrušen a nebude se mrhat státními penězi.* (It will certainly be crap, the project will likely be canceled and state money will not be squandered.)
generalization: *Projekt je blbost, zastavte jej a nemrhejte penězi.* (The project is crap, stop it and don't squander money.)
gossip: *Ten projekt je šílenost, měl by se okamžitě zastavit a neplýtvat na něm miliony z našich daní.* (The project is crazy, it should immediately stop and not to waste millions on it from our taxes.) **minimal change:** *Je to blbost, přerušte ten projekt, prosím, a mrhejte státními penězi.* (It's crap, interrupt the project, please, and squander state money.)
nonsense: *Zrušte státní peníze prosím a nemrhejte prostými blbostmi.* (Please cancel the state money and do not squander plain crap.)
nonstandard: *Je to kravina, práce by se měla zastavit a nemrhat státníma penězma.* (It's bullshit, work should stop and not squander government money.)
opposite: *Jedná se o promyšlený plán, je třeba ho realizovat a uvolnit peníze ze státní kasy.* (It is a well-thought-out plan, it needs to be implemented and the money released from the Treasury.)
paraphrase: *Je to hloupost, zastavte ten projekt a ušetřete státní peníze.* (It's stupid, stop the project and save state money.)
paraphrase: *Je to blbost, zastavte práce a neplýtvejte veřejnými prostředky.* (It's stupid, stop the work and don't waste public funds.)
past: *Byla to prostě blbost, projekt byl zrušen a nemrhalo se státními penězi.* (It was just crap, the project was canceled and state money weren't wasted.)
possibility: *Tento projekt může být blbost, mohl by se zrušit a nemělo by se zde mrhat státními penězi.* (This project can be stupid, could be canceled and state money shouldn't be squandered.)
simple: *Projekt je špatný, zastavte jej, neplýtvejte státními penězi.* (The project is bad, stop it, don't waste state money.)
ban: *Ten projekt nesmí být blbost, jeho realizace se nesmí zrušit a nesmí být na něj uvolněny státní peníze.* (The project mustn't be crap, its realization mustn't be canceled and state money mustn't be released for it.)

Table 4.6: Example of a single annotation.

Annotator	Annot.	Sent.	Imp.	Typos	Length	Time
armadillo	69	1035	0	9	10.3	12:32
wolverine	42	598	32	13	9.6	14:32
honeybadger	39	584	1	28	10.4	30:38
gorilla	31	448	17	16	9.8	16:55
porcupine	31	465	0	6	11.3	8:55
lumpfish	23	329	16	4	8.4	13:28
crane	22	319	11	15	9.2	15:30
meerkat	17	241	14	17	9.1	27:36
axolotl	8	116	4	11	10.1	24:02
bullshark	6	90	0	2	9.8	20:59
flamingo	3	45	0	8	11.3	11:37
capybara	2	30	0	0	7.6	25:06
total/avg	293	4,300	95	129	9.9	19:50

Table 4.7: Statistics for individual annotators (anonymized as armadillo, ..., capybara). *Annot.* indicates the total number of annotated seed sentences; *Sent.* the total number of sentences created; *Imp.* the count of the transformations marked impossible; *Length* the average sentence length; and *Time* the average time spent per annotation.

4.4.4 Round 2: Collecting Data

Sentence Transformations

We selected 15 types of modifications, they are presented in Table 4.5. We asked for two distinct paraphrases of each sentence because we believe that a proper sentence embedding should keep paraphrases close together in the vector space.

Several modification types were explicitly selected to constitute a thorough test of embeddings. In *different meaning*, the annotators should create a sentence with some other meaning using the same words as the original sentence. Another challenging transformation is *minimal change*, in which the meaning of the sentence should be significantly modified by only minimal alternation, or *nonsense*, in which words of the source sentence should be rearranged into a grammatically correct sentence without any sense. An example of the resulting annotations is in Table 4.6.

Seed Data

The source sentences for the annotations were selected from Czech data from Global Voices [Tiedemann, 2012] and OpenSubtitles[31] [Lison and Tiedemann, 2016]. We used two sources in order to have different styles of seed sentences, both journalistic and common spoken language. We considered only sentences with more than 5 and less than 15 words, and manually selected 150 of them for further annotation. This step was necessary to remove sentences that are as follows:

[31] `http://www.opensubtitles.org/`

- too unreal, out of this world, such as:
  "*Jedno fotonový torpédo a je z tebe vesmírná topinka.*"
  (One photon torpedo and you're a space toast.)

- photo captions, or incomplete sentences, e.g.:
  "*Zvláštní ekvádorský případ Correa vs. Crudo*"
  (Specific Ecuadorian case Correa vs. Crudo)

- too vague, overly dependent on the context:
  "*Běž tam a mluv na ni.*"
  (Go there and speak to her.)

Many of the intended sentence transformations would be impossible to apply to such sentences, and annotators' time would be wasted. Even after such filtering, it was still quite possible that the desired sentence modification could not be achieved for a sentence. For such a case, we gave the annotators the option to enter the keyword *IMPOSSIBLE* instead of the particular (impossible) modification.

This option allowed us to state that such a transformation is not explicitly possible. At the same time, most of the transformations are likely to lead to a large number of possible outcomes. As documented in Bojar et al. [2013b], a Czech sentence might have hundreds of thousands of paraphrases. To support some minimal exploration of this possible diversity, most of the sentences were assigned to several annotators.

Spell-Checking

Annotation is a challenging task and it is natural for annotators to occasionally introduce errors. Unfortunately, even a single typo can significantly influence the resulting sentences embeddings [Malykh et al., 2018]. To mitigate the influence of such typographical and grammatical errors on embedding performance, we applied Korektor [Richter et al., 2012], a statistical spellchecker and grammar checker, to all collected sentence variations. Of the 519 issues identified by Korektor, we manually reviewed each one and corrected 129 true errors.

Summary

In the second round of annotations, we collected a total of 293 annotation sets from 12 annotators. Following post-processing with Korektor, the dataset consisted of 4,262 unique sentences, including 150 seed sentences. Detailed statistics for individual annotators are provided in Table 4.7.

On average, the time required to complete one full annotation task, i.e., generating all 15 transformations for a given seed sentence, was approximately 20 minutes, although some annotators required up to 30 minutes. Of the 4,262 distinct sentences, only 188 were recorded more than once. In other words, the chance of two annotators producing the same output string is quite low. The most repeated transformations are by far *past*, *future* and *ban*. The least repeated is *paraphrase* with only one sentence appearing twice.

Consistently with the annotation guidelines, the *IMPOSSIBLE* option was rarely used (only 95 times, representing approximately 2% of all cases). Notably,

Transformations and their examples
seed: *Občas se mi na hlavě málo prokrvuje kůže.* (The skin on my head sometimes fills with little blood)
interpolation: *Kůže na hlavě se mi prokrvuje tak akorát* (The skin on my head fills with just the right amount of blood.)
transformation: *Občas se mi na hlavě hodně prokrvuje kůže* (The skin on my head sometimes fills with too much blood.)
extrapolation: *Nemám žádnou kůži na hlavě* (There is no skin on my head.)

Figure 4.7: Example from the third round of annotations. The annotator filled the interpolation and extrapolation to the seed and its transformation with *opposite meaning*.

only seven of the twelve annotators made use of this option, while the remaining five were able to produce all assigned transformations. The three transformation types most commonly deemed unfeasible were *different meaning* (using the same set of words), *past* (particularly when applied to sentences already in past tense) and *simple sentence*.

4.4.5 Round 3: Collecting Interpolations and Extrapolations

In the third round of annotation, we aimed to map the sentence transformations onto a linear continuum along the targeted dimension. Annotators were presented with a seed sentence together with one of its transformations and instructed to generate two additional sentences: an *interpolation*, representing a midpoint in meaning or style between the original and the transformation, and an *extrapolation*, which extends the transformation further away from the seed sentence in the same semantic or stylistic direction. An example of this annotation process is presented in Figure 4.7.

Of the 14 transformation types used in the second annotation round, not all were included in this phase. The reason was straightforward: certain transformation types are unsuitable for interpolation or extrapolation. For example, *paraphrases* are expected to preserve meaning that is identical or nearly identical to the original sentence; trying to interpolate between a sentence and its paraphrase would be unproductive. Similarly, there is the *non-sense* transformation, which is created by shuffling content words of a seed sentence, so the final sentence is grammatically correct, but has no meaning. There are no interpolations or extrapolations of nonsense.

Following a manual review, we selected six types of transformations that appeared to exhibit a linear and scalable nature: *formal sentence*, *future*, *generalization*, *nonstandard sentence*, *opposite meaning*, and *past*. No new types of transformation were introduced.

In total, we collected approximately 1,500 annotations from seven annotators, resulting in 2,749 unique sentences. We again applied Korektor and manually inspected 195 errors identified by Korektor and fixed 37 true errors. The final size of the Costra dataset is 6,968 sentences.

4.4.6 Round 4: Sentence Comparison

In the final round, we focused on obtaining additional comparisons among the sentences collected in the previous rounds, i.e., to densify the set of pairwise judgments. This enriched dataset allows us to test more rigorously whether the embedding space preserves the orderings implied by human judgments. A secondary objective was quality control – specifically, identifying potentially questionable sentence comparisons by evaluating annotator agreement.

We continued working with the same types of linearly comparable transformations as in the previous round: changes in tense (*future*, *past*), changes in style (*formal sentence*, *nonstandard sentence*), and significant semantic shifts (*generalization*, *opposite meaning*). Additionally, we merged the *gossip* category with *nonstandard* language, as gossip-like content in our dataset was frequently expressed using informal or nonstandard constructions and vice versa. We also reintroduced the *simple sentence* category, which we had previously considered too difficult for interpolation and extrapolation tasks; this time, we included it in comparisons.

Annotators were presented with a pairs of sentences together with criteria for comparison.[32] Not all pairs were comparable, some were nearly identical in meaning, while others completely unrelated, making them hard to compare. For every pair of sentences S_1 and S_2, the annotators had the following four options:

1. S_1 is more *general*/*formal*/in the *past*/*nonstandard*/... than S_2.
2. S_2 is more *general*/*formal*/in the *past*/*nonstandard*/... than S_1
3. S_1 and S_2 are **too similar**, for example: *Byl rozčilený a hodně mluvil.* (He was upset and talked a lot.) and *Ovlivněn silnými emocemi říkal ledacos.* (Influenced by strong emotions, he said all kind of things.) These sentences are so close in their meaning that it is almost impossible to select the more general one.
4. S_1 and S_2 are **too dissimilar**, for example: *Všechno zlé je pro něco dobré.* (Every cloud has a silver lining; lit. All bad is good for something.) and *V Asii jsou různá období.* (There are different seasons in Asia.) Neither of the sentences is a generalization of the other sentence, although they were both created as generalizations of the sentence *Bangladéšská monzunová sezóna přináší radost, problémy i pozoruhodné fotografie* (Bangladesh's monsoon season brings joy, problems and remarkable photos.)

We collected more than 25,000 pairwise comparisons from 7 annotators. We compute inter- and intra-annotator agreement using average pairwise Cohen's kappa Carletta [1996]. The scores are generally good, in line with other linguistic annotation tasks: inter-annotator raw agreement is 0.62 ($\kappa = 0.49$) and intra-annotator raw agreement is 0.77 ($\kappa = 0.7$).

We combine sentence comparisons obtained in the third and fourth rounds of the annotation. A pair of sentences can have multiple annotations in the

[32]Only for *opposite meaning* annotators were presented with three sentences: two candidates and a source sentence. They were asked which candidate is closer in meaning to the source sentence.

collection. We trust the annotation only if there is an option with a majority of votes. We keep 16,385 unique sentence pairs with human comparison, and 1,620 were disregarded because of a disagreement in annotators' judgments.

4.4.7 Dataset Summary

COSTRA consists of 6,968 unique Czech sentences derived from 150 manually selected seed sentences, collected over four annotation rounds. The dataset includes both sentence-level transformations and human judgments about their semantic relationships. Below we summarize the structure and contents of each round.

Round 1: Collecting Ideas. Annotators created 984 transformed variants based on 15 seed sentences. Each variant was labeled with one of 269 free-form transformation names (e.g., *ban*, *possibility*, *change of tense*), yielding a diverse pool of ideas that informed the design of fixed transformation types.

Round 2: Collecting Data. For each of the 150 seed sentences, annotators produced 14 specific transformation types:

1. Two *paraphrases*
2. *Different meaning*
3. *Opposite meaning*
4. *Nonsense*
5. *Minimal change*
6. *Generalization*
7. *Gossip*
8. *Formal*
9. *Nonstandard*
10. *Simple sentence*
11. *Possibility*
12. *Ban*
13. *Future tense*
14. *Past tense*

This resulted in 4,262 unique sentences (including the 150 seed sentences).

Round 3: Interpolations and Extrapolations. Seven annotators generated 2,749 new sentences by applying six *scalable* transformation types (those assumed to allow for gradation in meaning change):

1. *Formal*
2. *Nonstandard*
3. *Generalization*
4. *Opposite meaning*
5. *Future tense*
6. *Past tense*

Round 4: Pairwise Judgments. In the fourth round, annotators produced over 25,000 pairwise similarity judgments for sentence pairs drawn from the third round. Additionally, two more categories were included:

1. *Simple sentence* – initially excluded due to difficulty, later included
2. *Gossip* – merged with *nonstandard sentence* due to their similarity

These judgments assessed whether sentence A was more or less general/*formal*/in the *past*/*nonstandard* than sentence B, or whether the sentences were too similar or too dissimilar to compare meaningfully. We retain 16,385 comparisons with a majority agreement, and discard 1,620 pairs where no consensus was reached.

Quality Control. To minimize the influence of typographical and grammatical errors on the quality of the dataset, we ran the Korektor spellchecker [Richter et al., 2012] and manually verified flagged items:

- Round 2: 519 flagged issues, 129 confirmed and corrected
- Round 3: 195 flagged issues, 37 confirmed and corrected

Final Dataset Contents. The final Costra dataset includes:

- 6,968 unique Czech sentences
- 150 seed sentences with 13[33] labeled transformation variants each
- Scalable transformations with graded variants for semantic probing
- 16,385 pairwise human similarity judgments for evaluation and training

This structured dataset supports both intrinsic evaluation of sentence representations and the development of models that capture semantic relationships beyond surface-level lexical similarity. Our hope is that Costra will be useful in developing sentence embeddings for the Czech language. It is freely available at the following link:

`http://hdl.handle.net/11234/1-3248`

[33] *gossip* is merged with *nonstandard sentence*.

4.4.8 Sentence Embeddings Evaluation

We propose a multi-faceted evaluation of sentence embeddings using 2 paraphrases and 12 semantic transformation types for each seed sentence, grouped into six interpretable categories for conciseness. Each category probes different dimensions of meaning variation. We organize the evaluation into two parts on the basis of the nature of the underlying annotation and the assumptions made about the transformation structure.

1. Non-Scalable (Binary Comparison) Evaluation. The first part of our evaluation covers 5 types of transformations that do not naturally form linear or ordinal scales, but still allow for meaningful binary comparisons. For these, we measure how often embeddings correctly reflect that a paraphrase should be semantically closer to the seed sentence than any transformation that significantly changes the meaning of the seed. We create the following two categories:

- **Basic:** Paraphrases should be closer to the seed sentence than any transformation that significantly changes the meaning of the seed (*different meaning*, *minimal change*, *nonsense*).
- **Modality:** Paraphrases should be closer to the seed sentence than transformations that change sentence modality (*possibility*, *ban*).

Formally, for each paraphrase P and transformation T in a given category, we compute accuracy as the proportion of cases where

$$\text{sim}_c(\mathbf{v}_{\text{seed}}, \mathbf{v}_P) \; > \; \text{sim}_c(\mathbf{v}_{\text{seed}}, \mathbf{v}_T)$$

where sim_c denotes cosine similarity (unless stated otherwise), and $\mathbf{v}_\bullet$ are the sentence embeddings.

2. Scalable Evaluation. The remaining four categories represent remaining 8 scalable transformation types, for which we have collected human judgments on sentence pair similarity. These allow us to test whether the sentence embedding space preserves the relative semantic ordering implied by annotator judgments. The categories are:

- **Time:** Transformation of the seed sentence into its *future* or *past* variants.
- **Style:** Transformations involving changes in formality, nonstandard usage, or sentence simplification (*formal*, *nonstandard*, *simple*).
- **Generalization:** Abstraction of the seed sentence to a more general or less specific version.
- **Opposite Meaning:** Shifts that invert the core meaning of the seed sentence.

For these categories, we used the human-annotated pairwise comparisons from Costra. If annotators indicate that the sentences A, B, C satisfy $A \prec_h B$ and

$B \prec_h C$ along the target dimension, we count how often the embedding space respects the induced ordering by satisfying

$$\mathrm{sim}_c(\mathbf{v}_A, \mathbf{v}_B) > \mathrm{sim}_c(\mathbf{v}_A, \mathbf{v}_C) \quad \text{and} \quad \mathrm{sim}_c(\mathbf{v}_B, \mathbf{v}_C) > \mathrm{sim}_c(\mathbf{v}_A, \mathbf{v}_C)\,.$$

To make use of the options *too similar* and *too dissimilar*, we also verify that

$$\mathrm{sim}_c(\mathbf{v}_A, \mathbf{v}_B) > \mathrm{sim}_c(\mathbf{v}_B, \mathbf{v}_C)$$

for all triples (A, B, C) where annotations mark A and B as too similar and B and C as too dissimilar.

This two-part evaluation setup allows us to probe whether sentence embeddings not only capture intuitive semantic relationships but also preserve fine-grained semantic orderings across interpretable transformation dimensions. Each category yields a score between 0 (worst) and 1 (best), and the overall COSTRA score is calculated as the arithmetic mean across all categories.
In order to make the evaluation using COSTRA simply reproducible, we created an easy-to-use tool. It is available as a Python package ('costra') on PyPI, or directly from the following repository:

`https://github.com/barancik/costra`

Bednář et al. [2024] integrated COSTRA into a larger evaluation pipeline that also includes other tasks such as semantic textual similarity and relevance assessment. This comprehensive tool is publicly available at the following repository:

`https://github.com/seznam/czech-semantic-embedding-models`

4.4.9 Results

For our experiments in evaluating sentence embeddings using COSTRA, we used several state-of-the-art sentence embedding models, employing both Czech-specific and multilingual variants. The Czech encoders include `Czert` □ , `FERNET` □ , `RobeCzech` □ , `SimCSE-dist-cs` △ and `RetroMAE` △ . To provide a broader comparative analysis, we also experimented with multilingual sentence embedding models trained on datasets that contain Czech texts. These include `LaBSE` ■ , `XLM-R` ⬟ , and `mE5` ⬟ . In addition, we included two commercial text-embedding models: `TE3-small` ● and `TE3-large` ● from OpenAI. These proprietary models are optimized for general-purpose semantic similarity tasks and have been shown to perform competitively across a wide range of languages and domains. As control models, we employ `Random BERT` ■ and `Random`. These models are described in more detail in Section 4.2.

Table 4.8 reports the intrinsic evaluation results using the COSTRA dataset, with the overall COSTRA score calculated as the arithmetic mean for all six categories.

The most striking finding is that in the **Basic** and **Modality** categories, all trained models performed substantially worse than the `Random` baseline. The `Random` model achieves scores near 0.50, effectively acting as a neutral coin-flip. It has no opinion on the relationship between sentences, and subsequently correctly guesses the proximity of paraphrases roughly half the time by chance. In contrast,

name	basic	mod.	time	style	gen.	opp	costra	costra-
SimCSE-dist-cs △	0.20	0.35	**0.74**	0.63	0.73	**0.78**	0.57	**0.72**
TE3-large ●	0.26	0.35	0.72	0.62	0.74	0.77	**0.58**	0.71
mE5 ⬟	0.24	0.34	0.71	0.62	**0.75**	0.77	0.57	0.71
LaBSE ■	0.20	0.26	0.71	0.63	0.75	**0.75**	0.55	0.71
RetroMAE △	0.06	0.06	0.69	0.63	0.70	0.76	0.48	0.70
TE3-small ●	0.11	0.16	0.70	0.62	0.72	0.76	0.51	0.70
RobeCzech □	0.15	0.13	0.69	**0.65**	0.69	0.75	0.51	0.70
Random BERT ■	0.07	0.06	0.65	0.60	0.72	0.74	0.47	0.68
Czert □	0.31	0.35	0.66	0.64	0.69	0.69	0.56	0.67
XLM-R ⬟	0.16	0.11	0.65	0.61	0.67	0.68	0.48	0.65
FERNET □	0.33	0.38	0.65	0.61	0.63	0.68	0.54	0.64
Random	**0.49**	**0.52**	0.48	0.50	0.50	0.50	0.50	0.50

Table 4.8: This Table presents the results of intrinsic evaluation using the Costra dataset. The score ranges from 0 (worst) to 1 (best) in each category. Costra– score represents the mean score excluding the first two categories (**Basic** and **Mod.**), as these categories appear excessively challenging for all pretrained encoders evaluated.

state-of-the-art models like TE3-large ● (0.26) and mE5 ⬟ (0.24) score far below the coin-flip. This shows that these models are not merely "guessing" incorrectly; they are tricked by surface level similarity, because these categories compare paraphrases with low lexical overlap against "false friends" with high lexical overlap but different meaning. The reliance on surface-level word matching causes them to confidently assign higher similarity to the wrong pairs.

As a result, it is impossible to distinguish genuine model quality from noise in these categories. In order to keep the evaluation reliable, we report the Costra– score, which excludes these two problematic categories and averages performance in the four remaining categories: **Time**, **Style**, **Generalization**, and **Opposite**. However, the Costra– scores revealed only marginal differences between the models' performance. The *small* SimCSE-dist-cs △ model slightly leads the group; however, the model performs only marginally better than Random BERT ■ , suggesting limited success in capturing phenomena tested in Costra, such as linearity of time or generalization. Several trained models, including *large* multilingual XLM-R ⬟ or *base* Czech FERNET □ , even performed worse than Random.

The limited gains over Random BERT ■ suggest that phenomena such as temporal ordering, stylistic variation, and logical opposition are only weakly encoded in current embedding spaces. Overall, the results indicate that while modern sentence embeddings excel in general semantic similarity, they are still predominantly driven by surface-level heuristics. Therefore, the Costra dataset remains a challenging benchmark that highlights a persistent gap between surface-level similarity modeling and truly abstract semantic representation.

4.4.10 Conclusion

In this chapter, we introduce the COSTRA dataset, which addresses a gap in the intrinsic evaluation of Czech sentence embeddings by providing a challenging benchmark focused on complex semantic and stylistic transformations. Through a four-round annotation process, we constructed a dataset with 6,968 unique sentences and 16,385 human pairwise judgments, for testing of both binary semantic distinctions and fine-grained ordinal relations along dimensions such as meaning, time, style and generalization.

We use the COSTRA dataset to evaluate a representative set of state-of-the-art Czech-specific, multilingual, and commercial sentence embedding models. However, all models were outperformed by random baseline in categories that test susceptibility to surface-level similarity, and showed only marginal improvements on scalable categories compared to the untrained `Random BERT` ■ model These findings indicate that current embeddings rely on lexical overlap, with limited ability to capture abstract semantic structure or graded meaning shifts.

The lightweight Python COSTRA toolkit supports reproducible benchmarking of Czech sentence embeddings, and the language-agnostic construction protocol supports future extensions to other languages. In conclusion, COSTRA serves as a challenging probe for progress toward more robust and semantically meaningful sentence embeddings.

To what extent can modern sentence embeddings distinguish between paraphrases and sentences with high lexical overlap but different meanings? The results indicate that current models are largely unable to prioritize semantic content over surface-level similarity. In the **Basic** and **Modality** categories, which contains comparisons of paraphrases and sentences with high word-overlap but divergent meanings, all tested models, including state-of-the-art commercial variants such as `TE3-large` ●, do not outperform a random baseline.

How effectively do embedding spaces capture the ordinal progression of stylistic, temporal, and semantic transformations? Although models perform better than random on scalable dimensions (**Time**, **Style**, **Generalization** and **Opposite**), the improvement over a `Random BERT` ■ is not substantial. The model that performed the best (`SimCSE-dist-cs` △) achieved a COSTRA– score of 0.72, compared to 0.68 for an untrained `Random BERT` ■. This small gap demonstrates that phenomena such as the linearity of time or levels of abstraction are encoded only weakly in current vector spaces.

Do specialized Czech-language models provide a significant advantage over multilingual or general-purpose commercial models in complex semantic tasks? No, specialized Czech encoders do not show a clear superiority in this benchmark. The COSTRA– scores for Czech-specific models like `RetroMAE` △ (0.70) and `SimCSE-dist-cs` △ (0.72) are nearly identical to multilingual models like `mE5` ⬟ (0.71) and commercial models like `TE3-large` ● (0.71).

What is the relationship between model complexity and performance on the Costra benchmark? We do not observe any relation between model complexity and performance. All models perform comparably.

4.5 Intrinsic vs. Extrinsic Evaluation of Czech Sentence Embeddings

As mentioned in Chapter 4, machine translation evaluation has advanced considerably in recent years, moving beyond surface-level, string-matching metrics such as BLEU [Papineni et al., 2002] towards more sophisticated approaches based on neural networks and contextualized embeddings. Metrics like COMET [Rei et al., 2022b] and BLEURT [Sellam et al., 2020a] use sentence embeddings to better capture semantic similarity between translations and references and have achieved much stronger alignment with human judgments than traditional metrics.

At the same time, the rapid development of new embedding models presents MT researchers with a challenging choice. Multilingual models such as LaBSE [Feng et al., 2022] and XLM-RoBERTa [Conneau et al., 2019] demonstrate strong cross-lingual capabilities, while language-specific models claim superior performance for individual languages. However, for morphologically rich languages such as Czech, it remains an open question whether these specialized sentence embeddings provide advantages over multilingual alternatives when applied to MT evaluation.

Typically, embedding models are compared through *extrinsic evaluation*, i.e., directly using them as feature vectors for machine learning algorithms in downstream NLP tasks. This is the ultimate goal – to select the embeddings that perform the best at a given practical application such as machine translation evaluation. It serves well to choose the best method for a particular task but not as an absolute metric of embedding quality, as the performance of the embeddings does not correlate across different tasks [Bakarov, 2018].

Moreover, extrinsic evaluation is also computationally expensive and time-consuming, often requiring fine-tuning each candidate model. *Intrinsic evaluation* offers a more efficient alternative: if tasks such as Semantic Textual Similarity (STS) correlate reliably with downstream MT evaluation, they could serve as practical proxies for model selection. Furthermore, the two tasks are fundamentally similar, both STS and MT evaluation involve judging whether two sentences convey the same meaning despite differences in surface form. However, whether this alignment holds empirically remains an open question.

This chapter[34] explores how well intrinsic evaluation can approximate extrinsic evaluation in the context of Czech MT evaluation. Specifically, we evaluate Czech-specific and multilingual sentence embedding models using (i) semantic benchmarks such as Costra and STS datasets, and (ii) by fine-tuning them within the COMET framework for translation evaluation and quality estimation.

In particular, we examine the following research questions:

- **Can intrinsic evaluation serve as a reliable predictor of embedding performance in MTE/QE?**

[34]This chapter is based on Barančíková and Bojar [2025]. I was responsible for all experimental design, implementation, and evaluation. Ondřej Bojar contributed through consultations and co-writing of the published paper.

- **Is it better to use a Czech-specific or multilingual encoder for Czech sentence evaluation?**
- **Does model size matter after fine-tuning in MTE/QE?**

4.5.1 Related Work

Several studies have already questioned the suitability of STS for the embedding selection. For example, Reimers et al. [2016], Eger et al. [2019], and Zhelezniak et al. [2019] argue that, while STS can capture certain semantic similarities, it does not reliably predict how effective sentence representations will be for downstream tasks. These works show that STS tasks often encourage surface-level heuristics or oversimplified semantic similarity patterns that may not generalize to more complex applications like entailment or paraphrasing detection.

To address these limitations, new intrinsic evaluation methods such as EvalRank [Wang et al., 2022a] and SentBench [Xiaoming et al., 2023] have been proposed, both of which exhibit a stronger correlation with extrinsic evaluation measures. These benchmarks evaluate sentence representations through information retrieval, sentence ordering, and probing tasks, offering a more holistic view of embedding quality that aligns better with actual downstream task performance.

It is important to note that these previous experiments did not specifically focus on machine translation evaluation, which seems to be very close to STS, in the sense that they both compare pairs of sentences and evaluate their semantic similarity. Cífka and Bojar [2018] report a negative correlation between the translation quality of the Transformer models measured by BLEU and the semantic properties (assessed using STS) of the sentence embeddings derived from the Transformer model. In contrast, Libovický and Madhyastha [2019] demonstrate a strong positive correlation between STS performance and translation quality for both Transformer and RNN-based models.

Freitag et al. [2022] have advocated for the use of semantic-aware metrics such as BERTscore [Zhang et al., 2020] and COMET [Rei et al., 2020] in MT evaluation, showing that these outperform BLEU in correlating with human judgments. These models incorporate contextual embeddings and often exhibit closer alignment with human-perceived meaning, bringing MT evaluation closer to the goals of intrinsic semantic understanding.

4.5.2 Models

For our experiments, we used several state-of-the-art sentence embedding models, employing both Czech-specific and multilingual variants. The Czech encoders include `Czert` □ ,`FERNET` □ , `RobeCzech` □ , `RetroMAE` △ and `SimCSE-dist-cs` △ . To provide a broader comparison, we also evaluated several multilingual sentence embedding models trained on datasets containing Czech texts. These include `LaBSE` ■ , `XLM-R` ⬟ and `mE5` ⬟ . As a baseline model, we included a `Random BERT` ■ . For more details about the models, we refer to Section 4.2. **??** provides a brief overview of their architectures.

Embeddings	CNA	SVOB_IMG	SVOB_HL	mean
SimCSE-dist-cs △	88.56	**90.73**	**84.22**	**87.83**
mE5 ⬟	**89.25**	90.13	82.77	87.38
LaBSE ■	85.54	85.58	77.62	82.91
RetroMAE △	82.96	79.10	63.63	76.30
Czert □	84.10	77.04	63.24	74.79
RobeCzech □	76.35	71.24	60.64	69.41
FERNET □	78.22	63.59	54.56	65.46
Random BERT ■	65.56	63.70	52.17	60.48
XLM-R ⬟	68.48	51.70	45.19	55.12

Table 4.9: Results of intrinsic evaluation on three STS datasets. We used the best performing pooling method and cosine similarity as similarity function.

4.5.3 Intrinsic Evaluation

We first evaluate the embeddings on a series of semantic benchmarks to examine how accurately they capture various semantic properties of sentences. As the first dataset for intrinsic evaluation, we used the Costra– dataset [Barančíková and Bojar, 2020], which was described in detail in the previous section.

As established in Section 4.4.9, current sentence encoders struggle to move beyond surface-level heuristics. For the purposes of this comparative study, we focus on the Costra– aggregate score, which measures the models' ability to capture scalable semantic relations (**Time**, **Style**, **Generalization**, and **Opposition**) while excluding the "surface-similarity trap" of the **Basic** and **Modality** categories.

The results, summarized in the last column of Table 4.8, show that the distilled SimCSE-dist-cs △ model possesses the strongest intrinsic grasp of these relations (0.72), however, the differences among the trained models are only marginal. In fact, the best models performed only marginally better than the Random BERT ■ baseline, suggesting limited success in capturing the phenomena tested in the Costra dataset, such as linearity of time or generalization.

Table 4.9 presents the results of our evaluation of sentence embeddings on the Semantic Textual Similarity (STS) task. Performance is measured using the automated evaluation tool[35] provided by Bednář et al. [2024].

The Semantical Textual Similarity (STS) is evaluated on three datasets. The first dataset, CNA, comes from the test set of Sido et al. [2021a] and contains 1,100 sentence pairs drawn from Czech journalistic texts. The similarity labels were obtained by averaging nine independent human annotations, providing a high-quality reference for semantic equivalence judgments.

The remaining two datasets, SVOB-IMG and SVOB-HL, were created by Svoboda and Brychcín [2018] as Czech adaptations of the SentEval STS benchmark [Conneau and Kiela, 2018]. SVOB-IMG includes 850 sentence pairs derived from image captions, focusing on descriptive and contextually rich sentences. SVOB-HL consists of 525 sentence pairs from news headlines. Both datasets were translated from English into Czech and manually re-annotated for semantic

[35] https://github.com/seznam/czech-semantic-embedding-models

similarity.

We evaluated all models using the automated evaluation tool[36] provided by Bednář et al. [2024]. For precomputed sentence embeddings, it explores different embedding similarity metrics including cosine similarity, dot product, and Manhattan distance. Additionally, it applies various sentence embeddings pooling strategies and selects the highest average score as the final result.

Interestingly, the STS results are consistent with the findings from COSTRA, with `SimCSE-dist-cs` △ being the overall best performing model, followed by `mE5` ⬟ , `LaBSE` ■ and `RetroMAE` △ . `XLM-R` ⬟ , despite being a large and powerful multilingual encoder, may not be well-optimized for Czech-specific STS tasks, ranking last in the evaluation, performing even worse than `Random BERT` ■ . This result is in line with Litschko et al. [2022], who shows that multilingual encoders such as mBERT and `XLM-R` ⬟ often underperform on sentence-level semantic similarity tasks without task-specific fine-tuning.

4.5.4 Extrinsic Evaluation—MTE and QE

As our extrinsic evaluation, we selected two tasks: machine translation evaluation (MTE) and quality estimation (QE). These tasks are very similar – both evaluate the quality of a translation hypothesis. The main difference is that MTE relies on reference sentences, while QE does not. Nevertheless, both tasks may incorporate the source sentence as additional context. A more detailed discussion of these tasks is provided in Section 2.2.

Data

In the following experiments, we use datasets from the Workshop on Machine Translation (WMT), selecting data from English-to-Czech translations. These datasets include English source sentences, Czech hypotheses, Czech reference sentences, and the human translation quality scores collected using the Direct Assessment (DA) method [Graham et al., 2013] and subsequently `z`-normalized to take individual annotator bias into account by centering and scaling scores of each annotator.

Data from WMT17 to WMT19 [Bojar et al., 2017a, 2018, Barrault et al., 2019] were used to train the COMET estimators [Rei et al., 2020]. The validation of the models was performed on the WMT20 dataset [Barrault et al., 2020], and the performance of the models was tested using the WMT21 Akhbardeh et al. [2021] and WMT22 [Kocmi et al., 2022] datasets.

MTE Baseline Approach

Before fine-tuning the sentence embedding models for machine translation evaluation and quality estimation, we conducted a preliminary analysis to assess their default ability to evaluate translation quality. Specifically, we examined Pearson's correlation between human judgments and the cosine similarities computed between (i) a hypothesis and a reference translation and (ii) a hypothesis and a source sentence. We expected high cosine similarity for multilingual models,

[36] `https://github.com/seznam/czech-semantic-embedding-models`

Sentence	WMT21 test set			WMT22 test set			
Embeddings	$\rho_{H,R}$	$\rho_{H,S}$	$S_C(S,R)$	$\rho_{H,R}$	$\rho_{H,S}$	$S_C(S,R)$	$S_C(S_R,R_R)$
`mE5` ⬟	**0.29**	0.04	0.89	0.26	0.01	0.90	0.75
`RetroMAE` △	0.26	-0.10	0.76	**0.27**	**0.09**	0.76	0.69
`SimCSE-dist-cs` △	0.24	**0.13**	0.85	0.25	0.05	0.82	0.09
`Czert` □	0.20	-0.03	0.63	0.18	-0.06	0.62	0.52
`XLM-R` ⬟	0.17	-0.08	1.00	0.05	-0.10	1.00	0.99
`RobeCzech` □	0.15	-0.16	0.92	0.11	-0.06	0.91	0.89
`LaBSE` ■	0.11	0.03	0.89	0.19	0.06	0.88	0.31
`FERNET` □	0.07	-0.11	0.45	0.11	-0.03	0.40	0.35
`Random BERT` ■	0.06	-0.16	0.99	0.03	-0.20	0.98	0.98

Table 4.10: Results of the baseline MTE approach – using sentence embeddings for direct evaluation without fine-tuning. $\rho_{H,R}$ represents Pearson correlation between human quality assessments and the cosine similarity between the translation hypothesis and the reference translation, while $\rho_{H,S}$ shows the correlation between human judgments and the cosine similarity between the hypothesis and the source sentence. $S_C(S,R)$ represents cosine similarity between references and sources. The last column represents cosine similarity between randomly shuffled source and reference sentences averaged over 100 runs.

reflecting their ability to capture cross-lingual semantic relationships, whereas Czech-specific models—lacking such cross-lingual information—were anticipated to have random similarity scores.

Furthermore, we examined the intrinsic quality of the embedding spaces by measuring the cosine similarity between the source and reference embeddings. We also performed a random shuffling experiment designed to evaluate the discriminative ability of the embeddings.

The results presented in Table 4.10 reveal that even without fine-tuning, a slight correlation between human judgments and cosine similarity of hypotheses and references is observable in certain models—particularly `mE5` ⬟ , `RetroMAE` △ , and `SimCSE-dist-cs` △ . However, contrary to expectations, this does not hold for source sentences; no relationship was detected between human evaluation scores and the cosine similarity computed between a translated sentence and its source sentence, even among the multilingual models.

The analysis of the embedding space via similarity between source and reference sentences provides further insights. In line with our hypothesis, `XLM-R` ⬟ exhibits a near perfect similarity between the source and reference sentences, which indicates that the `XLM-R` ⬟ latent space is collapsed for the Czech language before task-specific training. In this state, the high-dimensional vectors are compressed into such a narrow cone that the model loses the resolution necessary to distinguish between semantic equivalence and random noise, causing even unrelated shuffled pairs to appear identical.

To further investigate this behavior, we repeated the experiment using random shuffle of source and reference sentences; see the last column of Table 4.10. The similarity remained perfect for both `XLM-R` ⬟ and `Random BERT` ■ even on shuffled pairs, indicating an overly invariant embedding space, where even pairs of semantically unrelated sentences tend to cluster together. This *over-smoothing*

reduces the model's capacity to distinguish subtle differences that are essential for evaluating translation quality. In such cases, even bad translations can receive high similarity scores, lowering the correlation with human judgment. This also explains the poor performance of `XLM-R` ⬟ in our intrinsic evaluation task, especially in STS (Table 4.9). More broadly speaking, it casts doubts on any results based on the direct similarity of `XLM-R` ⬟ embedding vectors in the Czech language, given that `XLM-R` ⬟ assigns similar vectors to all tested sentence pairs from WMT22.

Models fine-tuning for MTE and QE

For all sentence encoders, we fine-tuned two COMET-based estimators (CE, Rei et al., 2020), one for machine translation evaluation using reference sentences and the other for quality estimation without reference sentences. The COMET models use a dual-encoder architecture: the source sentence, reference translation, and hypothesis are each processed independently using transformer encoder models followed by two hidden layers of sizes 3072 (resp. 2048 for QE) and 1024.

We used the default training settings with the AdamW optimizer ($1.5 \cdot 10^{-5}$ for the regression layers and $1.0 \cdot 10^{-6}$ for the encoder) and a layer-wise decay of 0.95. To preserve encoder generalization, the embeddings were frozen for the first 0.3 epochs. Both models used mixed-layer pooling with a sparsemax-based transformation before pooling and were optimized with mean squared error loss (using a dropout of 0.1). Training was conducted over five epochs, and we selected the checkpoint with the highest Kendall's tau validation on a held-out validation dataset.

These settings were applied consistently across all models without extensive hyperparameter tuning. In total, we trained a total of 18 COMET estimators (i.e. number of encoders × 2 tasks). To avoid confusion with the original embeddings, we refer to a trained COMET estimator for given embeddings X as to $\mathrm{CE}_{\mathrm{MTE}}$(X) for machine translation with reference sentences and $\mathrm{CE}_{\mathrm{QE}}$(X) for the referenceless quality estimation metric (e.g., for the `Czert` □ embeddings, we use $\mathrm{CE}_{\mathrm{MTE}}$(`Czert` □) and $\mathrm{CE}_{\mathrm{QE}}$(`Czert` □), respectively).

Results of MTE and QE evaluation

We compare the performance of trained evaluation metrics at the system level with traditional string-matching MTE metrics. Specifically, we include BLEU [Papineni et al., 2002], TER [Snover et al., 2006a], and chrF [Popović, 2015], using their default configurations as implemented in SacreBLEU [Post, 2018]. Because of reproducibility, we report the SacreBLEU signature strings used in our experiments:

BLEU	nrefs:1|case:mixed|eff:no|tok:13a|smooth:exp|version:2.4.3
TER	nrefs:1|case:lc|tok:tercom|norm:no|punct:yes|asian:no|version:2.4.3
chrF	nrefs:1|case:mixed|eff:yes|nc:6|nw:0|space:no|version:2.4.3

Additionally, we employ Meteor-next [Denkowski and Lavie, 2010b], a metric that includes paraphrase support, on both system and segment levels.

Furthermore, we compute scores using the official pretrained COMET models for machine translation evaluation, namely `wmt22-comet-da` [Rei et al., 2022a],

	system-level		segment-level	
MTE metrics	2021	2022	2021	2022
CE_{MTE}(FERNET □)	**0.98**	**0.97**	0.60	0.47
wmt22-comet-da	0.97	0.93	**0.66**	**0.51**
CE_{MTE}(Czert □)	0.96	0.93	0.57	0.43
CE_{MTE}(XLM-R ⬟)	0.96	0.93	0.62	0.47
CE_{MTE}(RobeCzech □)	0.97	0.92	0.58	0.44
CE_{MTE}(mE5 ⬟)	0.96	0.92	0.59	0.46
METEOR	**0.98**	0.84	0.24	0.21
chrF2	0.97	0.84	-	-
CE_{MTE}(RetroMAE △)	0.91	0.82	0.43	0.34
CE_{MTE}(LaBSE ■)	0.89	0.79	0.56	0.45
CE_{MTE}(SimCSE-dist-cs △)	0.96	0.74	0.44	0.37
1-TER	0.95	0.60	-	-
BLEU	0.94	0.54	-	-
CE_{MTE}(Random BERT ■)	0.35	-0.35	0.23	0.22

	system-level		segment-level	
QE metrics	2021	2022	2021	2022
CE_{QE}(FERNET □)	**0.98**	**0.96**	0.60	0.46
wmt22-cometkiwi-da	0.95	0.91	**0.67**	**0.49**
CE_{QE}(XLM-R ⬟)	0.96	0.88	0.63	**0.49**
CE_{QE}(RobeCzech □)	0.97	0.87	0.58	0.39
CE_{QE}(Czert □)	0.95	0.86	0.57	0.39
CE_{QE}(mE5 ⬟)	0.93	0.76	0.59	0.45
CE_{QE}(LaBSE ■)	0.83	0.39	0.54	0.40
CE_{QE}(RetroMAE △)	0.64	0.15	0.39	0.23
CE_{QE}(Random BERT ■)	0.47	-0.19	0.26	0.20
CE_{QE}(SimCSE-dist-cs △)	0.12	-0.92	0.38	0.24

Table 4.11: Correlations between human scores and evaluation metrics, including both fine-tuned COMET-based metrics and traditional metrics, computed at the system and segment levels.

and for quality estimation, specifically wmt22-cometkiwi-da [Rei et al., 2022b]. These COMET models extend beyond a simple trained COMET estimator, as they incorporate an ensemble approach combining a COMET estimator trained on DA data (See Section 2.1.3) and sequence predictors trained on MQM annotations (See Section 2.1.4).

The results in Table 4.11 indicate a clear advantage for COMET-based evaluation metrics over traditional metrics in MTE. In the system-level analysis, the COMET variants CE_{MTE}(FERNET □) and CE_{QE}(FERNET □) achieved consistently remarkably high correlation outperforming even the official COMET ensemble metrics – wmt22-comet-da and wmt22-cometkiwi-da, which were the top-performing metrics at the segment level.

In contrast, classical metrics, although competitive in 2021, showed signifi-

cant drop in performance in 2022. CE_{MTE}(Random BERT ■) failed completely, highlighting the importance of using pretrained sentence embeddings. Interestingly, CE_{QE}(Random BERT ■) outperformed CE_{QE}(SimCSE-dist-cs △), despite SimCSE-dist-cs △ being the best performing encoder in intrinsic evaluation.

Another interesting observation is the minimal difference in correlations of the top performing embeddings between MTE and QE. CE_{MTE}(FERNET □) and CE_{QE}(FERNET □) have nearly identical correlation at both the system and segment levels, suggesting that these models may no longer rely heavily on reference translations. This finding is consistent with recent research showing that reference-free evaluation has become competitive with reference-based evaluation [Rei et al., 2021] or even outperforms it [Moosa et al., 2024].

4.5.5 Intrinsic vs. Extrinsic Evaluation

When comparing the results of MTE and QE with those of the intrinsic evaluation tasks, we can observe an interesting inversion. Although both evaluation approaches aim to capture semantic similarity, the performance of the embeddings changed significantly after fine-tunings. Specifically, XLM-R ⬟ and FERNET □ embeddings, which performed poorly in intrinsic evaluation, became the best performing MTE and QE metrics. In contrast, SimCSE-dist-cs △ , which dominated intrinsic evaluations, ranks among the worst performing metrics in MTE and QE. These results are in line with related research (Section 4.5.1), which shows that STS performance may not accurately predict effectiveness in downstream tasks.

There are several plausible hypotheses that might explain these discrepancies. Let us at least mention them here— unfortunately, their thorough testing is beyond the scope of this article.

First, XLM-R ⬟ and FERNET □ might perform poorly in intrinsic tasks because their representation spaces are not tuned for fine-grained semantic differences. However, when fine-tuned on a translation quality task, the model might learn to emphasize those aspects of the embedding space that are important for distinguishing translation quality.

The fine-tuning process in COMET-based evaluation may effectively reconfigure the XLM-R ⬟ embedding space, transforming its initially over-smoothed representations into task-specific features that are highly discriminative for translation quality. Although the raw XLM-R ⬟ embeddings appear largely clustered together (see Table 4.10), fine-tuning likely introduces transformations that reweight and separate dimensions relevant for capturing translation errors. In contrast, SimCSE-dist-cs △ embeddings, which are already optimized for intrinsic semantic discrimination, may leave less room for further adaptation to a new training objective.

We should also not forget about the different embedding sizes, which played an important role in the observed behavior. The *small* embeddings—SimCSE-dist-cs △ and RetroMAE △ —were among the worst performing COMET estimators. Large embeddings, such as those produced by XLM-R ⬟ , offer a higher-dimensional space that can capture more nuanced semantic and syntactic features. When fine-tuning with the COMET estimator—which adds two hidden layers with sizes 3072 (resp. 2048) and 1024—the richer representation provided by larger em-

beddings could allow the model to extract and emphasize the translation-specific signals more effectively.

The performance of `SimCSE-dist-cs` △ illustrates the risks of distillation and task-specific tuning. Despite strong intrinsic results, its poor performance on MTE/QE tasks suggests limited generalization. Optimized primarily for STS, `SimCSE-dist-cs` △ focuses on surface-level distinctions and may discard latent semantic structure retained by larger, less specialized models such as `XLM-R` ⬟ and `FERNET` □ . Although this residual structure appears as noise in intrinsic evaluations, it provides essential signal for downstream adaptation, which the COMET regression head can exploit once the embedding space is re-scaled during fine-tuning.

Interestingly, we can see not too much difference between the monolingual vs. multilingual embedding performances—they seem to be equally represented among the best performing embeddings in both intrinsic and extrinsic tasks. The size of the embeddings seem not to matter in the intrinsic tasks—the top 3 best performing embeddings (`SimCSE-dist-cs` △ , `LaBSE` ■ and `mE5` ⬟) are *small*, *base* and *large*, respectively.

The correlation analysis between different evaluation methodologies, visualized by heatmaps in Figure 4.8, reveals interesting patterns of how different evaluation methodologies relate to each other. These patterns provide valuable insight into the reliability and consistency of various embedding evaluation approaches.

The heatmaps highlight a strong alignment among all intrinsic tasks (Costra–, STS, and MTE baselines). Moreover, there is a strong correlation between the segment-level and the system-level metrics, indicating that aggregated segment scores provide reliable system-level insights. In particular, we observe strong correlations between segment-level metrics (*segment MTE* and *segment QE* showing correlations of 0.97 and 0.91 for 2021 and 2022 respectively), suggesting that these evaluation approaches capture similar aspects of translation quality despite their methodological differences.

However, one of the most striking findings is the weak and sometimes even negative correlation between intrinsic evaluation metrics (Costra–, STS) and the system-level quality estimation scores *system QE*. This discrepancy is particularly evident in the 2022 data, where Costra– shows a negative correlation (-0.52) with *system QE*, challenging the assumption that better semantic representation capabilities necessarily translate to improved MT evaluation performance.

These findings indicate that intrinsic measures, while useful for general semantic similarity, may not sufficiently reflect translation-specific nuances required for MTE or QE. Consequently, our results caution against using intrinsic semantic benchmarks as proxies for downstream MT evaluation and highlight fine-tuning capacity and model scale as dominant factors. Further research is needed to identify intrinsic evaluation methods that better capture the subtleties relevant to machine translation. Additionally, it would be valuable to explore in more detail the types of errors penalized in manual MT quality assessments to determine whether these errors predominantly concern sentence meaning or other aspects that should be preserved in translation.

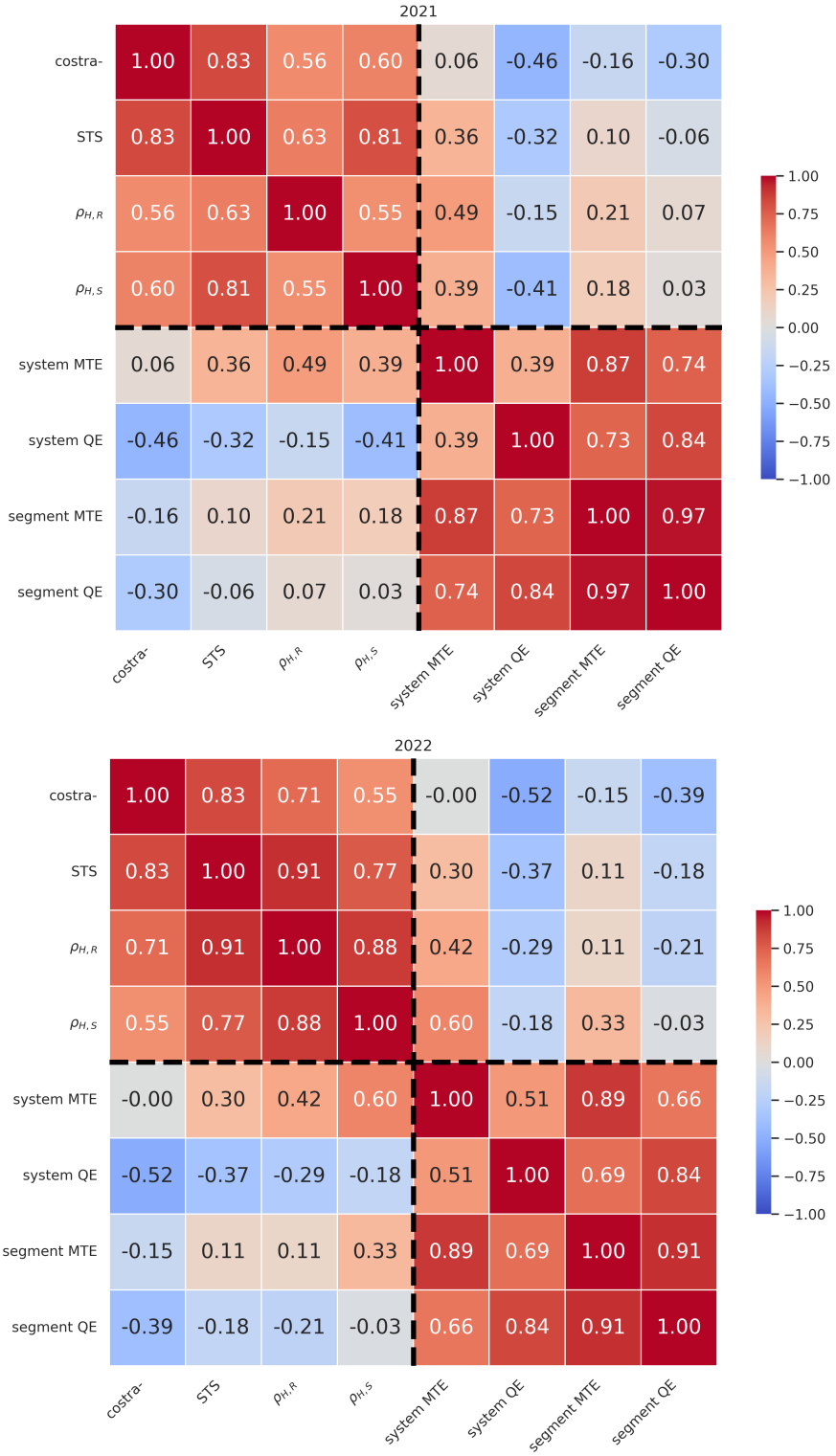


Figure 4.8: Correlation heatmaps for different method of embedding evaluations. The heatmaps are intentionally kept in a rectangular shape to emphasize the mismatch in correlation patterns between intrinsic evaluation (Costra–, STS, $\rho_{H,R}$, $\rho_{H,S}$) and extrinsic evaluation (system MTE/QE and segment MTE/QE).

4.5.6 Conclusion

We experimented with several evaluation methods for both Czech and multilingual sentence embeddings, considering intrinsic semantic tasks and downstream application in machine translation evaluation and quality estimation. We answered the following research questions:

Do intrinsic metrics (Costra, STS, baseline cosine) predict downstream MTE/QE performance? No, we observe a lack of correspondence between the two tasks for Czech data. Models that excel in STS/COSTRA (e.g., `SimCSE-dist-cs` △) do not translate into the best MTE/QE metrics after fine-tuning, while models with weak intrinsic scores (`XLM-R` ⬟ , `FERNET` □) become top performers once trained within the COMET framework. The correlations between intrinsic scores (Costra–, STS) and system-level QE are weak or negative, indicating that the tested intrinsic tasks are poor selectors for downstream MT evaluation.

Are Czech-specific encoders better than multilingual embedders for evaluation of Czech translation in fine-tuned COMET metrics? Not decisively. There are no strong differences in performance between language-specific and multilingual models. Both categories are comparably represented among the top-performing models in intrinsic and extrinsic tasks.

Does model size matter after fine-tuning for MTE/QE? Yes. In contrast to intrinsic tasks, fine-tuning embeddings for MTE/QE reveals that model size does matter, with the *small* embeddings consistently showing poor performance, while *base*/*large* encoders provide richer representations that the regression head can leverage.

Do references still provide clear benefits for the strongest models? Only marginally in our experiments. The top reference-based and reference-free estimators (CE_{MTE}(`FERNET` □) vs. CE_{QE}(`FERNET` □)) achieve nearly identical correlations at both system and segment levels, suggesting a reduced marginal utility of references when encoders are strong. The same holds for the official ensemble baselines.

We also observed stability of the correlations over time between the system-level COMET-based metrics, which provides encouraging evidence for the reliability of these evaluation approaches in contrast to traditional string-based metrics, which remained competitive in 2021 but degraded substantially in 2022.

Conclusion

This thesis examines two complementary approaches to automatic machine translation evaluation. The first part focuses on targeted paraphrasing of reference translations as a method to increase the reliability of traditional string-based evaluation metrics. The second part examines the behavior and limitations of modern sentence embedding models with respect to machine translation evaluation. These two approaches together demonstrate difficulties in measuring semantic equivalence in machine translation.

In the first part, we show that the reliability of string-based metrics such as BLEU can be improved by creating artificial reference sentences more similar to machine translation outputs. In particular, we find that simple substitution-based paraphrasing methods often work better than more complex approaches based on statistical or rule-based machine translation systems. This result demonstrates that carefully replacing individual words can be more effective than generating complete paraphrases that differ more strongly in structure. We also show that large, automatically created paraphrase resources can reduce evaluation quality by introducing noise. By filtering and moving paraphrase processing from evaluation to preprocessing, we obtain much better results. In addition, our experiments present that grammatical correctness matters: paraphrased references that were linguistically well-formed led to more reliable evaluation.

The second part of the thesis focuses on sentence embeddings as a more recent approach to semantic evaluation. We analyze the geometric properties of both English and multilingual sentence embedding spaces and introduce a new dataset for Czech sentence embeddings evaluation. Our experiments show that even strong, state-of-the-art embedding models still tend to rely on surface-level similarity and often fail to identify paraphrases. We also find that good performance on standard intrinsic evaluation tasks does not reliably predict how well an embedding model will perform when used for machine translation evaluation. This finding is especially interesting given that intrinsic evaluation and machine translation evaluation are both intended to measure semantic similarity.

Both parts of this thesis lead to a common conclusion: more complex methods do not automatically lead to better evaluation results. Simple string-based metrics, when supported by carefully constructed and grammatically sound paraphrases, can perform as well as or better than more complex approaches. In the same way, larger and more general sentence embedding models do not always outperform smaller or more specialized models when they are applied to machine translation evaluation. These findings show that evaluation quality depends strongly on how well a method is matched to the specific task, rather than on model complexity alone.

5. Appendix

5.1 Example of distributional similarity computation

To illustrate the computation of cosine similarity in the context of text analysis, we use a simple corpus to demonstrate how this measurement can be calculated to assess the similarity between word distributions. The process begins with constructing a co-occurrence matrix within sentences from a given corpus. Consider a corpus consisting of the following three sentences:

1. Birds fly high, birds fly fast.
2. Fly high, fly often.
3. Often birds sing loudly.

In the first step, we split sentences into words, remove punctuation, and lowercase all words. In this example, we focus on a window size of 1, which means that we are only considering immediate neighbors within a sentence. In this co-occurrence matrix (Table 5.1), word order is not taken into account, leading to a symmetrical structure. Each entry in the matrix represents the frequency with which each word appears adjacent to another, regardless of whether it appears before or after.

Word	**birds**	**fast**	**fly**	**high**	**loudly**	**often**	**sing**
birds	[0,	0,	2,	1,	0,	1,	1]
fast	[0,	0,	1,	0,	0,	0,	0]
fly	[2,	1,	0,	3,	0,	1,	0]
high	[1,	0,	3,	0,	0,	0,	0]
loudly	[0,	0,	0,	0,	0,	0,	1]
often	[1,	0,	1,	0,	0,	0,	0]
sing	[1,	0,	0,	0,	1,	0,	0]

Table 5.1: The co-occurrence matrix presented here is computed from the simple three-sentence corpus, using a context window size of 1 and ignoring word order. This matrix illustrates how frequently each word appears adjacent to every other word within the corpus.

From the matrix, we can derive context vectors for the words "fly" and "sing" based on their immediate neighbors:
Context(fly) = [2, 1, 0, 3, 0, 1, 0]
Context(sing) = [1, 0, 0, 0, 1, 0, 0]
We calculate the cosine similarity between these two vectors to quantify the similarity of their distribution.

$$
\begin{aligned}
\text{CS(fly,sing)} &= \frac{Context(fly) \cdot Context(sing)}{\|Context(fly)\|\|Context(sing)\|} \\
&= \frac{(2 \times 1) + (1 \times 0) + (0 \times 0) + (3 \times 0) + (0 \times 1) + (1 \times 0) + (0 \times 0)}{\sqrt{2^2 + 1^2 + 0^2 + 3^2 + 0^2 + 1^2 + 0^2}\sqrt{1^2 + 0^2 + 0^2 + 0^2 + 1^2 + 0^2 + 0^2}} \\
&= \frac{2}{\sqrt{15}\sqrt{2}} \\
&\approx 0.37
\end{aligned}
$$

In this example, the cosine similarity is approximately 0.37 suggesting relatively small degree of similarity between the contexts of word *fly* and *sing*.

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List of Abbreviations

ACL – Association for Computational Linguistics
biLM - bi-directional Language Model
BiLSTM - Bidirectional Long Short-Term Memory
BLEU - Bilingual Evaluation Understudy
BERT – Bidirectional Encoder Representations from Transformers
BLEURT - Bilingual Evaluation Understudy with Representations from Transformers
BP - Brevity Penalty
COMET - Crosslingual Optimized Metric for Evaluation of Translation
chrF - Character-level F-score
DA - Direct Assessment
DARPA - Defense Advanced Research Projects Agency
ESA - Error Span Annotation
GPT – Generative Pre-trained Transformer
IAA - Inter-Annotator Agreement
LDC – Linguistic Data Consortium
LLM - Large Language Model
MERT - Minimum Error Rate Training
Meteor - Metric for Evaluation of Translation with Explicit Ordering
MFAS - Minimum Feedback Arc Set
MLM - Masked Language Modeling
MultiNLI - Multi-Genre NLI
MRL - Matryoshka Representation Learning
MRPC - Microsoft Research Paraphrase Corpus
MQM – Multidimensional Quality Metrics
MT - Machine Translation
MTEB - Massive Text Embedding Benchmark
MTurk – Amazon Mechanical Turk
NLI - Natural Language Inference
NLP – Natural Language Processing
NMT - Neural Machine Translation
PDT - Prague Dependency Treebank
PPDB – Paraphrase Database
Prism – Probabilistic Reference-based Interpretation of Semantic Meaning
QE – Quality Estimation
RoPE - Rotary Positional Embeddings
RR - Relative Ranking
SMT - Statistical Machine Translation
SQM – Scalar Quality Metric
SNLI - Stanford Natural Language Inference
SOTA - State-of-the-Art
STS - Semantic Textual Similarity
TER - Translation Error Rate
TERp – Translation Error Rate Plus
UniTE – Unified Translation Evaluation

WMT - Workshop on Statistical Machine Translation
XLM-R – Cross-lingual Language Model – RoBERTa

List of publications

Publications Directly Related to the Thesis

1. Petra Barančíková and Ondřej Bojar. (2025). *Intrinsic vs. Extrinsic Evaluation of Czech Sentence Embeddings: Semantic Relevance Doesn't Help with MT Evaluation.* In *Proceedings of Machine Translation Summit XX: Volume 1* (pp. 265–275). Geneva, Switzerland: European Association for Machine Translation.

2. Jiří Bednař, Jakub Náplava, Petra Barančíková, and Ondřej Lisický. (2024). *Some Like It Small: Czech Semantic Embedding Models for Industry Applications.* In *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(21), 22734–22742.

3. Petra Barančíková and Ondřej Bojar. (2020). *Costra 1.1: An Inquiry into Geometric Properties of Sentence Spaces.* In *Proceedings of the 23rd International Conference on Text, Speech and Dialogue* (pp. 135–143). Cham, Switzerland: Springer.

4. Petra Barančíková and Ondřej Bojar. (2020). *COSTRA 1.0: A Dataset of Complex Sentence Transformations.* In *Proceedings of the Twelfth Language Resources and Evaluation Conference* (pp. 3535–3541). Marseille, France: European Language Resources Association.

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(LREC'14). Reykjavik, Iceland: European Language Resources Association.

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11. Petra Barančíková. (2013). *Lexical Paraphrasing for Improvement of MT Evaluation.* In *WDS'13 Proceedings of Contributed Papers* (pp. 7–11). Prague, Czechia: Matfyzpress.

Other Publications

1. Václava Kettnerová, Markéta Lopatková, Anna Vernerová, and Petra Barančíková . (2020). *Towards a Semi-Automatic Detection of Reflexive and Reciprocal Constructions and Their Representation in a Valency Lexicon.* In *Proceedings of the Twelfth Language Resources and Evaluation Conference* (pp. 3136–3144). Marseille, France: European Language Resources Association.

2. Václava Kettnerová, Markéta Lopatková, Eduard Bejček, and Petra Barančíková . (2018). *Enriching VALLEX with Light Verbs: From Theory to Data and Back Again. The Prague Bulletin of Mathematical Linguistics, 111*, 29–56.

3. Petra Barančíková and Václava Kettnerová. (2018). *Paraphrases of Verbal Multiword Expressions: The Case of Czech Light Verbs and Idioms.* In *Multiword Expressions at Length and in Depth: Extended Papers from the MWE 2017 Workshop* (pp. 35–59). Berlin, Germany: Language Science Press. ISBN 978-3-96110-123-8.

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